

HIRING MANAGER INTERVIEW PACK

Walmart Marketplace SFS R-2596147 War Book

Req R-2596147

Senior Manager, Technology Operations

Pre-Purchase Delivery Experience · Seller Fulfillment Solutions

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Hiring manager interview · Tuesday, September 9, 2026 · 30 minutes

CONFIDENTIAL INTERVIEW PREP · NOT FOR DISTRIBUTION

R-2596147

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What you do with this on Tuesday

Walmart Marketplace SFS – The R-2596147 War Book

Senior Manager, Technology Operations • Pre-Purchase Delivery Experience Hiring-manager interview • Tuesday Sep 9, 2026 • 30 minutes Rajesh Rishiyur Chandrasekaran • rrishiyur@gmail.com • (318) 533-2815

Companion artifacts: `Walmart_WBR_R2596147_Solver_Model.xlsx` (live WBR + Excel Solver + HM One-Pager) • `model_reference.py` (verified math) • interview Q&A bank (Part 9, including Amazon LP / dashboard / bar-raiser sections 11–16).

How to use this book

Thirty minutes with a hiring manager is not a content problem — it is a selection problem. This book over-prepares you on purpose.

Parts 1–3 are a classroom, not a memo. They walk one physical order — Maya’s blender — from the product page to the porch, then teach every formula, every output metric, and every Amazon mechanism as something that happened (or failed to happen) to that box. Read them so you can *derive* O1–O7 on a napkin, not recite them. Memorize nothing except the six numbers in Part 5 and the three story nuggets in Part 7. Walk in with one goal: **be the only candidate who talks about the delivery promise as an underwriting business, with the math to back it.** Everything else is depth-on-demand.

****HM lens.**** An ex-Amazon hiring manager is not grading whether you know Maya’s story. They are grading whether you can **work backwards from a mechanism**: named owner, grain, window, denominator, and what you would do this week. After you understand the blender, the test is: can you draw Tuesday’s one-pager (Part 8.5) and survive “what did YOU do?” (Q-bank §11).

A note on every number in this book

Three kinds of numbers appear. They are not interchangeable.

1. **WBR sample (Wk 28–Wk 33, Mon Jul 6 – Sun Aug 16, 2026).** From the team’s own deep-dive workbook `Walmart_WBR_R-2596147.xlsx`. These are **illustrative sample series** for teaching the WBR grammar — not live Walmart production data. Said once, here, and it covers every O1–O7 / I1–I10 figure that follows.
2. **Canonical 12-lane model** in `model_reference.py`. Also illustrative, but internally consistent. Policy A/B/C/D numbers (including “uniform 1.10-day buffers → 7.6% fast share”) are computed from that file. Reuse them; do not re-invent them.
3. **Toy / ILLUSTRATIVE examples** (1,000 product-page views, 100 orders, a 60/40 carrier-vs-warehouse split). Labeled **ILLUSTRATIVE** at the point of use. They exist so you can check the arithmetic on paper. They are not Rajesh resume metrics and not Walmart actuals.

No resume number in this book was invented for the interview. Experience figures live in Part 7 and the Q&A bank, unchanged.

Recommendation – SFS FY27 Q2 QBR

This is not Walmart recommending you. It is the live scoring sheet. The *SFS FY27 Q2 QBR notes* (internal takeaways from the discussion) are what Kyle, Anil, Gagan, Aaron, and Rahil just argued. Tuesday's hiring manager already lives in this agenda. **Do not recite the notes.** Answer every question as if you already work this P&L.

Every figure in this section is **from those QBR notes** — not the WBR sample, not `model_reference.py`, and not a resume metric. Use them to aim. Do not volunteer that you read an internal notes dump.

****Play this Tuesday.**** Five moves, all from the QBR: 1. ****SWW is the default, not a program.**** Kyle: position Ship with Walmart as the default way to win on Walmart seller-fulfilled, not another optional program. Activation is still seller-by-seller (SAM / BD / channel partners). The ask is to reduce dependence on manual selling motions. Gagan: awareness is the barrier; ShipStation + SellerCloud + marketing / activation are the tailwinds. Aaron: SWW rate differentiation currently leans on FedEx — channel-partner FedEx access is an exec topic for the FedEx Top-to-Top. 2. ****Seller P&L, not conversion theater.**** Expedited converts better (3.5% vs. 2.4%) but sellers face ~\$18/order shipping. Leadership wants seller-level ROI, break-even, and profitability — not conversion alone. Expedited adoption moved ~7% → ~10% in Q2; Gagan's Q3 co-funding promotions let sellers **feel** speed before they commit. Aaron: many sellers still pick the cheapest option regardless of conversion lift — that is why Walmart Shipping's future third-party ground model is the long-term play (faster **and** lower-cost, for margin-sensitive sellers). 3. ****Tighten the promise without a national pad.**** Kyle asked whether additional buffers can come out of promise calculations. Rahil's next step is Neuron 1.0 — a neural-network model that jointly optimizes lag and transit, so promises can tighten without sacrificing reliability. Kyle also wants ****bucket migration**** (orders moving into faster delivery bands over time), not only average C2P. International already showed OTD and speed can improve together (shorter promises, better performance). Domestic OTD was hit by a tech bug during carrier onboarding onto SSS — own that distinction; do not paper over it. 4. ****Scan compliance is the foundation.**** Aaron: it underwrites delivery promises, refund accuracy, fraud prevention, and lost-in-transit investigations. Scan compliance ~75%, up ~600 bps YoY. When they ask what you would instrument this week, this is a named foundational capability — not a side metric. 5. ****Scale past pilots; China is an escalation problem.**** Kyle and Anil: the quarter validated years of foundational investment (\$1.84B GMV, +24.5% YoY, +7.5% vs. AOP; five of six exec goals green; OI ~\$28M above plan). The challenge now is accelerating successful pilots into scaled capabilities and naming where executive support is needed. China Post outage: monitoring worked, escalation channels did not — there is now a formal carrier escalation SOP. Gagan: China-origin carrier relationship management remains an unresolved gap despite long-running discussions.

What the QBR actually said

Source: *SFS FY27 Q2 QBR notes* — “Below are the key takeaways and action items from the discussion.” Names below are as they appear in the notes. Transcribed so you can check the recommendation against the primary, not so you can quote it.

Executive summary

- **SFS delivered a strong Q2:** \$1.84B GMV (+24.5% YoY, +7.5% vs. AOP), five of six executive goals green, and Operating Income ~\$28M above plan. Leadership viewed the quarter as validation that foundational investments made over the past several years are translating into measurable improvements across performance, returns, cancellations, and adoption.
- **Leadership's primary challenge was scaling beyond pilots.** Kyle and Anil emphasized accelerating successful pilots into scaled capabilities and identified where executive support is needed.
- **Kyle asked the team to position SWW (Ship with Walmart) as the default way to win on Walmart seller fulfilled,** rather than another optional program. Requires heavy seller-by-seller activation through SAM, BD, or channel partners.
- **Seller profitability — not conversion alone — was a key strategic focus.** While expedited offers convert better (3.5% vs. 2.4%), sellers face materially higher shipping costs (~\$18/order). Leadership asked the team to quantify seller-level ROI, break-even points, and profitability rather than relying on conversion metrics alone.
- **The long-term opportunity is faster, lower-cost shipping.** Many sellers prioritize shipping cost over speed today, reinforcing the importance of Walmart Shipping's future third-party ground model as a scalable solution for margin-sensitive sellers.

SWW adoption & growth strategy

- SWW achieved its Q2 adoption milestone and enters Q3 with positive momentum; however, Kyle asked the team to explore reducing dependence on manual selling motions to drive seller adoption.
- Gagan identified seller awareness as a primary adoption barrier and expects ShipStation and SellerCloud integrations, combined with stronger marketing and activation, to become meaningful adoption tailwinds.
- Aaron highlighted a key strategic dependency: SWW's rate differentiation currently relies heavily on FedEx. Channel-partner access to FedEx rates was elevated as an executive discussion topic for the upcoming FedEx Top-to-Top.

Speed, promise & seller economics

- Leadership agreed that faster delivery improves conversion, but Kyle emphasized that sellers ultimately care about profitability. The team was encouraged to provide seller-specific economics, showing when increased conversion justifies higher shipping spend.
- Rahil shared that expedited adoption has increased from ~7% to ~10% in Q2 and that expedited offers convert significantly better, while Gagan described planned Q3 co-funding promotions designed to let sellers experience the benefit of speed before committing longer term.
- Aaron reinforced that many sellers still select the cheapest shipping option regardless of conversion lift, leading to broad alignment that Walmart Shipping's future lower-cost speed solution is strategically important.

Delivery performance & promise optimization

- International performance improved substantially while reducing promise length, demonstrating that OTD and speed can improve simultaneously. Domestic OTD was negatively impacted by a tech bug introduced during carrier onboarding onto SSS (Simplified Shipment Settings).

- Kyle asked whether additional buffers can be removed from promise calculations. Rahil described the ability of Neuron 1.0 (a new neural-network model) to jointly optimize lag and transit time as the next major step to further tighten promises without sacrificing reliability.
- Kyle also requested a more detailed view of how orders are migrating into faster delivery buckets over time rather than relying solely on average C2P metrics.

Carrier management & China operations

- Aaron highlighted scan compliance as a foundational capability because it supports delivery promises, refund accuracy, fraud prevention, and lost-in-transit investigations. Scan compliance improved approximately 600 bps YoY to ~75%.
- Following the China Post outage, leadership concluded that monitoring worked but escalation channels did not. The team has implemented a formal carrier escalation SOP.
- Gagan described China-origin carrier management as a continuing gap and requested additional support for local carrier relationship management, noting that the issue has remained unresolved despite long-running discussions.

The notes also listed **Other Topics**; the screenshot did not include that body.

Part 1 – First principles, rebuilt

Maya buys a blender

Tuesday, 3:00 PM Pacific. Maya in Sacramento opens Walmart.com, searches “blender,” and lands on a \$29 personal blender sold by a Dallas kitchen shop she has never heard of. The product page says **Get it by Thursday**. She believes the page. She taps Buy.

That sentence — *Get it by Thursday* — is the entire product this role operates. Not the blender. Not the Dallas warehouse. The **sentence**. Everything in this book is about whether Thursday is a honest sentence, a lucky sentence, or a lie; why Walmart is the one who pays when it is a lie; and how a weekly review can make the sentence both faster *and* more often true.

The blender is the protagonist. Every later issue, metric, and Amazon mechanism comes back to Maya’s blender, or to a nearby cousin: a rural mountain order, a peak-week order, a seller who lied about how fast they pack.

Words for the walk

You do not need marketplace vocabulary to start. You need pictures. Each term below is a picture first; the jargon is just the label on the picture.

Term	Picture, then the label
Marketplace	Walmart.com as a mall. Walmart owns the mall and the signs. Other companies own many of the shops. Maya thinks she is buying from Walmart.
3P seller	Third-party seller. The Dallas kitchen shop. Not a Walmart warehouse employee.
SFS vs WFS	SFS (Seller Fulfilled Shipping): the Dallas shop picks, packs, and hands the box to UPS. WFS (Walmart Fulfillment Services): the seller sends inventory into a Walmart warehouse and Walmart ships — Amazon’s FBA equivalent. This role’s hard case is SFS: Walmart prints the date, someone else moves the box.
PDP	Product Detail Page. The blender’s page. Search, cart, and checkout are the other “surfaces” that can print a date.
Promise	The printed date itself. “Get it by Thursday.” A prediction, a commitment, and a conversion tool at the same time.
EDD	Estimated Delivery Date. The calendar day inside the promise. Thursday.
Handling time	Hours the box sits in Dallas after Maya pays, before UPS has it. Picking off a shelf, taping a box, waiting for the 4 PM pickup.
Transit	Hours the box is in the carrier network. Dallas → Sacramento on a truck and a plane.
Buffer	Padding Walmart adds because handling and transit will not hit their predictions exactly. Insurance days.
Scorecard	The seller’s report card. Rolling 30 days of on-time delivery, late shipments, tracking, cancellations. Fall below a floor and listings get suppressed.
WISMO	“Where is my order.” Maya’s chat/call when Thursday becomes a question mark.

Term	Picture, then the label
SSS	Simplified Shipping Settings. Walmart's program: <i>let us pick the transit settings; we will absorb carrier-delay blame on your scorecard</i> . Adoption is a choice by thousands of independent shops.
GMV	Gross Merchandise Value. Dollars of goods sold, before fees. The size of the book being underwritten.
Conversion (CVR)	Of people who viewed the blender page, the share who bought. Views that became orders.
On-time vs promise accuracy	Two different clocks. Promise accuracy (O1) asks: did Maya get it by the Thursday <i>printed at purchase</i> ? On-time delivery (O2) asks: did this seller's orders arrive by the <i>scorecard's</i> Expected Delivery Date, over 30 rolling days? Same porch. Different referee.

What the customer sees

Maya never sees Dallas. She never sees UPS's hub plan. She sees four screens, and each one may print a date:

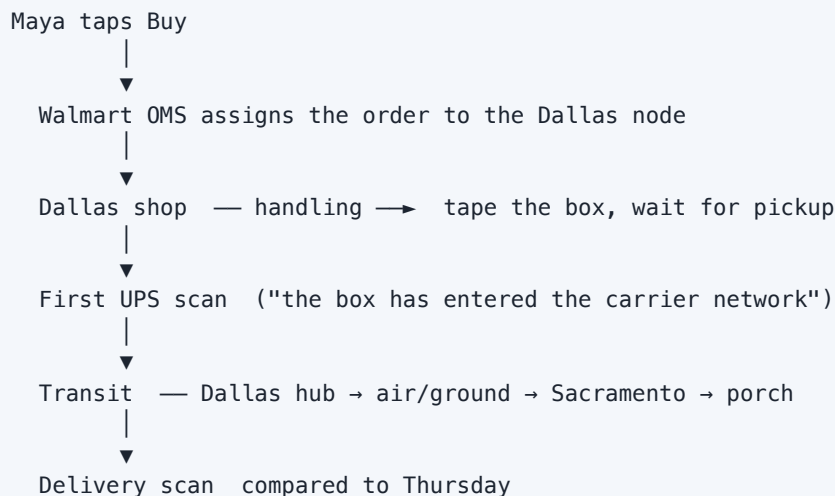
1. **Search.** A tiny "arrives Thu" under the price.
2. **PDP.** "Get it by Thursday." This is where conversion is won or lost.
3. **Cart.** The date again, next to the \$29.
4. **Checkout.** The moment of commitment. If this date disagrees with the PDP, trust starts leaking before the box exists.

From Maya's chair the date is a fact, like the price. She plans Thursday dinner around a blender that does not yet exist in California.

****The blender this week.**** One human, one \$29 SKU, one printed Thursday. The rest of the system is invisible to her — and that invisibility is the design. Walmart took responsibility for a physical event it will not perform.

Who actually moves the box

After checkout, three companies have to cooperate, and only one of them is Walmart.



- **The seller** owns inventory truth, pick/pack speed, and getting the box onto a truck.
- **The carrier** (UPS, FedEx, USPS, regional) owns pickup and transit.
- **Walmart** owns the date Maya saw, the product page, the refund when Thursday fails, and the brand she will or will not trust next time.

Walmart does not own the warehouse. Walmart does not drive the truck. Walmart owns the sentence.

Why Walmart is on the hook (this is the insurance)

If Thursday becomes Saturday, Maya does not email Joe's Kitchen Supply LLC. She opens the Walmart app. The refund comes from a Walmart flow. The next search happens on Walmart.com, or it happens on Amazon.

So the printed date is not a display string and not an ML feature. **It is an insurance policy Walmart writes on every order, on execution it does not own.**

An insurer does four jobs. Map them onto this role and the job description stops being a list of programs:

Insurer job	In this role
Price the risk	Set the buffer (how many extra days to add) per shipping lane.
Decide who gets coverage	Speed-program eligibility — which items may even show a 1- or 2-day badge.
Reduce the hazard	Seller handling programs, carrier pickup discipline, SSS adoption.
Grow the book without blowing the loss ratio	More GMV on fast, accurate promises (SSS, fast-promise penetration) without letting promise accuracy fall through the floor.

The **loss ratio** here is broken promises per order. The **premium** Maya pays is not a dollar line — it is the extra wait. A slower promise is a higher premium: she is less likely to buy. A faster promise is a lower premium: more conversion, more risk.

That framing is the interview thesis. Every issue in Part 2, every Amazon mechanism in Part 3, and every proposal in Part 4 hangs off it.

****Systems punchline — shifting the burden.**** Padding the buffer treats the **symptom** (broken promises this week) by moving the cost somewhere tickets are not filed (orders never placed). That is the archetype called **shifting the burden**: the symptomatic fix (more padding) is fast and locally rewarding; the fundamental fix (shrink the variance of handling and transit) is slower and belongs to other teams. Unmanaged, the system always chooses padding.

The atomic timeline of one order

Use round numbers a human can check on paper. Maya orders **Tuesday 3:00 PM PT**.

The Dallas shop has configured **handling = 1 day**. Translation: “we will tender this to UPS by Wednesday 3:00 PM.” That is a *declaration*, not a measurement. (Part 2, Family B is mostly about declarations that are optimistic.)

The promise engine's model says **predicted transit = 2 days** for Dallas → Sacramento on this service.

Walmart adds **buffer = 1 day** because both numbers will be wrong some of the time.

```
promised_EDD = t_order + handling_config + transit_predicted + buffer
              = Tue 3pm   + 1 day         + 2 days         + 1 day
              = Saturday 3pm
```

In English: start at order time, add the seller's promised warehouse time, add the model's expected truck time, add a pad. The page prints the resulting calendar day.

Walk it on a clock:

Step	Clock math	Running total
Order	Tuesday 3:00 PM PT	Tuesday
+ handling 1 day	Wednesday 3:00 PM — box should be in UPS's hands	+1
+ predicted transit 2 days	Friday 3:00 PM — predicted porch time	+3
+ buffer 1 day	Saturday 3:00 PM — printed EDD	+4

The page says **Get it by Saturday**. (We used Saturday rather than Maya's original Thursday so the arithmetic is obvious. The *structure* is identical when the page says Thursday: the engine simply used smaller handling, transit, or buffer. California Intra in the 12-lane model does this for real: $0.30 + 0.55 + 0.54 = 1.39$ days.)

The box actually arrives at:

```
actual_delivery = t_order + handling_actual + transit_actual
```

Subtract the two equations. The promise is **kept** if and only if the combined miss is smaller than the pad you bought:

```
promise kept ⇔ (handling_actual - handling_config)
               + (transit_actual - transit_predicted)
               ≤ buffer
```

Call the left-hand side ϵ (combined execution error). Then: kept iff $\epsilon \leq b$.

In English: the seller may run late, the truck may run late, or both. The buffer has to cover the *sum*. A 1-day buffer cannot cover a 1-day warehouse miss *plus* a 1-day truck miss.

Whose chapter each term is – the three anchors

The equation is not only the team's operating model. It is also the outline of the career walking into the room. Three of its four terms are chapters already owned, one at a time, in three different seats. Say the mapping in the first two minutes; every story after it then lands as evidence instead of anecdote.

Term	What it decides	Anchor chapter	The number behind it
buffer — the promise days	how many days may be printed in front of a customer	Manhattan Associates — distribution order management. Order allocation and promising across 3 mega-DCs and 100 stores. The estimated lead time we could commit to was an <i>output</i> of sourcing each order from the best node through allocation: choose the node, and you have chosen the promise.	99% order fill rate held through peak · ~\$8M of sales protected on one Cyber Monday · structural stockouts down 20%
transit_predicted	the model's expected days inside the carrier network — and how far into the tail the promise sits	Elementum. Carrier-brochure lead times replaced with measured transit distributions across 200 lanes , promise set at the tail (p99), plus a per-lane drift flag so a widening spread triggered review instead of a silent re-plan.	99% lead-time accuracy held · 580,000 units and ~\$1.5M of safety stock released, because measured spread came in tighter than assumed
handling_config — seller processing	whether the box actually left the seller inside the declared window, and whether the first scan proves it	Flexport shipment-visibility program. The product's entire job was “did the origin party tender the freight, and did the first milestone events arrive clean” — I3 wearing a different logo.	7 pilots → 68 enterprise customers behind published gates · cost-to-serve \$0.66 → \$0.03 per container · four systems into Snowflake, data accuracy 60% → 90%
t_order	when the customer taps Buy	the customer. Nobody's lever.	—

Term one, in this book's arithmetic. Manhattan's lesson is that node choice moves **handling + transit_predicted** *before* a single buffer-day is spent. Take the two extremes of the 12-lane model:

```

source from the near node:  h 0.30 + t̂ 0.55  =  0.85 days
source from the far node:   h 1.20 + t̂ 3.40  =  4.60 days
difference                  =  3.75 days of promise, decided by allocation

```

What it means: allocation hands the promise engine its starting point. A buffer policy spends tenths of a day; a sourcing decision spends whole days. **What you'd do:** ask on day one whether node selection and buffer policy are ever solved together. At most marketplaces they are not, and that seam is where free speed hides.

Term two, worked — because “promise at the tail” is arithmetic, not a slogan. p99 is **cumulative**: the smallest number of days d such that $P(L \leq d) \geq 0.99$ — “99 of every 100 shipments have already arrived by day d .” It is **not** “the day on which 1% of shipments land.” **ILLUSTRATIVE** 100-shipment lane, carrying the right-skewed shape freight always has:

Transit days d	shipments	cumulative	$P(L \leq d)$
9	5	5	0.05
10	14	19	0.19
11	21	40	0.40
12	21	61	0.61 ← p50

Transit days d	shipments	cumulative	$P(L \leq d)$
13	15	76	0.76
14	9	85	0.85
15	6	91	0.91 $\leftarrow$ p90
16	3	94	0.94
19	3	97	0.97
22	3	100	1.00 $\leftarrow$ p99

mean = $1,250 / 100 = 12.50$ days $\sigma = 2.64$ days
 promise at the mean 12.50 $\rightarrow$ 61 of 100 beat it – two in five arrive late
 promise at p90 15 $\rightarrow$ 91 of 100
 promise at p99 22 $\rightarrow$ 100 of 100 $\leftarrow$ what Elementum shipped
 normal shortcut: $12.50 + 2.326 \times 2.64 = 18.64$ days

What it means: the gap between p90 (15 days) and p99 (22 days) is **seven days**, all of it in a tail a point estimate cannot see — and the normal shortcut understates the real tail by $22 - 18.64 = 3.4$ days. A model can be excellent at the middle and useless for promising. **What you'd do:** ask DS for calibrated quantiles rather than a point, and audit calibration by checking whether actuals land inside the predicted p90 about 90% of the time — because that is the assumption the Φ term below silently makes.

Term three, priced. The shipment-visibility number that matters here is not the customer count. It is the unit economics of measuring truth at scale:

cost-to-serve cut: $(\$0.66 - \$0.03) / \$0.66 = 0.955 \rightarrow 95\%$ reduction
 book growth: $68 / 7 = 9.7\times$, over the same period

What it means: the book grew roughly $10\times$ while unit cost fell 95%. That only happens if the adoption gates keep bad data out of the platform — growth and cost move in opposite directions otherwise. **What you'd do:** the same gate logic is the honest answer to “how do you grow SSS.” Enrollments without 30-day retention are not adoption (I10), the same way a connected feed was never an onboarded lane.

And the three terms do not add — they multiply. That is why one person holding all three is not an org-chart convenience:

$O1 \approx I3 \times I4 \times \Phi(b / \sigma)$
 Wk 33: $0.938 \times 0.915 \times 1.1115 = 0.954 \rightarrow 95.4\%$

$\Phi(z)$ is the standard-normal CDF — the share of a bell curve lying below z , taught two sections down; $\Phi(1.645) = 95.0\%$. The residual 1.1115 sits above 1 on purpose: $I4$ is measured at *lane* grain rather than order grain, and buffer quietly recovers some $I3$ misses. Part 2's $O1$ section derives that in full. Own handling alone and you have $I3$. Own the model alone and you have σ . Own the pad alone and you have b . **The customer lives at the product.**

Three Saturdays (the same math, three physical stories)

Same order. Same Saturday promise. Three worlds.

	Seller	UPS	Combined error ϵ	$\epsilon \leq 1$ day?	Maya's porch	Customer promise (O1)	Seller scorecard (O2)
A. Clean	Tenders Wed 3 PM (on config)	2 days, as predicted	$0 + 0 = 0$	Yes	Saturday	Kept	On-time
B. Buffer does its job	Tenders 8 hours late (Wed 11 PM)	2 days, as predicted	$8/24 + 0 = 0.33$ day	Yes	Saturday night / still Saturday	Kept	May ding — seller missed ship-by / handling window
C. Buffer is not enough	Tenders Wed 3 PM (on config)	+2 days (hub delay)	$0 + 2 = 2$ days	No	Sunday–Monday	Broken	May stay green if the scorecard excuses carrier delay (SSS) or uses a seller-only clock

Scenario B arithmetic. Eight hours is $8/24 = 1/3 \approx 0.33$ days. The inequality is $0.33 \leq 1.00$, which is true. Maya is fine. The buffer absorbed a warehouse miss. The scorecard may still punish the Dallas shop for missing its own handling window — that is O2, taught in Part 2. **Two clocks, one porch.**

Scenario C arithmetic. Seller perfect, carrier +2 days, buffer 1 day. $2 > 1$, so the promise breaks. Maya is one day (or more) late. If she is on SSS, Walmart may have promised the seller “we will not ding your scorecard for carrier delay.” That does **not** un-break Saturday for Maya. O1 has no excusal clause. The customer’s calendar does not care whose fault it was.

****The blender this week.**** Saturday was never a measurement of Dallas or of UPS. Saturday was a ***policy***: $1 + 2 + 1$. Scenario B is why buffers exist. Scenario C is why buffers cannot be a substitute for shrinking the error. And the O1/O2 split is already visible: Maya can be fine while the seller is dinged, or the seller can be fine while Maya is not.

σ – how wildly that combined error jumps around

ϵ is not a constant. Some weeks Dallas is sharp and UPS is boring. Some weeks a hub consolidates and ϵ jumps by a day. For one **lane** (origin region $\times$ destination region $\times$ service, e.g. Dallas–Sacramento Ground), treat the week-to-week combined error as a bell curve centered at zero with spread σ (sigma):

$$\epsilon \sim \text{Normal}(\text{mean} = 0, \text{standard deviation} = \sigma)$$

In English: zero is “exactly as predicted.” σ is “how fat the surprises are, in days.” A metro same-day lane in the 12-lane model has $\sigma = 0.22$ days (about 5 hours of typical surprise). Mountain Rural has $\sigma = 1.60$ days (more than a day and a half of typical surprise). That is a **7 $\times$** difference in wildness. A single buffer cannot be right for both.

If errors are roughly bell-shaped, the chance the buffer covers a typical surprise is a known table lookup: $\Phi(b / \sigma)$, the standard-normal curve evaluated at “how many sigmas of pad you bought.”

$$P(\text{this lane's model/execution noise is covered}) = \Phi(b / \sigma)$$

In English: divide your buffer by the lane’s wildness. Look up that z-score on a normal table. That percentage is how often a *typical* surprise still lands inside the pad — *assuming the seller and carrier did the thing the model was told they would do*. Seller misses and carrier SLA misses sit outside this term. They get their own factors in Part 2 (I3, I4).

A number you can feel: if $\sigma = 0.5$ day and $b = 1$ day, then $z = 1 / 0.5 = 2$. Two sigmas on a normal curve is $\Phi(2.00) = 97.7\%$. A one-day pad on a half-day-wild lane covers about 97.7% of typical errors. Not 100. The remaining 2.3% are the right-hand tail — the ugly surprises.

Now the table the interview needs. Hold the buffer **fixed at 1.10 days** (the naive uniform policy in `model_reference.py`) and change only σ . Φ computed from the standard normal CDF (`NormalDist().cdf` in Python):

Lane personality	σ (days)	$z = 1.10 / \sigma$	$\Phi(z)$	What 1.10 days of pad actually buys
Tight metro (West-metro-ish)	0.30	1.10 / 0.30 = 3.67	100.0% (99.99%)	Almost never breaches from <i>this</i> noise. You paid a full day of slower promise for protection you did not need. Speed left on the table.
Medium (Pacific NW-ish)	0.70	1.10 / 0.70 = 1.57	94.2%	Below a 95% accuracy floor before seller or carrier misses even arrive.
Chaotic (Mountain Rural)	1.60	1.10 / 1.60 = 0.69	75.4%	About one in four typical surprises already exceeds the pad. You are underwriting known losses.

Same 1.10-day buffer. Coverage from ~100% to 75%. That is why **uniform buffers fail**: they over-protect the tight lane (Maya in Sacramento sees a slower date than the physics required, and sometimes does not buy) and under-protect the wild lane (a cousin in the mountains sees a date the trucks cannot keep, and files WISMO).

****Systems punchline — one policy, heterogeneous stock.**** Buffer is a single **policy** applied to many **stocks** (lanes) whose variance differs 7×. Policy resistance shows up as: tighten the uniform pad, and mountain orders break; loosen it, and California conversion dies. The system is not refusing to cooperate. The policy is written at the wrong grain.

The silent tax (padding fails quietly; cutting fails loudly)

Two ways to miss the economic optimum, and they are not symmetric.

Cut the buffer. Saturday becomes Friday. More Mayas buy (conversion up). More Saturdays break (accuracy down). Broken Saturdays generate **WISMO**, refunds, and “I don’t trust Walmart delivery” — tickets, chats, a visible red on the dashboard. Cutting fails **loudly**.

Pad the buffer. Saturday becomes Monday. Accuracy looks great. Fast-promise share falls. Conversion falls. **Nobody files a ticket for the order that was never placed.** Padding fails **silently**. Every ops review is built from tickets and reds. The silent path therefore wins by default. That is the bias.

Work it with round numbers. **ILLUSTRATIVE.** 1,000 people view the blender PDP.

Promise shown	Conversion	Orders that exist
2-day (“Thursday”)	7.0%	$1,000 \times 0.070 = 70$
4-day (“Saturday”)	5.5%	$1,000 \times 0.055 = 55$
Silent tax		15 orders never exist

Those 15 Mayas did not have a bad delivery. They had no delivery. Customer Care has nothing to count. The WBR’s WISMO line does not move. Finance sees a quieter week and may even praise accuracy. The 15 orders are the premium Maya refused to pay.

The equilibrium is an economic decision, not a modeling one:

$$\text{EV}(\text{fast promise}) = (\text{extra orders} \times \text{margin}) - (\text{extra breaches} \times \text{cost_of_breach})$$

$$\text{cost_of_breach} \approx \text{WISMO contact} + \text{refund/return risk} + \text{trust} / \text{LTV decay}$$

The 12-lane model prices that second term at **\$12 per breach** and contribution margin at **\$6.50 per order** (assumptions in `model_reference.py`, not Walmart actuals). The team that treats buffer as a data-science parameter will lose to the team that treats it as a **priced risk position**. That is the business-acumen edge to bring into the room.

The system on one page

Before the metrics, see the machine as stocks, flows, delays, and loops. Maya’s blender is one particle moving through it.

STOCKS (things that sit and accumulate)

seller inventory truth	unshipped orders	in-transit packages
customer trust	SSS-enrolled GMV	buffer-days of "insurance capital"

FLows (things that move per day)

orders in $\rightarrow$ first scans $\rightarrow$ deliveries $\rightarrow$ kept/broken

└ cancellations

SSS enrollments $\rightarrow$ enrolled GMV

scorecard dings → suppressions → seller exits

DELAYS

30-day rolling scorecard (a degrading seller burns ~30 days before the floor bites)

weekly model refresh (a Monday hub consolidation is invisible until next week's fit)

cutoff cliffs (order 4:01 PM, lose a calendar day)

LOOPS

R+ fast accurate promise → conversion → seller GMV → better ops / SSS → faster promises

B- broken promise → WISMO + distrust → conversion down → less GMV to invest

flowchart LR

```
PDP["PDP promise"] --> CVR["Conversion"]
```

```
CVR --> ORD["Orders"]
```

```
ORD --> H["Handling / first scan"]
```

```
H --> T["Transit"]
```

```
T --> PORCH["Porch vs EDD"]
```

```
PORCH -->|kept| TRUST["Customer trust"]
```

```
PORCH --> |broken| WISMO["WISMO"]
```

TRUST $\rightarrow$ CVR

WISMO --> TRUST

```
SSS["SSS enrollment"] --> H
```

$$SSS \rightarrow T$$

```
BUF["Buffer policy"] --> PDP
```

BUF --> PORCH

Two loops, always both running:

1. **Reinforcing (the flywheel you want).** Faster + more accurate promises → higher conversion → more seller GMV → the seller can afford SSS, better nodes, tighter handling → better inputs → still faster promises. Amazon's FBA is this loop with inventory *inside* one company. Walmart SFS has to spin the same wheel while the inventory sits in someone else's building.
2. **Balancing, destructive (the flywheel in reverse).** One bad promise week → scorecard reds → suppressed listings → seller churn → thinner assortment → worse promises. Trust is the non-renewable stock. GMS lost to a slow promise comes back next quarter; trust lost to a broken one compounds.

Amazon's promise flywheel runs on FBA: Amazon holds the inventory, so promise, pick, pack, ship, and delivery are one company's problem. **Walmart SFS inverts this: the hard case — seller-fulfilled — is the default case.** Walmart owns the promise and the customer. The seller owns the execution. The interface is configuration data (handling time, ship nodes, carrier choices) that sellers set optimistically and update rarely.

****Systems punchline — delays write the weekly drama.**** The 30-day scorecard is a delay: today's miss is a 1/30th ghost on O2. The weekly model refresh is a delay: a Tuesday hub event is mis-priced until the next fit. Cutoff is a delay that looks like a cliff: one minute past 4 PM and EDD jumps a full day. Every "why didn't the metric move" argument in a WBR is usually a delay, not a disagreement about physics.

Why a uniform 1.10-day buffer leaves fast share at 7.6%

This is the Part 1 closer, using the canonical 12-lane numbers — not a new model.

Promise days on a lane = handling + predicted transit + buffer. A lane is **fast** if that sum is ≤ 2 days.

Under **Policy A** (naive uniform $b = 1.10$ days everywhere):

Lane (from <code>model_reference.py</code>)	$h + t$	+ 1.10 buffer	Promise days	Fast (≤ 2)?
California Intra / Same-Day	0.85	1.95	1.95	YES
Northeast Metro / 2-Day	1.30	2.40	2.40	no
West Metro / Ground	1.45	2.55	2.55	no
Mountain Rural / Ground	4.60	5.70	5.70	no

Only California Intra clears the 2-day badge. One lane out of twelve. Order-weighted, that is **7.6% fast share** at **95.8% accuracy** (Policy A in Part 5). You did not fail at math. You spent the same 1.10 days of pad on a 0.22- σ lane that needed maybe half a day, and on a 1.60- σ lane that needed more than a day — and the 2-day cliff only rewarded the first.

Policy D (the profit-optimum buffers in the same file) gives California Intra $b = 0.54$, so $0.85 + 0.54 = 1.39$ days, **FAST**, and Mountain Rural $b = 2.46$, so $4.60 + 2.46 = 7.06$ days, **not fast**. Fast is a mix of lanes that *can* clear 2 days after an honest pad, not a vibe applied to the catalog.

That is the underwriter's job in one table: **do not buy the same insurance policy for a 20-minute commute and a mountain pass.**

The one-sentence thesis (say a version of this in the room)

"The delivery promise is an insurance policy Walmart writes on sellers it doesn't control — my job is to price it lane by lane, grow the book through SSS, and run the weekly review that keeps the loss ratio honest. I've owned each term of that promise separately — the promise days at Manhattan, the transit tail at Elementum, the handling and first-scan telemetry at Flexport; this role is the first seat where all three sit on one page."

****HM lens — Ownership + Are Right A Lot.**** If they stop you after this sentence, they will ask: *who owns k, and what evidence would make you change it?* Answer: one published dial, shadow four weeks, promote on Promise Yield; peak is a different σ regime (Part 5, proposal 10). Do not add a second thesis.

Part 2 – Every issue, taught through the blender

Fifty-five failure modes in six families. You will never recite this list in the room. Any question the hiring manager asks lives somewhere in it, and knowing the tree is what makes the answers land as ownership rather than theory.

How this part is taught. Each family is one mechanism, shown on Maya's blender (or a cousin) with arithmetic, then the remaining issues as named ways that same mechanism breaks. Before the families, though: **every output metric, derived.** If you can teach O1–O7 back with a napkin example, the issue tree will attach itself.

The WBR grammar (from the Flexport LSA pattern this team already uses): **outputs** are what the business feels; **inputs** are levers a named owner can move this week. Inputs without outputs are hobby. Outputs without inputs are weather.

Sample-week snapshot (illustrative workbook, Wk 28 → Wk 33):

Line	Metric	Target	Wk 28	Wk 32	
O1	Promise accuracy	≥ 95%	93.8%	95.0%	95.4%
O2	On-time delivery (scorecard, 30d)	≥ 90%	89.2%	90.6%	91.3%
O3	Fast-promise penetration (≤ 2-day)	+2 pts / qtr	41.5%	44.2%	45.1%
O4	Late shipment rate	≤ 5%	5.8%	5.2%	4.6%
O5	Seller-initiated cancellation	≤ 2%	2.4%	1.9%	1.8%
O6	WISMO contacts per 1,000 orders	−10% QoQ	12.4	11.5	10.8
O7	Conversion lift on fast-promise items (A/B)	measured	1.8%	2.3%	2.4%

O1 – Promise accuracy

Picture. Of every blender (and every other seller-fulfilled item) that actually reached a porch this week, what share arrived on or before the date printed at purchase?

Definition (WBR): % of *delivered* orders with actual delivery date ≤ promised EDD shown at checkout. Target ≥ 95%. Denominator is delivered orders — cancelled orders have left the room (that is a different defect; see O5 and D8).

$$O1 = (\text{delivered on or before printed EDD}) / (\text{delivered orders})$$

In English: among boxes that showed up, how often was the sentence true?

Worked example, Wk 33. Use 10,000 delivered orders so the 95.4% is a count you can feel. **ILLUSTRATIVE denominator; the 95.4% is the sample week.**

	Count	Share
Delivered this week	10,000	100%
Arrived on/before printed EDD	9,540	95.4% = O1
Late to the printed date	460	4.6%

Check: $9,540 / 10,000 = 0.954$. $10,000 - 9,540 = 460$. $460 / 10,000 = 0.046$.

What a human does that moves O1. Anything that makes ϵ smaller, or makes b a smarter match to ϵ : Dallas tenders on time (I3), UPS meets the transit the model used (I4), the model's typical error is small (I5), the pad is calibrated to that error (I6), the seller is on SSS so transit settings are not a hopeful guess (I1).

The planning identity – and the honest Wk 33 catch

The WBR's planning decomposition, assuming three *independent sequential gates at order grain*, is:

$$O1 \approx I3 \times I4 \times \Phi(b / \sigma)$$

In English:

- **I3** — seller handling adherence: % of orders whose **first carrier scan** lands inside the configured handling window. Wk 33 = **93.8%**. Picture: Dallas got Maya's box onto a truck by Wednesday 3 PM, and UPS scanned it.
- **I4** — carrier SLA adherence: % of **lanes** meeting published transit this week. Wk 33 = **91.5%**. Picture: this origin–destination–service, as a lane, hit its brochure days.
- **$\Phi(b/\sigma)$** — the buffer / model term from Part 1. Picture: leftover noise after seller and carrier did what they were supposed to, still smaller than the pad.

Multiply the two measured gates for Wk 33:

$$I3 \times I4 = 0.938 \times 0.915 = 0.85827 \rightarrow 85.8\%$$

If every I3 miss *and* every I4 miss killed Maya's Saturday, O1 could not exceed 85.8%. Actual O1 is **95.4%**. Implied third factor:

$$\text{implied } \Phi = O1 / (I3 \times I4) = 0.954 / 0.85827 = 1.1115$$

1.11 is not a probability. A teacher who hides that is not teaching. The product exceeding 1 is the lesson:

1. **I4 is lane-grain, not order-grain.** "91.5% of lanes met SLA" is not "8.5% of packages were late." A red rural lane can be 1% of volume. Treating I4 as an order survival rate double-counts geography as if it were packages.
2. **Buffer recovers misses (Scenario B).** A late first scan of 8 hours can still make Saturday. I3 counts a miss; O1 counts a keep. That recovery is *why* O1 (95.4%) can sit **above** I3 (93.8%).
3. **The three gates are not independent absorbing states.** The planning identity is a flashlight: it tells you which *family* to open when O1 goes red. It is not a conservation law.

So what is Φ doing on Wk 33? It is the residual “buffer plus grain plus recovery” term. Holding that residual **fixed** (same buffers, same recovery behavior, same mix), you can still run a sensitivity — and that sensitivity is what a Tuesday WBR is for.

If I3 drops 3 points and I4 is flat, O1 still falls

Hold I4 = 91.5% and the Wk 33 residual 1.1115. Drop I3 from 93.8% to **90.8%** (Dallas had a bad week; buffers unchanged).

$$O1' = 0.908 \times 0.915 \times 1.1115 = 0.9235 \rightarrow 92.3\%$$

95.4% – 92.3% = **3.1 points**, through the 95% floor, with no change in buffer policy. A red O1 week that “the model didn’t cause.” Seller Ops owns I3. The decomposition named the owner in one line.

Wk 32 was a live version of this shock. Hub consolidation in the Southeast: I3 fell 93.1% → 92.4%, I4 fell 90.8% → 89.6% (below the 90% floor), O1 only 95.2% → 95.0%. The customer metric barely moved because buffer absorbed the bump — while **O4** (seller clock, no customer pad) blipped to 5.2% and went red. Different clocks, same physical week. We will use that again.

****Systems punchline — a keep is not a virtue if the pad did all the work.**** O1 can be green because Dallas got better, or because you bought more buffer. Only the decomposition (and later, Buffer Waste Index in Part 4) tells you which. Green without cause is how padding wins.

****HM lens — Dive Deep.**** When they say “walk me through the data pull,” this is the grain: delivered-order week, cancels out of the denominator, I4 at *lane* not package, implied Φ allowed to exceed 1. The follow-up they love: “output green, input red.” Wk 32 is the answer — I4 89.6% red, O1 still 95.0%; O4 5.2% told the truth.

I6, at first appearance. % of lanes whose buffer sits between the 50th and 90th percentile of that lane’s transit error. Wk 33 = **70.2%**, climbing toward an 80% intra-quarter target. Picture: a human (Promise DS) looked at the error histogram and put the pad inside a sanity band — not yet the optimum of Part 5, just “not insane.” I6 is a *band compliance* lever, not a proof the pad is priced.

O2 – On-Time Delivery (the seller’s clock)

Picture. Forget Maya’s Thursday for a moment. Ask a different referee: over the last 30 days, what share of this seller’s orders arrived by the **Expected Delivery Date the scorecard uses**? That number is the seller’s report-card on-time rate. Policy floor $\geq 90\%$. Fall under it and listings start to die.

Definition (WBR): % orders delivered on/before Expected Delivery Date, **rolling 30-day** window. Owner: Seller Performance.

Why both O1 and O2 exist — the blender.

- **O1** is the customer-facing insurance outcome. Did the printed sentence come true? No excusal. No 30-day smear. This week’s deliveries, this week’s truth.

- **O2** is the *seller contract*. It is used to punish or protect the Dallas shop. It rolls 30 days (a delay). SSS can move carrier-caused lateness off the seller (an excusal). It may score against a ship-by + transit-SLA clock that is **not** the padded Thursday Maya saw.

Same porch. Two questions: “Was Maya told the truth?” vs “Should we punish Dallas?”

ILLUSTRATIVE. 100 orders, one week, round numbers.

What happened	Orders	O1 (Maya's Thursday)	O2 (seller scorecard, simplified to this week)
Clean: seller on time, UPS on time	88	kept	on-time
Scenario B: seller 8 hours late, buffer saves Thursday	5	kept	miss (handling window missed)
Scenario C: seller perfect, UPS hurricane, no excusal	3	miss	miss
Scenario C with SSS excusal: seller perfect, UPS hurricane, carrier delay absorbed	4	miss	on-time (excused)
Totals	100	kept = 88+5 = 93 → O1 93%	on-time = 88+4 = 92 → O2 92%

Read the two divergence cells out loud:

- **5 orders:** Dallas was late; Maya still got Thursday. **O1 kept, O2 miss.** Buffer did its customer job and the scorecard still told the truth about the seller.
- **4 orders:** Dallas was perfect; a hurricane ate the truck; SSS excused the seller. **O1 miss, O2 green.** Maya was failed; the shop was not. That is the accountability-transfer SSS sells — and it is why O1 cannot be replaced by O2. If you only watched the scorecard, you would declare the hurricane a non-event.

Wk 33 sample: O1 = 95.4%, O2 = 91.3%. A four-point gap is the fingerprint of (a) 30-day smear on O2, (b) seller-clock dings that buffers hid from Maya, (c) mix of SSS excusal. Do not “reconcile” them into one number. Ask which clock the hiring manager is holding.

****The blender this week.**** If Dallas scanned late and Saturday still happened, Maya’s metric is green and the shop’s metric may be red. If UPS drowned and SSS is on, the shop’s metric may be green and Maya’s is red. Quoting one without the other is how teams talk past each other for a quarter.

O3 – Fast-promise penetration

Picture. Of all seller-fulfilled orders, what share were shown a promise of **2 days or faster** at display time? Not “did we deliver fast.” **Did we even offer fast?**

Definition (WBR): % of seller-fulfilled orders displaying a ≤ 2 -day delivery promise. Target **+2 points per quarter**. Wk 28 = 41.5%. Wk 33 = **45.1%**. Climb in six weeks: 45.1 – 41.5 = **+3.6 points**.

$$O3 = (\text{seller-fulfilled orders with displayed promise} \leq 2 \text{ days}) / (\text{all seller-fulfilled orders})$$

A single order is fast only if the *sum* clears two days:

```
fast ⇨ eligible AND (handling + transit_predicted + buffer ≤ 2)
```

In English: the item is allowed to wear the badge (I8), and the engine's arithmetic actually fits inside two days. Eligibility without physics is a costume. Physics without eligibility never prints.

Two lanes from `model_reference.py` , Policy D buffers (canonical):

Lane	Handling h	Transit t	Buffer b	h + t + b	Fast?
California Intra / Same-Day	0.30	0.55	0.54	1.39	YES ($1.39 \leq 2$)
Mountain Rural / Ground	1.20	3.40	2.46	7.06	no ($7.06 > 2$)

Check: $0.30 + 0.55 = 0.85$; $0.85 + 0.54 = 1.39$. $1.20 + 3.40 = 4.60$; $4.60 + 2.46 = 7.06$.

California can be in the 45.1%. Mountain Rural cannot, not at an honest pad. **Fast share is a mix, not a vibe.** If someone "raises O3" by promising 2 days into 1.60- σ mountain zips, they did not raise O3. They borrowed from O1, with interest payable in WISMO.

I8, at first appearance. % of assortment eligible for expedited / same-day programs. Wk 33 = **56.8%** (up from 52.0% in Wk 28, +4.8 points). Picture: a Product owner marked the blender's size / weight / node coverage as speed-eligible. I8 is the *permission* layer. O3 is permission $\times$ physics.

I1, at first appearance. % of seller-fulfilled **GMV** on Simplified Shipping Settings. Wk 28 = 38.5%. Wk 33 = **45.3%**. Climb: $45.3 - 38.5 = +6.8$ points in six weeks (~1.4 points/week in the callouts). Picture: a human at the Dallas shop clicked enroll (or a new-seller default-on pilot enrolled them). Walmart now picks transit settings and absorbs carrier-delay accountability. Past 50% GMV, the majority of the book is finally on the contract. At 45.3% it is not. Most promise risk still rides on manual configuration.

I10, at first appearance. SSS seller activations per week that are still retained at 30 days. Wk 31 = 245 (holiday staffing dip). Wk 33 = **305**. Picture: onboarding throughput. I1 is the stock (GMV share); I10 is the enrollment flow that fills it, net of drop-off.

****Systems punchline — a threshold metric invites cliff-chasing.**** O3 counts " ≤ 2 days," not "how fast." A lane at 2.06 days (West Metro under Policy D: $1.45 + 0.61 = 2.06$) is a full miss. A lane at 1.99 is a full hit. Teams will spend a quarter shaving $2.05 \rightarrow 1.95$ and ignore 7-day mountain chaos. Watch the whole promise distribution, or O3 will hypnotize the roadmap. (Family F3; Part 4's Promise Yield is the antidote.)

O4 – Late Shipment Rate

Picture. The Dallas shop's **ship-by** clock, not Maya's porch. Of orders this week, what share were shipped *after* the ship-by date? Floor $\leq 5\%$. Above 5% is a hard red, even if Maya still got Saturday.

```
04 = (orders shipped after ship-by) / (orders)
```

Wk 33 = 4.6% (green). **Wk 32 = 5.2%** (red, the blip).

Relate to I3, and do not pretend they are complements.

	Wk 33	Complement / miss
I3 first-scan inside handling window	93.8%	$100 - 93.8 = 6.2\%$ not in window
O4 shipped after ship-by	4.6%	miss = 4.6%

6.2% is not 4.6%. Gap = **1.6 points**. Different clocks:

- **Ship-by** is a calendar deadline on the order (“ship by Wed”). It can be met by creating a label, in dirty systems, without a carrier scan.
- **First-scan-in-window (I3)** is a *physical telemetry event*: UPS’s scanner said the box exists in the network. Harder to fake, not impossible (Family B4).

A package can miss I3 (scan came at 3:40 PM on a 3:00 window) and still count as shipped-on-time, or the reverse (label on time, pickup missed, first scan Thursday). O4 is the policy floor the seller feels. I3 is the better input for O1 because Maya’s buffer is racing a *scan*, not a PDF label.

The Wk 32 story. Regional carrier hub consolidation, Southeast. I4 (lane SLA) dropped to 89.6%, under the 90% floor. Pickups slipped. O4 printed **5.2%**. Action item #1: recalibrate buffers on 42 Southeast lanes — shipped Wk 33; O4 back to 4.6%. Notice what the team did: they padded (symptomatic) *and* re-planned pickup windows (fundamental). The WBR is honest only if both stay on the log.

****The blender this week.**** If Dallas misses Wed 3 PM ship-by by two hours and UPS still makes Saturday, O4 can go red while O1 stays green. That is not a data bug. O4 is trying to change *Dallas’s behavior* before Maya has to notice. A 5% floor is a leading fire alarm, not a restatement of O1.

O5 – Cancellation Rate

Picture. Maya never gets a box. Dallas cancels: “out of stock,” “can’t fulfill,” “pricing error.” Floor $\leq 2\%$ seller-initiated. Wk 28 = 2.4% (red). Wk 33 = **1.8%** (green).

$$O5 = (\text{seller-initiated cancellations}) / (\text{orders})$$

Why this is worse than late. A late blender is a broken Saturday. A cancelled blender is a broken *purchase*. No tracking page, no porch, often a re-shop on Amazon. Trust drain per event is higher; WISMO here is “why did you take my money.”

Tie to I7 (inventory-feed truth), at first appearance. I7 = % of ship-from **nodes** whose inventory feed is $\geq 98\%$ accurate. Wk 33 = **88.9%** (up from 84.0% in Wk 28). Picture: the Dallas node’s on-hand file says “1 blender.” Reality is 0. The PDP still offered Thursday. Maya paid. Then cancel.

ILLUSTRATIVE. 100 orders placed against phantom stock at a dirty node. 4 cancelled by the seller.

$$\text{cancel rate on that slice} = 4 / 100 = 4\%$$

4% is double the 2% floor. The pre-purchase team inherited a *data* defect as a *customer* defect. Action item #4 in the sample WBR is exactly this: onboard remaining ~11% of nodes to a near-real-time feed ($100 - 88.9 \approx 11$ points of nodes still below the 98% bar, matching the “remaining 11%” note).

Late is a delay. Cancel is a stock that was never there. You cannot buffer your way out of a phantom unit.

O6 – WISMO per 1,000 orders

Picture. Maya types “where is my order” into chat, or calls. Count those contacts, scale to 1,000 orders. Target: **–10% quarter-over-quarter**. Wk 28 = 12.4. Wk 32 = 11.5 (the hub week — contacts ticked *up* while O1 stayed flat). Wk 33 = **10.8**.

$$O6 = (\text{WISMO contacts} / \text{orders}) \times 1000$$

In English: of a thousand Mayas, how many had to ask?

WISMO is not a personality metric. It is mostly **broken promises plus tracking opacity**. Back-of-envelope reconstruction of 10.8, using Wk 33’s 4.6% late-to-promise ($1 - 0.954$) so the metric feels *caused*:

ILLUSTRATIVE contact rates; 10.8 and 4.6% are sample.

Source	Arithmetic	Contacts per 1,000
Late to printed EDD	4.6% of orders × 20% of late customers contact	$0.046 \times 0.20 \times 1000 = 9.2$
“In the dark” (on-time or not-yet-due, but no usable scan / anxiety)	remainder to hit 10.8	$10.8 - 9.2 = 1.6$
Total		10.8

Check: 46 lates per 1,000 delivered-equivalent; 20% of 46 = 9.2. 1.6 leftover contacts from tracking gaps is $1.6 / 954 \approx 0.17\%$ of the on-time Mayas — a small anxiety tax, enough to matter at marketplace volume.

Wk 32’s 11.5 vs Wk 33’s 10.8: $(11.5 - 10.8) / 11.5 = 6.1\%$ relative drop, matching the sample WoW of –6.1%. The hub week raised opacity (missed scans, I4 red) even where buffers saved O1 — which is why O6 can worsen while O1 looks calm.

****Systems punchline — information delay is a defect.**** A package can be physically fine and still generate WISMO if Maya cannot *see* it. Scan quality (Family C4) is not a carrier-ops nicety. It is how you drain the 1.6 without spending buffer.

O7 – Conversion lift on fast-promise items

Picture. Run an A/B test. Same blender, two promises: a fast one vs a slower control. Did the fast badge actually make more Mayas buy?

Definition (WBR): “%, Conversion lift on fast-promise items (A/B).” Sample path: 1.8, 1.9, 2.1, 2.2, 2.3, 2.4 across Wk 28–33. Target: measured (no floor). Owner: Product/DS.

Honest reading of the unit. The column is labeled % and the values are 1.8 → 2.4. Other % lines in this WBR are rates (95.4 means 95.4%). For a *lift*, 2.4 could mean **+2.4 percentage points of CVR** (6.0% → 8.4%) or a **+2.4% relative lift** (6.0% → 6.144%). A +2.4 point CVR move would be an extraordinary A/B for a delivery badge; the six-week crawl 1.8 → 2.4 looks like a **relative lift of a few percent**, which is the conservative teaching reading. The workbook does not print “pp” vs “relative.” **Treat Wk 33 as a 2.4% relative lift, and say so.** If the team later clarifies it is percentage points, every GMS implication scales up.

Relative-lift napkin (the conservative read):

control CVR	= 6.000%	(illustration)
treatment CVR	= 6.000% × 1.024	= 6.144%
extra CVR	= 0.144 percentage points	= 0.00144 absolute

Check: $6 \times 0.024 = 0.144$; $6 + 0.144 = 6.144$.

GMS illustration — labeled ILLUSTRATIVE, not WBR, not resume.

The user-facing sketch uses a rounder extra CVR of 0.2 points (0.002), which is *larger* than a 2.4% relative lift on a 6% base:

Sketch	Sessions	Extra CVR	AOV	Weekly GMS
Round 0.2-pt extra (ILLUSTRATIVE)	1,000,000	0.002	\$35	$1e6 \times 0.002 \times 35 = \$70,000$
2.4% relative on 6% CVR	1,000,000	0.00144	\$35	$1e6 \times 0.00144 \times 35 = \$50,400$

Either way the order of magnitude is tens of thousands of dollars **per million sessions per week**, which is why O7 is not a science fair. It is the price of a buffer-day.

I9, at first appearance. Promise-engine **p99 latency** in milliseconds. Target ≤ 50 ms. Wk 33 = **41 ms**. Picture: Maya’s PDP asked “what date do I print?” and the engine answered in 41 milliseconds at the slow tail. When it *misses* its budget, the page falls back to a conservative default (often +1–2 days). That fallback is a silent conversion tax wearing an infra costume — Family A10. Cache hit rate is the twin lever (not a separate line in this sample workbook).

****The blender this week.**** O1–O7 are one story: print a date (O3, O7), keep it (O1), score the seller fairly (O2, O4), don’t take money for a ghost (O5), don’t make Maya hunt (O6). Inputs are the hands on those valves.

Family A – Promise generation (13 issues)

The engine that *wrote* Saturday. If this family is sick, every other family is debugging the wrong patient.

Worked example A — wrong loss function / tail blindness (A1, A2). Promise DS trains a transit model to minimize **MAE** (mean absolute error) — I5, Wk 33 = **0.72 days** (target ≤ 0.8). MAE loves the middle of the

error distribution. A promise is a *tail* commitment: Maya's Saturday breaks at p90+, not at the median.

ILLUSTRATIVE histogram of 10 Dallas→Sacramento transits, days: 1.8, 1.9, 2.0, 2.0, 2.1, 2.1, 2.2, 2.4, 2.8, **4.0**. Median = 2.1. Mean absolute error vs a 2.1-day prediction is small. The 4.0-day trip is +1.9 days — already larger than a 1-day buffer. A model can **improve I5 and break more Saturdays**. Fix: quantile loss at the percentile implied by the accuracy target (for a 95% floor, think p95, not the mean). Track p90 error and buffer-exceedance per lane, not MAE alone.

Worked example B — uniform / heuristic buffers (A3, A4). I6 says “put buffer between p50 and p90 of transit error.” That is a sanity band, not an optimum. Part 1's Φ table is the proof: the same 1.10-day pad is 100% coverage on $\sigma = 0.30$ and 75% on $\sigma = 1.60$. Policy A's **7.6% fast share** is this family, quantified.

The remaining issues in this family are named failure modes of the same engine:

- **A5. Cold start.** New seller, new lane, no history. Without hierarchical shrinkage (lane → region → national prior), Maya's first order from a new shop gets either a reckless date or an absurd Monday. Stock: no data. Policy: need a prior.
- **A6. Seasonality regime breaks.** σ is not stationary. In Nov–Dec carrier networks saturate; σ inflates ~1.4–1.5× (model assumption, Part 5 / Q4 tab). A July pad silently breaks the accuracy floor in November. Peak is a different system.
- **A7. Model staleness.** Weekly refresh misses mid-week hub events. Couples to C2. Staleness SLO per lane, event-triggered re-fit.
- **A8. Promise inconsistency across surfaces.** Search says Thursday, PDP Friday, checkout Saturday. Every hop that *re-computes* instead of carrying a promise token is a trust leak and a WISMO generator. Maya was told three sentences.
- **A9. No re-validation at checkout.** The PDP quote can go stale (cutoff passed, inventory moved). The moment of commitment needs the freshest quote *or* a held token plus an explicit change.
- **A10. Latency fallbacks are a silent revenue tax.** If 2% of PDP views miss the p99 budget and fall back to a +2-day promise, those views pay the Part 1 silent tax forever. **ILLUSTRATIVE:** 1,000 views, 20 fall back; at 5.5% vs 7.0% CVR that's 1.1 vs 1.4 orders on the fallback slice — 0.3 orders lost per 1,000 views, invisible. I9 is a revenue metric in infra clothing.
- **A11. Cache staleness.** The twin of A10: a cached Thursday that outlived the 4 PM cutoff is a manufactured breach.
- **A12. Cutoff-time cliffs.** EDD jumps a full day at the cutoff hour. “What would 90 more minutes of cutoff at the top 20 nodes buy?” is the cheapest speed lever in the system (Part 4, proposal 6).
- **A13. Eligibility/promise mismatch.** I8 says the blender is speed-eligible; the real node cannot support two days. Penetration is celebrated; O1 pays two quarters later.

****Systems punchline — the model is a delay wrapped in a policy.**** A weekly fit is a delay. A uniform pad is a policy that ignores stock variance. MAE is a loss function that cannot see the tail the customer lives in. Family A is not “the ML team.” It is how the underwriter prices.

Family B – Seller execution (10 issues)

Dallas, the warehouse Walmart does not enter.

Worked example A — optimistic handling (B1), the blender’s original sin. The shop declared handling = 1 day to win the buy box (faster promise → O7). They hit it 85% of the time. **ILLUSTRATIVE.** 100 orders:

Count	
Declared 1-day handling	100
Actually tendered within 1 day	85
The 15 misses	become Walmart’s ϵ , Maya’s risk

That 15% is **adverse selection, not bad luck**. Sellers who declare honestly look slower and lose the badge. Sellers who declare hopefully look fast and externalize the miss. Amazon’s Automated Handling Time exists to kill this (Part 3): observed distributions replace declarations. **I2** (first appearance): % of items with ML-recommended (auto) lag time. Wk 33 = **56.5%**, up from 44.0%. That is the Walmart-side start of the same idea — not yet default.

Worked example B — inventory-feed lies (B5) → O5. Already derived: phantom stock → cancel. I7 is the lever. Wrong-node routing (B6) is the multi-warehouse cousin: the blender is in Fort Worth, the order routes to Dallas, cancel or a slower lane.

The remaining named modes:

- **B2. Day-of-week blindness.** A “1-day” shop that does not work weekends is a 3-day shop for Friday 4 PM orders. Maya orders Friday, promise assumes Saturday tender, the shop is closed until Monday. Handling must be order-hour $\times$ day, or Friday is a structured miss.
- **B3. Capacity spikes.** 100 orders/day handling says nothing about the 400-order day after a Lightning Deal. Volume-conditional risk is usually unmodeled.
- **B4. Gaming the first scan.** Print a label (or induce a phantom scan) without tendering. I3 looks green; the box sits. Detect via scan-to-movement gaps.
- **B6. Wrong-node orchestration.** Multi-node sellers; routing picks an empty node or a farther node. Silent speed loss, occasional cancel.
- **B7. Holiday/closure calendars.** Vacation mode, local holidays, weather. Configured handling becomes fiction.
- **B8. SSS adoption stall — THE strategic issue.** At Wk 33, **45.3%** of GMV is on SSS; the majority of promise risk still rides on manual configuration. Sellers resist because adoption trades *control* (their carrier rates, their handling declarations) for a *benefit* (accountability transfer) that is real but weakly priced and weakly communicated. This is incentive design, not feature marketing. Part 3’s OTDR-protection bundle is the reference mechanism; Part 4 proposal 4 is the transplant.
- **B9. The long tail can’t execute.** Small sellers structurally cannot hit fast-promise standards. Pushing eligibility onto them (chasing O3) converts to O1 breaches. Honest answer: fast band for proven cohorts; standard band + WFS path for the tail.
- **B10. Enforcement lag.** 30-day rolling scorecards: a degrading seller does ~30 days of damage before floors bite. Real-time control loop (Part 4, proposal 5). Amazon moved to listing-level surgical deactivation in Feb 2026 for this reason.

****Systems punchline — the delay is the enforcement.**** B10's 30-day window is why a bad Dallas week is still "yellow" on Tuesday. B8 is a reinforcing loop you have not priced: until SSS's benefit is a dollar figure on the seller's dashboard, the stock of enrolled GMV fills slowly, and Family A keeps pricing noise that SSS would have removed.

Family C – Carrier execution (8 issues)

UPS does not work for the Dallas shop. UPS does not work for Maya. UPS works a network.

Worked example A — lane-level SLA variance (C1) and the Wk 32 hub (C2). National carrier on-time of "97%" can be **82% in the Southeast after a hub consolidation** (the deep-dive doc's own example). Averages hide bimodality. Wk 32: I4 = 89.6% (red vs 90% floor), O4 = 5.2%. The sample callout names the event. Weekly-refresh models (A7) learn this a week late unless someone event-triggers a re-fit.

Worked example B — missed pickup (C3) on the blender. Dallas is perfect: box on the dock at 2:50 PM. UPS does not show. First scan is Thursday. I3 goes red, O4 goes red, the *seller* looks late. Without a separate attribution bucket, sellers eat carrier misses and **trust in the scorecard dies** — which kills SSS's sales pitch (B8). Absorbing this risk *is* what SSS is supposed to sell.

Remaining modes:

- **C4. Scan data quality.** Missing origin scans make on-time tenders look late; missing delivery scans make delivered packages look lost. Measurement noise becomes seller punishment. VTR-style floors (99% valid tracking on Amazon SFP) belong on the carrier side too. This is the 1.6 WISMO "in the dark" term.
- **C5. Peak capacity caps.** Carriers cap volume in Q4 exactly when volume peaks. Uncapped promise generation writes checks the network will not cash. Couples to A6.
- **C6. Weather / force-majeure attribution.** A hurricane week charged to seller scorecards destroys program legitimacy. Excusal must be principled and auditable (Amazon's June 2025 multi-carrier disruption protection is the reference).
- **C7. Service-level mis-selection.** Cheapest service that *usually* makes Saturday is a hidden breach factory. The choice should be promise-aware — SSS's whole reason to exist.
- **C8. Rural / last-mile structural gaps.** Some zips are structurally 2σ worse (Mountain Rural $\sigma = 1.60$). Promising them at metro speed is underwriting known losses. Pull them from the fast band or route to WFS (Part 4, proposal 9).

Family D – Data and measurement (9 issues)

If you cannot score the sentence you printed, you are not underwriting. You are decorating.

Worked example A — attribution (D3) is the peace treaty. Late blender: Dallas, UPS, or an optimistic Saturday? Without clean factor attribution, the WBR is a blame negotiation. The O1 flashlight (I3 vs I4 vs residual) is the treaty. The sample action item #5 is to *automate* that decomposition so Tuesday does not start with a spreadsheet argument.

Worked example B — mix-shift / Simpson's paradox (D4). ILLUSTRATIVE. Two lanes, two weeks:

	Week 1 volume	Week 1 accuracy	Week 2 volume	Week 2 accuracy
California Intra	50	96%	80	95% (worse)
Mountain Rural	50	92%	20	91% (worse)
Portfolio	100	$0.5 \times 96 + 0.5 \times 92 = \mathbf{94.0\%}$	100	$0.8 \times 95 + 0.2 \times 91 = \mathbf{94.2\%}$

Portfolio **improved** (94.0 → 94.2) while **every cohort got worse**. Volume fled to the easy lane. Every output metric needs a mix-adjusted twin (Part 4, proposal 8) or the WBR will celebrate geography.

Remaining modes:

- **D1. Promise-at-display not logged.** If the EDD Maya saw is not captured with model version and buffer, O1 cannot be scored honestly. The single most load-bearing log line in the system.
- **D2. “Delivered” ≠ delivered.** Carrier delivery scan vs porch reality (theft, wrong address, INR). O1 can read 95% while customers experience 91%.
- **D5. Rolling-window smearing.** O2’s 30-day window lags inflections by weeks. Pair every rolling metric with a 7-day panel. Wk 32’s hub is a smear on O2 (90.6%) and a spike on O4 (5.2%).
- **D6. Definitional drift.** Seller Perf’s OTD, Promise’s accuracy, Care’s lateness: three clocks (order date vs ship date vs promise date). One glossary, one owner per definition, or every meeting is a units war. You just learned why O1 ≠ O2.
- **D7. Unweighted defects.** A missed \$500 TV ≠ a missed \$9 case. Dollar-weight breaches or the program optimizes for trinkets.
- **D8. Denominator games.** Cancels quietly leave the OTD denominator; suppressed listings leave the penetration denominator. Metric governance is auditing denominators.
- **D9. Severity blindness.** 1 day late and 6 days late are the same binary miss. WISMO and refunds scale with *days late given late*. Track the lateness distribution.

****Systems punchline — the map is not the territory, but a wrong map is a policy.**** D4 (mix) and D5 (delay) make green happen without improvement. D1 (unlogged promise) makes improvement unprovable. Measurement defects become operating defects because the WBR **is** the control loop.

Family E – Program and organization (9 issues)

The engine can be right and the company still cannot steer.

Worked example — no single-threaded owner (E1) plus WBR as theater (E2). Promise DS owns the model, Eng owns the engine, Seller Ops owns handling, Carrier PM owns lanes, Product owns eligibility — and O1 is owned by a committee, which means it is owned by nobody between Tuesdays. A WBR that reads numbers aloud without pre-wired cause, owner, and date is theater. The sample file already shows the *shape* of the fix: Call Outs name the hub, Action Items have owners and dates, one item is DONE with O4 as the verify metric. This role **is** E1’s fix, if run that way.

Remaining modes:

- **E3. Action-item rot.** A log without dates, owners, and a verified-by-metric close is a diary.

- **E4. Trade-offs made by default.** The speed-vs-accuracy dial gets set by whoever last tuned the model. It should be an explicit, versioned policy (the k-dial of Part 5).
- **E5. No experimentation gate.** Buffer and display changes without A/B on conversion *and* breach cost — flying blind on O7, the one elasticity that prices everything.
- **E6. Adoption under-resourced.** I10 competes with shiny features. Adoption is a grind unless GMS value is priced (proposal 4).
- **E7. Eligibility expansion celebrated, execution debt deferred.** +2 pts/qtr on O3 is an activity target. Hit it with B9's long tail and O1 pays later (A13).
- **E8. Cross-timezone / global cadence.** Canada / Mexico / international: different carriers, customs, data grains. A US-tuned WBR exported naively reads garbage.
- **E9. Hypercare absent on launches.** Speed programs ship like features (announce, move on) instead of go-lives (entrance criteria, hypercare, exit criteria). Same-day especially.

Family F – Economics and strategy (6 issues)

Worked example — unpriced elasticity and unpriced breach (F1, F2). Without ΔCVR per promise-day by category and price band, every buffer decision uses a national average that is wrong everywhere (elasticity is often $\sim 3\times$ higher on impulse SKUs than on considered purchases). Without a dollar number on a broken Saturday (contact + refund risk + LTV), the accuracy-vs-speed argument is theological. The 12-lane model's \$6.50 margin and \$12 breach are **assumptions to be attacked by Finance**, not truths. Put a number on it; fight about the number.

Remaining modes:

- **F3. Threshold-metric gaming.** Covered under O3: West Metro at 2.06 days is a full miss. Promise Yield (Part 4) pays on fast *and* kept.
- **F4. SFS / WFS boundary unmanaged.** Some SKU $\times$ geography cells cannot win on SFS. Converting them to WFS is better for Maya and for the seller, but the funnel crosses org lines (proposal 9).
- **F5. Competitive expectation ratchet.** Amazon's July 2026 SFP speed floors (40% of standard-size **page views** ≤ 1 day, 75% ≤ 2 days) reset customer expectations for everyone. Walmart's fast band is graded on Amazon's curve whether it likes it or not. Note the grain: **page views**, not orders — demand-weighted, where conversion happens.
- **F6. Trust is the only non-renewable resource.** GMS lost to a slow promise comes back next quarter; trust lost to a broken one compounds. Any strategy that trades trust for penetration is borrowing at loan-shark rates.

The shortlist – where the damage concentrates

If the hiring manager asks “what would you actually worry about”: **B8** (SSS adoption stall), **B1** (optimistic handling), **A3/A4** (heuristic uniform buffers), **A6/C5** (peak regime break), **D3** (attribution), **E1** (no single-threaded owner), **F1/F2** (unpriced elasticity and breach cost), **B10** (enforcement lag). Nine issues, on the order of 80% of the expected loss.

Part 3 – How Amazon would attack this (the other underwriter)

The interviewer's team already borrows Amazon's WBR grammar (input levers vs output metrics). The differentiation is knowing the *rest* of the machine — the mechanisms Amazon built around its seller-fulfilled program — and which ones transplant into Walmart's anatomy. Program facts below are verified against Amazon's published **2025–2026 SFP / OTDR** rules.

Maya's blender, listed on Amazon too

Same \$29 blender, same Dallas shop, same Maya in Sacramento. On Amazon the shop is in **Seller Fulfilled Prime (SFP)** or it is not. If it is, Maya sees a Prime date. The shop must clear published floors, including (current published bar) **OTDR 93.5%, valid tracking $\geq 99\%$, cancellations $< 0.5\%$** . Walmart's analogous floors in the sample WBR are $O2 \geq 90\%$, $O4 \leq 5\%$, $O5 \leq 2\%$ — looser on cancel, different grain on tracking.

The interesting difference is not the floor. It is **what the shop has to give up to get protection when the truck fails**.

The headline: Amazon converted an adoption problem into an insurance contract

Walmart's SSS pitch is: "let us manage your transit settings; we'll absorb carrier-delay accountability." Sample adoption: 38.5% → 45.3% of GMV in six weeks — still not the majority. Amazon's equivalent is not a pitch. It is a priced bundle with teeth.

A seller's late delivery does **not** count against On-Time Delivery Rate **only if all three are true at once**:

1. **Automated Handling Time (AHT)** enabled — Amazon computes the shop's real handling speed from observed scan history. Kills B1 (self-reported optimism).
2. **Shipping Settings Automation (SSA)** on the SKU's template — Amazon picks the transit promise. Kills C7 (cheap-service roulette) and the "I set transit hopefully" version of B1.
3. The label was bought through **Amazon Buy Shipping** as "**OTDR Protected**." The scan trail is the one Amazon trusts.

Cede all three controls → full accountability transfer. Keep any one → keep all the risk. Amazon does not beg sellers to adopt. It makes adoption the only rational hedge. It still refuses to excuse a **late warehouse handoff** — the shop's own controllable stays the shop's problem.

ILLUSTRATIVE 60/40 split (not Amazon-published; a teaching sketch). Dallas has **4% late deliveries** this month.

Whose fault (illustrative)	Share of the 4%	Points
Carrier (hub, weather, last mile)	60%	$0.60 \times 4.0 = 2.4$ points
Warehouse handoff (Dallas was actually late)	40%	$0.40 \times 4.0 = 1.6$ points

Without the bundle, the scorecard eats all **4.0 points**. OTDR-like on-time $\approx 96.0\%$. With the bundle, the 2.4 carrier points are excused; only **1.6 points** remain. On-time $\approx 98.4\%$. Same physical month, different contract.

If the shop had been at **8% late** (4.8 carrier + 3.2 warehouse): without protection 92.0% (under a 93.5% OTDR floor — listings in danger); with the bundle 96.8% (clears). **That gap is the product.** SSS that only toggles a transit setting, without observed handling and without a protected label, is selling a fraction of this contract and wondering why I10 stalls.

Enforcement grain matches the variance grain: **OTDR < 90% now deactivates the offending listings, not the whole account** (Feb 2026) — surgical, at the SKU where the miss lives. (The 93.5% is the SFP performance bar; the 90% is the listing-deactivation tripwire — two thresholds, one machine.)

Transplant to SSS: fold an AHT-equivalent into SSS (observed handling, not declared — Walmart’s B1 fix; I2 is the start), make scorecard protection conditional on the **full** bundle, price the benefit in Seller Center as projected GMV from the fast badge (O7 × that seller’s sessions), and enforce at item/lane grain rather than account grain. Part 4, proposal 4 packages this.

The mechanism table

#	Walmart issue (Part 2)	Amazon mechanism	What Tech Ops copies	Blender one-liner
1	B1 optimistic handling	Automated Handling Time — observed distributions replace declarations	Fold AHT-equivalent into SSS as default; declared handling becomes a scored prediction	Dallas stops typing “1 day”; Amazon already watched the 85% hit rate
2	B8 SSS adoption	OTDR-protection bundle (AHT+SSA+Buy Shipping) — all-or-nothing transfer	Rebuild SSS as a priced insurance contract, not a settings toggle	The 60/40 sketch: protection is the product
3	B10 / E-lag	Listing-level targeted deactivation (Feb 2026); weekly SFP reviews	Andon at the seller-lane cell: auto-widen buffer or pull fast badge within 24h, restore on recovery	Kill the 30-day delay; don’t nuke the whole shop for one SKU
4	F5 / O3 target design	SFP speed floors as % of page views per size tier (40% / 1d, 75% / 2d standard from Jul 2026), reviewed weekly	Define fast-promise penetration on page views (demand-weighted), not only orders	Measure the sentence where Maya converts, not where Dallas shipped
5	B9 long-tail execution	SFP trial gate: 100 packages / 30 days at full standards, badge withheld during trial, 3 attempts/yr, no graduations before peak	Earn-the-badge trial per cohort; graduation published; peak enrollment frozen	Don’t put a brand-new shop on Maya’s Thursday
6	A12 cutoff cliffs	Zip-code-level promise inputs (Sep 2026 tool): granular cutoffs and weekend schedules	Node-level cutoff/weekend as an SSS onboarding field	90 more minutes at Dallas can buy Maya a whole calendar day
7	C6 force majeure	Multi-carrier disruption protection (≥2 carriers affected ⇒ metric excusal, formal process)	Principled, auditable excusal — protects scorecard legitimacy, which protects SSS’s pitch	Hurricane week must not look like Dallas failed
8	E1 committee ownership	Single-threaded owner doctrine	This role claims O1+O3 outcome ownership; everyone else owns inputs	Saturday has a name next to it

#	Walmart issue (Part 2)	Amazon mechanism	What Tech Ops copies	Blender one-liner
9	E2 WBR theater	WBR with pre-wired reds and “the ask” ; COE (correction-of-error) post-mortems, blameless, mechanism-focused	Reds arrive with cause+owner+date; accuracy incidents $\geq X$ bps get a COE that changes a mechanism	Wk 32 hub: not “try harder” — pickup windows + 42-lane buffer re-solve
10	E5 unpriced changes	Weblab culture — nothing ships without an experiment	Buffer/display/eligibility gated by A/B with conversion and breach cost (fixes F1 too)	O7 is not optional decoration
11	D9 severity blindness	Defect taxonomy + DPMO — count and classify every defect	Lateness severity in the WBR; dollar-weighted breach lines (fixes D7)	Sunday vs next Friday are not the same miss
12	F2 unpriced breach	“Cost of a defect” doctrine	Publish the breach-cost model; let Finance attack it; survivor sets the buffer dial	\$12 in the 12-lane file is a stake in the ground, not a truth
13	A5 cold start	New-seller probation defaults — conservative promises until evidence accumulates	Hierarchical priors; new cells start padded, earn tighter buffers	First blender from a new shop does not get a heroic Thursday
14	F4 SFS/WFS boundary	FBA as the default escape valve	Promise-to-WFS conversion funnel: rank cells where SFS structurally loses	Mountain Rural at 7.06 days is a fulfillment-mode problem, not a buffer problem

Speed floors, page views, and why Walmart should care

Amazon’s July 2026 SFP floors are **percent of page views**, not percent of orders: 40% of standard-size page views promised in ≤ 1 day, 75% in ≤ 2 days, reviewed weekly. That is O3 with two upgrades: (1) a 1-day bar, not only 2-day; (2) demand-weighted grain.

ILLUSTRATIVE. 100 blender PDP views. 70 are California-like (can be 1.39 days). 30 are mountain-like (7.06 days). Order-weighted O3 might look fine if Californians convert and order more. Page-view O3 asks: what did the other 30 Mayas *see*? If they saw 7 days, Amazon’s 75%-at-2-days bar fails even if most *orders* were fast. Customer expectation is set at the view, not at the shipment.

What NOT to copy (say so – it reads as judgment, not worship)

- **Amazon’s leverage came with FBA’s scale behind it.** Amazon could make demands of sellers because FBA was a real alternative: send us the blender, we will ship it. Walmart’s WFS is younger and smaller. SSS must win more on carrots (priced GMV upside, absorbed carrier risk) and less on sticks, especially with strategic sellers. Copy the **contract structure**. Tune enforcement aggression down a notch until WFS is a credible alternative at scale.
- **Walmart’s ~4,600-store network is an asymmetric asset Amazon cannot match.** Store-adjacent same-day, returns density, local courier integration are Walmart-native levers. An Amazon-clone roadmap would leave them on the table. Maya in Sacramento can, in some cells, be a *store* problem rather than a Dallas-to-porch problem.
- **Don’t import the churn tolerance.** Amazon accepts seller churn as a cost of standards. Walmart Marketplace is still in assortment-growth mode; a churned seller is a strategic hole in the mall.

Enforcement thresholds should be solved for Walmart's seller-lifetime economics, not Amazon's.

****Systems punchline — copy the feedback design, not the costume.**** Amazon did not win SFP with a prettier badge. It coupled **observed reality** (AHT) to **promise authority** (SSA) to **label telemetry** (Buy Shipping) to **surgical enforcement** (listing-level) to **demand-weighted goals** (page-view floors). That is one control loop. SSS as a settings toggle is a lever disconnected from its loop. The transplant is to reconnect it — then use stores and WFS as Walmart-only stocks Amazon does not have.

****The blender, end of Part 3.**** On Walmart, Maya's Thursday is an insurance policy priced with a buffer, sold by a mall that does not own the warehouse. On Amazon, the same shop can buy a real transfer of carrier risk only by surrendering handling, transit, and label. The interview is about whether you see that difference as a ****mechanism gap**** (fixable — proposal 4) rather than as "Amazon is Amazon." Parts 4–8.5 turn the gap into moves, math, stories, a Tuesday one-pager, and a 30-minute clock.

****HM lens — Invent and Simplify / Have Backbone.**** Copy Amazon's **contract** (observed handling + transit authority + protected label + listing-level Andon + page-view speed floors). Do not copy FBA-backed churn tolerance. If they ask "disagree with me on SSS," your backbone line is: SSS as a settings toggle is not insurance; without the bundle it will stall near half of GMV and you will pad buffers to hide it.

Part 4 – Ten proposals (the innovation section)

These are the ideas to spend sparingly — one or two in the room, chosen to match whatever the hiring manager says is red. Each is stated as: the move, why it wins, how it's measured, and the honest failure mode. All ten are consistent with the issue tree and the verified math in Part 5.

Proposal 1 – Promise Yield: one number for the whole trade

The move. Grade the pod on **Promise Yield = fast-promise penetration × promise accuracy** — the fraction of orders that got a fast promise AND kept it. Today those two metrics live on separate lines with separate owners, and they trade against each other by construction: either can be juiced by bleeding the other invisibly. **Why it wins.** It makes the speed-accuracy trade un-gameable at the KPI layer, because both factors have to survive the same multiplication.

The arithmetic, self-contained.

PY = O3 (fast-promise penetration) × O1 (promise accuracy)

six-week sample average: $0.436 \times 0.952 = 0.4151 \rightarrow 41.5\%$

Wk 33 specifically: $0.451 \times 0.954 = 0.4303 \rightarrow 43.0\%$

4-quarter target: 50.0%

Now price the two cheats it kills. **ILLUSTRATIVE**, starting from Wk 33's 45.1% / 95.4%:

Move	O3	O1	PY	Verdict
Wk 33 baseline	45.1%	95.4%	$0.451 \times 0.954 = 43.0\%$	—
Pad nationally: buy 1.5 accuracy points by slowing promises 6 points	39.1%	96.9%	$0.391 \times 0.969 = 37.9\%$	−5.1 pts — caught
Cliff-chase: badge 5 more points of lanes that cannot execute, accuracy pays 2	50.1%	93.4%	$0.501 \times 0.934 = 46.8\%$	+3.8 pts, but O1 breaches the 95% floor → hard RED regardless of PY
Real: σ down, so buffers shrink at constant coverage	48.6%	95.4%	$0.486 \times 0.954 = 46.4\%$	+3.4 pts, honestly

What it means: padding shows up as a *loss* on the north star even though it makes the accuracy line prettier — which is exactly the silent tax made visible. Cliff-chasing does raise PY, which is why PY alone is not enough: it rides **under** the hard accuracy floor, and the floor is what stops it. **What I'd do:** publish PY as the north star and the four policy floors as guardrails on the same header line, so neither can be traded for the other without someone noticing in the room.

Measure. Weekly, order-weighted and page-view-weighted (Amazon defines its SFP speed floors on page views — demand-weighted is the honest grain). **Failure mode.** Composite metrics can hide which factor moved; always decompose one line below — print O3 and O1 beside PY, never PY alone.

Proposal 2 – σ -proportional buffers with one governed dial

The move. Replace the per-lane heuristic (buffer somewhere in the p50–p90 band) with **$b = k \cdot \sigma$ per lane cohort**, where k is a single portfolio dial solved weekly against the accuracy floor ($k = 1.645$ hits a 95% floor exactly under normal errors). **Why it wins.** It converts buffer-setting from a hundred ML micro-decisions into ONE auditable policy instrument that this role owns — the underwriter’s dial. σ improvements from DS flow into speed automatically and visibly, which also fixes the political problem of DS work being invisible to the business.

The arithmetic, self-contained. Model the combined execution error on a lane as $\varepsilon \sim \text{Normal}(0, \sigma^2)$, so the chance the pad covers the noise is $\Phi(b/\sigma)$, the standard-normal CDF. Pick the coverage you want, then read k off the curve:

$$k = \Phi^{-1}(\text{target coverage}) \quad k = \Phi^{-1}(0.95) = 1.6449$$

$$b_{\text{lane}} = k \times \sigma_{\text{lane}}$$

Φ bookmarks: $\Phi(1.00) = 84.1\%$ $\Phi(1.645) = 95.0\%$ $\Phi(2.00) = 97.7\%$ $\Phi(2.33) = 99.0\%$

One multiplication per lane. On the 12-lane model’s extremes (σ spans 7 $\times$):

California Intra	$\sigma = 0.22 \rightarrow b = 1.645 \times 0.22 = 0.36$ days
Mountain Rural	$\sigma = 1.60 \rightarrow b = 1.645 \times 1.60 = 2.63$ days

Against the naive uniform 1.10 days everywhere, the same protection budget lands completely differently:

Lane	σ	Uniform $b = 1.10 \rightarrow \Phi(1.10/\sigma)$	σ -proportional $b = 1.645\sigma \rightarrow \Phi(1.645)$
California Intra	0.22	$\Phi(5.00) \approx 100\%$ — three times the pad it needed	0.36 d $\rightarrow$ 95.0% , and 0.74 days handed back to speed
Mountain Rural	1.60	$\Phi(0.69) = 75.4\%$ — one surprise in four already breaks the promise	2.63 d $\rightarrow$ 95.0% , honestly priced

Portfolio result, order-weighted, from `model_reference.py` :

Policy A (uniform $b = 1.10$):	95.8% accuracy, 7.6% fast share
Policy B ($b = 1.645 \times \sigma$):	95.0% accuracy, 39.6% fast share
fast-share gain:	$39.6 - 7.6 = 32.0$ points, for 0.8 points of accuracy

What it means: this is not a better model, it is the same total protection re-allocated by measured risk — and it is worth 32 points of the fast band. The 0.8 accuracy points given up were over-protection California never needed. **What I’d do:** run it in shadow on ten lanes first (proposal 7), because the whole gain is only as good as the σ estimates underneath it.

Measure. k published in the WBR header every week; fast share and accuracy move together or the model’s σ estimates are wrong — self-auditing. **Failure mode.** Garbage σ estimates make k confidently wrong; gate with backtests per lane before cutover (shadow mode, proposal 7).

Proposal 3 – Buffer Waste Index: the “speed for free” backlog

The move. Track **BWI = mean days delivered early on kept promises**, per lane cohort, ranked by $BWI \times \text{volume}$. **Why it wins.** Every early day is buffer the customer paid for in promise speed and didn’t need. Nobody tracks over-delivery because it files no tickets — it is the silent half of the padding asymmetry. The ranked list IS the buffer-trimming roadmap, pre-prioritized, zero new models needed.

The arithmetic, self-contained.

```
BWI_lane = mean( promised_EDD - actual_delivery ) over KEPT promises only
           (breaches are excluded: a negative "early day" is a different defect)
ranking key = BWI_lane × weekly orders on that lane
```

ILLUSTRATIVE. Ten kept deliveries on one lane, days early: 2, 1, 2, 3, 1, 2, 2, 1, 3, 2.

$$BWI = (2+1+2+3+1+2+2+1+3+2) / 10 = 19 / 10 = 1.9 \text{ days early, on average}$$

Now separate the waste from the insurance. That lane’s measured σ is 0.30 days, so an honest 95% pad is:

```
honest b    = 1.645 × 0.30 = 0.49 days
actual pad = 1.10 days (uniform policy)
waste      = 1.10 - 0.49 = 0.61 days of promise speed bought and never used
coverage lost by trimming:  $\Phi(1.10/0.30) = \Phi(3.67) \approx 100.0\% \rightarrow \Phi(1.645) = 95.0\%$ 
```

Rank three lanes by waste $\times$ volume to see why the index is a backlog and not a report:

Lane (ILLUSTRATIVE volumes)	BWI	Orders/wk	BWI $\times$ volume	Rank
Tight metro	1.9 d	120,000	228,000 order-days	1
Mid- σ regional	0.8 d	90,000	72,000 order-days	2
Mountain rural	0.4 d	15,000	6,000 order-days	3

What it means: the biggest pile of recoverable speed sits on the lanes that look *healthiest* — high volume, low σ , always green — which is precisely why nobody is looking there. The mountain lane, the one everybody argues about, has almost no waste to harvest. **What I’d do:** hand the ranked list to Promise DS as the quarter’s speed backlog, and trim it by re-solving k for the whole portfolio rather than surgically editing the top lane, so the accuracy floor is never breached lane by lane.

Measure. New input line in the WBR rolling into O3; target: portfolio BWI down 0.3 days in two quarters with O1 flat. **Failure mode.** Early delivery is partly intentional slack for carrier variance; trim by re-solving k , not by ad-hoc lane surgery.

Proposal 4 – Seller Promise Insurance: the SSS adoption unlock

The move. Rebuild the SSS offer as Amazon rebuilt theirs (Part 3.1) — a priced insurance contract with an all-or-nothing bundle: (a) SSS transit automation, (b) **observed handling time** (Walmart computes it from the seller’s scan history — the seller stops declaring), (c) participating label purchase. Full bundle $\rightarrow$ carrier

lateness excused from the seller's scorecard AND the fast badge where the math clears. Partial adoption → no protection. And show every non-enrolled seller a personalized line in Seller Center: **"SSS would raise your projected GMV by \$X/month"** — computed from the badge's measured conversion lift on their assortment. **Why it wins.** Today SSS asks sellers to give up control for an abstract benefit. This makes the benefit concrete (dollars, on their dashboard), the trade explicit (protection for control), and the mechanism self-selecting: sellers who trust their own ops keep control and risk; sellers who don't, convert. Amazon ran this exact play and it drove SSA/Buy Shipping to default status. **Measure.** I1 (% GMV on SSS) and I10 (activations/week + 30-day retention), with a new line: % of enrolled GMV under full-bundle protection. **Failure mode.** Label-purchase requirements can antagonize sellers with negotiated carrier rates; tier the requirement (protection scales with bundle depth) rather than cliff it, at least until WFS is a credible alternative.

Proposal 5 – Real-time promise Andon (kill the 30-day lag)

The move. Per seller-lane cell, run an EWMA/CUSUM control chart on 7-day breach rate. Tripwire (sustained $2\times$ baseline, $n \geq 30$): **auto-widen that cell's buffer or pull its fast badge within 24 hours**, notify the seller with the specific defect evidence, auto-restore when the control chart recovers. **Why it wins.** Policy floors on 30-day windows let a degrading seller burn customers for weeks (issue B10). This is a control loop, not a policy review — and it protects sellers too: surgical cell-level action instead of account-level scorecard damage. Amazon converged on the same grain in Feb 2026 (listing-level deactivation). **The arithmetic, self-contained — and it is the reason the design is CUSUM and not a threshold.** Baseline breach rate on a healthy cell is the portfolio's 4.6% late-to-promise ($1 - O1\ 95.4\%$). The tripwire is $2\times$ baseline = **9.2%**.

cell of $n = 30$ orders: expected breaches = $30 \times 0.046 = 1.38$
tripwire at $2\times \Rightarrow$ fire when breaches $\geq 30 \times 0.092 = 2.76$, i.e. ≥ 3

Ask how often a *perfectly healthy* cell throws 3 or more breaches by luck alone. Binomial($n = 30$, $p = 0.046$):

$P(0) = 0.954^{30} = 0.244$
 $P(1) = 30 \times 0.046 \times 0.954^{29} = 0.352$
 $P(2) = 435 \times 0.046^2 \times 0.954^{28} = 0.246$
 $P(\geq 3) = 1 - 0.244 - 0.352 - 0.246 = 0.158 \rightarrow \sim 16\% \text{ false-trip rate}$

Same test at $n = 100$ (mean 4.6, $sd = \sqrt{(100 \times 0.046 \times 0.954)} = 2.10$, fire at ≥ 10):

$z = (9.5 - 4.6) / 2.10 = 2.33 \rightarrow P(\text{false trip}) \approx 1\%$

What it means: a single-week $2\times$ threshold on a 30-order cell is wrong about one time in six — that is not an Andon, that is a random seller-punishment generator. The $n \geq 30$ floor only guarantees the *signal exists*; it does not make one week's reading trustworthy. **What I'd do:** keep $n \geq 30$ as the floor, but fire on accumulated evidence — EWMA/CUSUM across weeks, which reaches the same $2\times$ conviction at an effective sample large enough to hold false trips under the 5% budget, and auto-restores when the chart recovers. Publish the false-trip budget alongside the tripwire, so sellers can see the standard they are being judged against.

Measure. Detection-to-action lag (target $< 24h$ from sustained signal), false-trip rate ($< 5\%$), customer breaches avoided (counterfactual from pre-Andon baseline). **Failure mode.** Noise-tripping punishes small sellers; the n

≥ 30 floor and CUSUM (not single-week thresholds) are the guard — with the 16% figure above as the proof of why.

Proposal 6 – Cutoff arbitrage: buy speed with a calendar, not a model

The move. For each top node × carrier, map the **promise cliff** (the hour where EDD jumps a day) against the intraday order-volume curve. Where volume is dense just after the cliff, buy later pickups / later cutoffs — a carrier negotiation and a seller-onboarding field, not an ML project. **Why it wins.** For orders in the recovered window, promise speed improves a full day at zero model risk and zero accuracy cost. Amazon's Sep 2026 zip-level promise tool is this same insight productized (granular cutoffs feeding the promise). Likely the cheapest speed points on the roadmap.

The arithmetic, self-contained. A cutoff is a step function, not a slope: an order at 3:59 PM gets today's ship date, an order at 4:01 PM gets tomorrow's — a full calendar day for two minutes of clock. So the value of moving the cutoff is (sessions in the recovered window) × (conversion at the faster promise – conversion at the slower one).

Price the promise-day using the 12-lane model's conversion curve (`model_reference.py` : $CVR_0 = 7.2\%$ at a 1-day promise, speed elasticity $\beta = 0.16$ per promise-day, 2-day badge bonus $\gamma = 0.11$):

```
CVR(d) = 0.072 × e(-0.16 × (d - 1)) × (1 + 0.11 × badge(d))

d = 3 days: 0.072 × e(-0.32) × 1.000 = 0.072 × 0.7261 = 5.229%
d = 2 days: 0.072 × e(-0.16) × 1.055 = 0.072 × 0.8521 × 1.055 = 6.472%
            (the ×1.055 is half the badge bonus – d = 2 sits on the badge threshold)

gain per session moved 3 days → 2 days: 6.472% – 5.229% = 1.243 points
```

ILLUSTRATIVE node. 100,000 weekly PDP sessions fall in the 4:00–5:30 PM window that a 90-minute cutoff extension would recover:

```
extra orders = 100,000 × 0.01243 = 1,243 orders/week
at $6.50 contribution margin = 1,243 × 6.50 = $8,080/week
annualized = $8,080 × 52 ≈ $420K/year, from ONE node
accuracy cost = zero – b, σ, and the model are untouched
```

What it means: this is the only speed lever in the book that does not spend accuracy to buy penetration, because it changes the *starting clock* rather than the pad. Every other proposal trades something. **What I'd do:** rank the top 20 nodes by sessions-in-the-recovered-window before touching a model, because the ranking is a SQL query against order timestamps and the fix is a carrier pickup negotiation — weeks, not quarters.

Measure. Orders/week crossing into the faster promise band × measured conversion delta; rank nodes by recoverable volume. **Failure mode.** Later cutoffs squeeze seller pick-pack windows — B2's day-of-week reality applies; pilot on SSS-enrolled nodes where Walmart controls the config.

Proposal 7 – Shadow challenger: the counterfactual WBR line

The move. Always run one challenger buffer policy in shadow — computed and logged per order, never displayed. The WBR carries a standing line: challenger vs champion on accuracy, speed, and modeled profit.

Promotion requires beating the champion on the frontier for 4 consecutive weeks. **Why it wins.** Makes policy evolution continuous and safe instead of episodic and political; the WBR debates measured counterfactuals instead of opinions. It is the promise-engine version of the mock-cutover discipline — rehearse against reality before you commit (and it's how I ran migrations at Sephora: three mocks, published tolerance, then go). **The arithmetic, self-contained — why four weeks and not two.** The promotion bar is a sequence of independent weekly coin-flips against the champion. A challenger with *no real edge* wins any given week with probability 0.5, so:

$$P(\text{worthless challenger clears the bar}) = 0.5^w$$

w = 2 weeks: $0.5^2 = 0.250 \rightarrow 25.0\%$ false promotion — unusable

w = 3 weeks: $0.5^3 = 0.125 \rightarrow 12.5\%$

w = 4 weeks: $0.5^4 = 0.0625 \rightarrow 6.25\% \leftarrow$ the bar

w = 5 weeks: $0.5^5 = 0.031 \rightarrow 3.1\%$, at the cost of five weeks of latency

What it means: four weeks buys a ~6% false-promotion rate, stated out loud rather than assumed away — and the second half of the guard is sample size: each “week” must be hundreds of thousands of orders per lane cohort, not thirty, or the weekly comparison is itself a coin flip and the 6.25% is optimistic. **What I'd do:** publish the bar *and* its false-positive budget before the first shadow week, exactly the way I published a 0.5% reconciliation tolerance before mock cutover one at Sephora, so promotion is a measured decision instead of an enthusiasm.

Measure. Challenger win-rate; time-from-idea-to-promoted-policy. **Failure mode.** Shadow scoring needs D1 (promise-at-display logging) airtight first — which is worth doing anyway.

Proposal 8 – Mix-adjusted twins for every output metric

The move. Report every output metric twice: raw, and mix-adjusted (fixed cohort weights from a base quarter). The delta between the two IS the mix effect, printed as its own line. **Why it wins.** Kills Simpson's paradox — the WBR stops celebrating “improvements” that are really volume migrating to easy lanes, and stops flogging teams for “degradation” that is really growth into hard geographies. One analyst-week to build; permanent honesty dividend.

The arithmetic, self-contained.

```
raw metric    = Σ (this week's cohort weight × this week's cohort rate)
twin          = Σ (BASE quarter's cohort weight × this week's cohort rate)
mix effect    = raw - twin          ← printed as its own line
```

ILLUSTRATIVE. Two lanes, base-quarter weights 50 / 50, and a week where volume migrates to the easy lane:

	Wk 1 volume	Wk 1 accuracy	Wk 2 volume	Wk 2 accuracy
California Intra	50	96%	80	95% (worse)
Mountain Rural	50	92%	20	91% (worse)

Wk 1 raw	= $0.50 \times 96 + 0.50 \times 92 = 94.0\%$	
Wk 2 raw	= $0.80 \times 95 + 0.20 \times 91 = 94.2\%$	← the WBR celebrates +0.2
Wk 2 twin	= $0.50 \times 95 + 0.50 \times 91 = 93.0\%$	← base weights, honest read
mix effect	= $94.2 - 93.0 = +1.2$ points	← geography, not performance
true move	= $93.0 - 94.0 = -1.0$ point	← every cohort got worse

What it means: the portfolio improved while *every single cohort degraded*. One extra line on the page turns a fake +0.2 into a real -1.0 and names the cause as mix. **What I'd do:** ship the twin in week one of owning the review — it is a weighted-average query, not a program, and it changes the conversation immediately because nobody can argue with their own cohort rates.

Measure. Itself — the mix line. **Failure mode.** Base-period weights go stale; re-base quarterly with a documented policy.

Proposal 9 – The SFS→WFS conversion funnel

The move. Score every SKU × geography cell: can SFS structurally hit the fast band here (σ , node distance, seller capability), or would WFS win? Publish the ranked migration list — cells where the promise gap × demand is largest — as a standing artifact handed to the WFS/Marketplace org. **Why it wins.** It's the honest answer to B9/F4: some cells shouldn't be fixed, they should be moved. Tech Ops generating a revenue-ranked funnel for an adjacent org is exactly the "serve the whole business, not the silo" behavior a Senior Manager is hired for — and it converts an awkward internal boundary into a managed pipeline. **Measure.** GMS uplift on converted cells; promise-gap closure per conversion. **Failure mode.** Reads as SFS surrendering volume; frame targets on total marketplace promise quality, not SFS share retention.

Proposal 10 – Peak regime: a seasonal k, declared in advance

The move. Treat Q4 as a different statistical regime, because it is: σ inflates ~1.45× when carrier networks saturate. Re-solve the k-dial for peak σ , publish the peak promise policy (and its speed cost) to the business in October — before Black Friday, not during. **Why it wins.** Announcing the trade (slightly slower promises, accuracy held) in advance converts a November firefight into an October memo. Amazon freezes SFP trial graduations before peak for the same reason — regime changes are managed, not discovered.

The arithmetic, self-contained — this is the failure that hides, because nobody changes anything. Take a lane sitting at $\sigma = 1.20$ days in July, buffered honestly:

summer: $b = 1.645 \times 1.20 = 1.98$ days → coverage $\Phi(1.98 / 1.20) = \Phi(1.65) = 95.0\%$

Now freeze b and let peak inflate σ by the model's ~1.45×:

peak σ	= $1.20 \times 1.45 = 1.74$ days	
coverage	= $\Phi(1.98 / 1.74) = \Phi(1.14) = 87.3\%$	← 7.7 points lost, buffer untouched
honest peak buffer	= $1.645 \times 1.74 = 2.86$ days	← +0.88 days of promise, declared

The same mechanism drives the non-seasonal version — a carrier consolidates and σ drifts from 1.20 to 1.90 with b still frozen at 1.98:

coverage = $\Phi(1.98 / 1.90) = \Phi(1.04) = 85.1\%$ ← ~10 points gone silently
 re-solved b = $1.645 \times 1.90 = 3.14$ days

What it means: nobody edited a config and the lane still lost 8 to 10 points of coverage, because the pad is denominated in days while the risk is denominated in sigmas. A July policy is a November breach with a delay fuse on it. **What I'd do:** re-solve k against peak σ and publish the peak promise policy — the wider buffers, the named lane classes, and the fast-share cost — in **October**, alongside a drift flag per lane that pulls a lane from the fast band automatically when its σ blows out mid-peak. The success metric is zero emergency buffer interventions between Thanksgiving and Christmas.

Measure. Peak-season O1 vs floor; zero emergency buffer interventions between Thanksgiving and Christmas. **Failure mode.** Over-padding peak kills the quarter's penetration target — which is why the dial is solved, not guessed.

Part 5 – The math that wins the room

Everything here is computed and verified in `model_reference.py` and reproduced live in the workbook's Solver tab. Twelve illustrative lane cohorts (~1M orders/week), errors $\varepsilon \sim N(0, \sigma^2)$ per cohort, σ from 0.22 days (metro same-day) to 1.60 days (mountain rural).

5.1 The four-policy table (the six numbers to memorize)

Policy	Accuracy	Avg promise days	Fast share ($\leq 2d$)	Weekly profit
A. Naive uniform buffer (1.10d everywhere)	95.8%	3.09	7.6%	\$5.69M
B. σ -proportional, one dial ($b = 1.645\sigma$)	95.0%	2.75	39.6%	\$5.94M
C. Min-days optimum (analytic KKT)	95.8%	2.81	22.4%	\$6.00M
D. Profit optimum (what Excel Solver finds)	97.8%	2.93	22.9%	\$6.07M

The six numbers, derived rather than recited. Every row is one multiplication per lane and one normal-table lookup, so you can rebuild the table on a whiteboard if challenged:

```

promise days on a lane    d = handling + transit_predicted + buffer
fast                     d ≤ 2.0 days
coverage on a lane        Φ(b / σ)                (ε ~ Normal(0, σ²))
portfolio accuracy        order-weighted mean of Φ(b/σ)
fast share                order-weighted share of lanes with d ≤ 2.0

Policy A:  b = 1.10 on every lane                (one number, no measurement)
Policy B:  b = k × σ,  k = Φ⁻¹(0.95) = 1.6449 (one dial, measured σ)

```

Worked on the two extremes, which is where the whole story lives:

Lane	σ	$h + \tau$	$b = 1.10 \rightarrow \text{days} \cdot \text{coverage}$	$b = 1.645\sigma \rightarrow \text{days} \cdot \text{coverage}$
California Intra	0.22	0.85	$0.85 + 1.10 = \mathbf{1.95 \text{ d, FAST}}$ · $\Phi(5.00) \approx 100\%$	$1.645 \times 0.22 = 0.36 \rightarrow 0.85 + 0.36 = \mathbf{1.21 \text{ d, FAST}}$ · $\Phi(1.645) = 95.0\%$
Northeast Metro	0.28	1.30	$1.30 + 1.10 = \mathbf{2.40 \text{ d, not fast}}$ · $\Phi(3.93) \approx 100\%$	$1.645 \times 0.28 = 0.46 \rightarrow \mathbf{1.76 \text{ d, FAST}}$ · 95.0%
West Metro	0.30	1.45	$1.45 + 1.10 = \mathbf{2.55 \text{ d, not fast}}$ · $\Phi(3.67) \approx 100\%$	$1.645 \times 0.30 = 0.49 \rightarrow \mathbf{1.94 \text{ d, FAST}}$ · 95.0%
Mountain Rural	1.60	4.60	$4.60 + 1.10 = \mathbf{5.70 \text{ d, not fast}}$ · $\Phi(0.69) = 75.4\%$	$1.645 \times 1.60 = 2.63 \rightarrow \mathbf{7.23 \text{ d, not fast}}$ · 95.0%

Read the pattern: under A, three metro lanes are drowning in unused pad and *still* miss the 2-day badge by tenths of a day, while the mountain lane is underwritten at 75%. Under B, the pad moves from where it was wasted to where it was missing — and three metro lanes cross into the fast band. Order-weighted across all twelve cohorts that is:

fast share:	7.6% → 39.6%	gain = 39.6 - 7.6 = 32.0 points
accuracy:	95.8% → 95.0%	cost = 0.8 points, all of it over-protection given back
profit:	\$5.69M → \$5.94M	gain = \$250K/week ≈ \$13M/year

Three sentences of meaning:

1. **B vs A — the free lunch.** Reallocating the same total protection in proportion to each lane's measured risk buys **+32 points of fast-promise penetration at the same 95% accuracy commitment** and roughly +\$250K/week. No new models — just stop buffering uniformly. This is the 60-second whiteboard story.
2. **D vs A — the economics.** With a breach costing ~\$12 (contact + refund risk + trust) against a \$6.50 margin, the profit optimum holds **97.8% accuracy — above the 95% floor**. The constraint isn't even binding. Translation for the room: *either the floor is set below what the economics want, or the breach-cost estimate is too high — and finding out which is a governance decision this role should force*. A candidate who can say "your target and your economics disagree, here's how to tell which one is wrong" is doing the Senior Manager job in the interview.
3. **B vs D — the honest tension.** The KPI-maximizing policy (B) and the profit-maximizing policy (D) are different policies. B chases the ≤2-day threshold; D buys accuracy because breaches are expensive. Which to run is not a math question — it's a question about what fast-promise penetration is *for* (conversion now vs trust compounding), and it should be decided explicitly, once, at leadership level, not implicitly by whoever tunes the model.

5.2 The equimarginal principle (30 seconds, plain English)

At the true optimum, **the last buffer-day on every lane must buy the same amount of accuracy** — formally, $\phi(b/\sigma)/\sigma$ equal across lanes (verified to machine precision: spread < 1e-15).

Where that condition comes from, in three lines. Minimize total promise days subject to a portfolio accuracy floor. The Lagrangian derivative on lane i is:

$$\begin{aligned} d/db [b_i - \lambda \cdot \Phi(b_i/\sigma_i)] &= 1 - \lambda \cdot \phi(b_i/\sigma_i)/\sigma_i = 0 \\ \Rightarrow \phi(b_i/\sigma_i) / \sigma_i &= 1/\lambda \quad - \text{the same constant on every lane} \end{aligned}$$

ϕ is the normal *density* (the height of the bell curve), not Φ the cumulative — density is the marginal return on one more buffer-day, cumulative is the coverage you already own.

Plug the extremes in at $k = 1.645$, where $\phi(1.645) = 0.1031$:

California Intra (σ 0.22):	$0.1031 / 0.22 = 0.469$ accuracy per buffer-day
Mountain Rural (σ 1.60):	$0.1031 / 1.60 = 0.064$ accuracy per buffer-day
ratio:	$0.469 / 0.064 = 7.3\times$

What it means: at equal sigmas of protection, a buffer-day spent on California is worth **7.3× more accuracy** than the same day spent on the mountain lane. The optimizer therefore does *not* pour more days into the chaotic lane — it takes days *away* and spends them where they convert into kept promises. **What I'd do:** when someone proposes "add half a day nationally," ask what $\phi(z)/\sigma$ that buys per lane; the answer is almost always that the money is being spent where it buys the least.

Consequences:

- Low- σ lanes get thin buffers — speed harvested where prediction is good.
- High- σ lanes get *fewer sigmas* of protection, not more — protecting them is expensive per accuracy point.
- Past a σ threshold, the interior solution vanishes: **stop buffering the lane and pull it from the fast band** (or route it to WFS — proposal 9). Buffering a chaotic lane into compliance is underwriting a known loss.

Spoken version: “Buffer is a portfolio of risk capital. Spend it where a day of padding buys the most kept promises — and where it buys almost nothing, stop promising fast and fix the lane instead.”

5.3 Where the model is honest about itself

Say these before the interviewer does: errors aren’t perfectly normal (freight tails are right-skewed — use the empirical quantile per lane in production; normal is the whiteboard version); σ is estimated, not known (bad σ makes k confidently wrong — hence shadow mode first); the conversion elasticity (β) and breach cost are assumptions to be measured by A/B and priced with Finance — *the model’s job is to make the argument quantitative, not to be right to the third decimal*. The Q4 tab shows the same optimum breaking under peak σ — policy is regime-conditional.

Part 6 – The 90-day plan (say it in 45 seconds, expand on request)

Days 1–30 — Learn the machine, touch nothing. Sit the WBR; map each metric to its owner and its decomposition; meet Promise DS, Seller Ops, Carrier PM, Product, and five sellers (two enrolled in SSS, two refused, one churned — the refusals teach the most). Deliverable: the issue tree of Part 2 rebuilt with *their* numbers, and the one metric the room trusts least identified. **Days 31–60 — Own the review.** Take the output block and the action log. Every red decomposes to a named input, owner, and date before the meeting, not during. Ship the two one-week analytics wins: mix-adjusted twins (proposal 8) and the Buffer Waste Index (proposal 3) — both are queries, not programs, and both change conversations immediately. **Days 61–90 — Move one number visibly.** Pick by expected value from what 1–30 revealed; default candidate: σ -proportional buffer pilot on 10 lanes in shadow mode (proposals 2+7), promoted if it wins 4 straight weeks. Alternates: SSS enrollment funnel fix at the top drop-off, or cutoff arbitrage on the top 5 nodes. **The closing line:** *“You shouldn’t have to teach me the operating system — only your numbers. My 90-day deliverable is one output metric moved by a mechanism the WBR can audit.”*

Part 7 – The story pack, re-aimed at this room

Eight stories re-targeted from the Apple v11 pack. Same canonical numbers, new aim: every nugget now lands on a pre-purchase-delivery-experience nerve. Speak the nugget; the mapping line is for your head, not the room.

Three of these stories are not anecdotes — they are the promise equation, term by term. Lead with this mapping and everything after it reads as evidence:

promised_EDD = t_order + handling_config + transit_predicted + buffer		
story 7.2	story 7.1	story 7.8
Flexport shipment	Elementum:	Manhattan DOM:
visibility – did the	200 lanes of	estimated lead time /
shipper actually	measured transit	promise days out of
ship in the window,	distributions,	sourcing the order
and did the first	promise set at	from the best node
scan prove it (I3)	the tail (p99)	through allocation

The other five stories (7.3–7.7) are how the machine gets *run* — the WBR, adoption gates, cost-math roadmaps, promising governance, and cutover discipline. Do not confuse the two piles in the room: 7.1 / 7.2 / 7.8 say *I have owned your equation*; 7.3–7.7 say *I can operate the org around it*.

7.1 Flagship: “Promise at the tail, not the brochure” (*Elementum* · v11-F4) – the transit_predicted term

Walmart-tuned nugget: “The exact problem this role owns — I’ve solved it once at smaller scale. Our inventory plan ran on carrier-brochure lead times: single optimistic numbers, never measured. I replaced them with measured transit distributions across 200 lanes — mean, spread, and tail per lane — and moved every promise to the tail, not the average. Result: 99% lead-time accuracy held, and because measured spread was *smaller* than assumed on most lanes, 580,000 units and \$1.5M of safety stock came free. Then a drift flag per lane so a widening spread triggered review instead of silent re-planning.” **Why it maps:** this is buffer calibration (I5, I6) in miniature — σ measured per lane, promise set at a quantile, monitoring for regime drift. It also carries proposal 2’s logic: measured spread beats assumed spread, and sometimes reality is *faster* than the padding.

Say “p99” correctly, because a hiring manager who models will check. p99 is **cumulative**: the smallest number of days d such that $P(L \leq d) \geq 0.99$ — “99 of every 100 shipments have already arrived by day d .” It is **not** “the day on which 1% of shipments arrive.” **ILLUSTRATIVE** 100-shipment lane with the right-skew freight always has:

Transit days d	shipments	cumulative	$P(L \leq d)$
9	5	5	0.05
10	14	19	0.19
11	21	40	0.40

Transit days d	shipments	cumulative	$P(L \leq d)$
12	21	61	0.61 $\leftarrow$ p50
13	15	76	0.76
14	9	85	0.85
15	6	91	0.91 $\leftarrow$ p90
16	3	94	0.94
19	3	97	0.97
22	3	100	1.00 $\leftarrow$ p99

mean = $1,250 / 100 = 12.50$ days $\sigma = 2.64$ days
 promise at the mean 12.50 $\rightarrow$ 61 of 100 beat it – two in five arrive LATE
 promise at p90 15 $\rightarrow$ 91 of 100
 promise at p99 22 $\rightarrow$ 100 of 100 $\leftarrow$ what I shipped
 normal shortcut: $12.50 + 2.326 \times 2.64 = 18.64$ days, vs empirical 22

What it means: promising at the mean is close to a coin flip; the seven days between p90 and p99 are the entire promise, and the normal approximation understates that tail by $22 - 18.64 = 3.4$ days. **What I'd do here:** set buffers off the empirical per-lane quantile in production and keep the normal curve for the whiteboard — which is also the honest caveat on Part 5's whole model.

Drill-down ready: why the tail and not the mean (“a promise at the mean is a coin flip”); right-skewed freight distributions; drift flag design.

7.2 “Scaling adoption gate by gate” (Flexport · v11-A1) – the `handling_config` term

Walmart-tuned nugget: “SSS adoption is a wave-onboarding problem, and I’ve run that machine: I scaled a logistics visibility platform from 7 pilot customers to 68 — wave by wave, with six published graduation criteria a wave had to clear before boarding (data-quality $\geq 90\%$ vs source of truth, milestone accuracy $\geq 80\%$, verification SLA, engagement floors, stability, CSAT). Adoption wasn’t a sales push; it was an operational gate the product had to keep earning. Cost-to-serve fell from \$0.66 to \$0.03 a container while the base grew $10\times$.”

Why it maps: the product’s whole job was answering *did the origin party actually tender the freight, and did the first milestone events arrive clean* — which is I3, first carrier scan inside the configured handling window, wearing a different logo. I1 / I10 (SSS GMV share, activations + retention) is the same adoption motion: earn it with proven reliability, gate expansion on evidence, keep unit economics falling as the book grows. Also feeds proposal 4’s “trial gate” logic.

The two numbers, worked, because they are only impressive together:

cost-to-serve cut: $(\$0.66 - \$0.03) / \$0.66 = 0.955 \rightarrow 95\%$ reduction
 book growth: $68 / 7 = 9.7\times$ over the same period

What it means: unit cost falling 95% while the book grows $\sim 10\times$ is only possible if the gates keep bad data out — a platform that onboards everyone sees cost-to-serve rise with scale, not fall, because every dirty feed

becomes a manual exception forever. **What I'd do here:** apply the identical test to SSS. An enrollment that is not retained at 30 days (I10) is not adoption, the same way a connected feed was never an onboarded lane.

7.3 "The WBR is the product" (Flexport · v11-A2/A3)

Walmart-tuned nugget: "I've run the input-lever/output-metric weekly review this team uses — at Flexport, on the same WBR pattern. When a week went red, the review didn't discuss the red; it arrived pre-decomposed — which input moved, whose lever, what correction, verified by the same number next Monday. And when the reporting itself was the bottleneck — four systems, one person, hours of manual pulls — I consolidated it into Snowflake and took system accuracy from 60% to 90%. The cadence only works if the data under it is boring." **Why it maps:** the role's operating rhythm, literally. Also the credibility base for taking over the output block in days 31–60.

7.4 "Re-sequencing a roadmap with cost math" (Flexport · v11-B1/B2)

Walmart-tuned nugget: "When leadership is set on a sequence and the data says otherwise: our launch plan ran operator-first, 18 months. I costed both sequences on cost-to-serve — partner-first cut total cost ~50% versus ~25%, and shipped value in 6 months instead of 18. I walked the math into leadership, flipped the roadmap, and we banked a \$5M cost avoidance and 66% less engineering effort. I didn't win the argument with seniority — I won it with a two-column table." **Why it maps:** the buffer-policy and SSS-investment debates are exactly this — competing sequences, resolvable only by pricing both. Pairs with Part 5's B-vs-D governance point.

7.5 "Promising is a policy problem before it's a config problem" (Sephora · v11-E1)

Walmart-tuned nugget: "The promise engine can only promise what the inventory model says is sellable — and I've owned that layer. On S/4HANA I redesigned the inventory operating model across stores, DCs, dotcom, and comingle nodes: every stock state — available-to-sell, reserved, in-transit, damaged — got one definition, one owner, and one rule about who may promise against it, signed by the business *before* configuration. Most oversell 'bugs' are vocabulary bugs. Fill rate, OTD, and oversell were computed on definitions everyone had signed." **Why it maps:** inventory-feed accuracy (I7) and cancellations (O5) are the same disease — promises made against fuzzy availability. Also demonstrates governance instinct (E-family issues). **Where this story does NOT belong:** the opener and "why this role." S/4HANA was an execution and governance program, not one of the three promise-equation terms — lead with Manhattan, Elementum, and Flexport shipment visibility (7.8 / 7.1 / 7.2). Bring S/4HANA when the question is about inventory truth, oversell, cancellations, or how you get a definition signed before anyone configures anything. It is a strong answer in the right question and a diluted one in the wrong question.

The arithmetic behind O5, so the story lands with a number. Cancellations are the failure buffer cannot fix:

05 = seller-initiated cancellations / orders	floor $\leq$ 2%
--	-----------------

ILLUSTRATIVE. 100 orders placed against phantom stock at a node whose feed is below the 98% accuracy bar; 4 are cancelled by the seller:

cancel rate on that slice = $4 / 100 = 4\%$ → 2× the policy floor

What it means: you cannot buffer your way out of a unit that was never there. A late blender is a broken Saturday; a cancelled blender is a broken purchase, with no tracking page to look at and a re-shop somewhere else. A data defect arrived as a customer defect. **What I'd do here:** exactly what the S/4HANA redesign did — sign the vocabulary first (available-to-sell, reserved, in-transit, damaged: one definition, one owner, one rule about who may promise against each), then gate feeds on a freshness-and-accuracy toll before a node's inventory is allowed to back a fast promise.

7.6 "Tolerance published before mock one" (Sephora · v11-D1/D2)

Walmart-tuned nugget: "How I de-risk cutovers: two DCs from legacy RP to JDA WMS in eight months, on time, on budget. The mechanism was three mock cutovers against a 0.5% reconciliation tolerance published *before* mock one — misses went 7% → 1.8% → 0.3%. The go/no-go read the scorecard, not the mood. It unblocked a \$20M ROI pipeline sitting behind the migration, with a 4-week hypercare and numeric exit criteria." **Why it maps:** shadow-mode promotion (proposal 7), peak-regime declarations (proposal 10), and E9's launch discipline are all this pattern: rehearse against a published number, then commit.

7.7 "Robots at 30%: protecting a launch from its own automation" (Sephora · v11-C1/C3)

Walmart-tuned nugget: "On the Utah unified-DC program — two warehouses into one omni facility on BlueYonder plus Locus robotics — week two showed the robots below the 98% pick-accuracy gate we'd set. I throttled them to ~30% of volume, kept the launch green on manual process, re-scoped, and returned them to full volume in week three once they cleared the bar. Processing efficiency ended +15%, zero customer incidents. The gate existed so that exact decision needed no debate." **Why it maps:** supervisory-flavored program leadership (the JD's 2-year requirement) + gates with teeth (E9) + the Andon instinct of proposal 5: degrade gracefully, restore on evidence.

7.8 "Cyber Monday, 4 AM" (Manhattan · v11-M1) – the **buffer** term

Walmart-tuned nugget: "Peak isn't abstract to me. As a distribution-OMS consultant — order allocation and promising across three mega-DCs and a hundred stores — the estimated lead time we could put in front of a customer came out of sourcing each order from the best node through allocation: choose the node, and you've chosen the promise. I took the 4 AM Cyber Monday call when allocation backed up. The noise said capacity; the root cause was a scheduler that had died quietly at the daylight-saving change. I triaged by promise-at-risk, rebalanced across the three DCs, and we held a 99% fill rate through peak and protected roughly \$8M in sales — with structural stockouts down 20% over the engagement. Since then I treat every peak as a different regime with its own plan — which is exactly how I'd run the Q4 promise policy here." **Why it maps:** this is the buffer-and-promise-days term of the equation, owned end to end in a closed network where both ends were visible. Plus A6/C5 peak issues, proposal 10, and incident-command credibility.

The arithmetic that makes it this role's story and not a war story. Allocation moves **handling + transit_predicted** *before* one buffer-day is spent. On the 12-lane model's extremes:

```
source from the near node:  h 0.30 + t̂ 0.55 = 0.85 days
source from the far node:   h 1.20 + t̂ 3.40 = 4.60 days
difference                   = 3.75 days, decided by node choice alone
compare: the entire σ-proportional buffer policy moves lanes by 0.36 to 2.63 days
```

What it means: the sourcing decision spends whole days; the buffer policy spends tenths. Nobody optimizes them together because they sit in different orgs. **What I'd do here:** on day one, ask whether node selection and buffer policy are ever solved on the same page — at most marketplaces they are not, and that seam is the cheapest speed in the building after cutoff arbitrage.

Failure story on deck (*if asked — v11-G1*): the Elementum UAT that crashed under concurrent load because I never wrote performance entrance criteria — five business users, ten scenarios, sponsor escalation, missed sign-off. The rebuild: every UAT I've run since has explicit performance criteria in the entrance gate, which is precisely why the Locus 98% gate existed. Own it fast, name the mechanism it built, stop talking.

Part 8 – The 30-minute room

8.1 The clock

Minutes	What happens	Your job
0–2	Rapport, their framing	Listen for the word they use for their pain (“accuracy,” “adoption,” “peak” ...) — that word picks your Part 4 proposal
2–5	“Tell me about yourself”	The 90-second TMAY (Part 9 §1), spoken as the promise equation term by term. Land: buffer / promise days = Manhattan DOM (allocation chose the node, the node chose the promise; 99% fill, ~\$8M protected) · transit_predicted = Elementum (200 lanes measured, promise at the p99, 99% LT accuracy, 580K units and ~\$1.5M freed) · handling_config = Flexport shipment visibility (did the shipper ship in the window, did the first scan prove it; 7→68, \$0.66→\$0.03). Do not open with S/4HANA — that is the execution story, saved for governance questions
5–20	Their questions	Stories 7.1–7.4 are the defaults. Every answer: mechanism + number + “here’s how it maps to your promise/SSS/WBR.” One answer should include the Part 5 free-lunch line
20–25	Your questions	The two below — both are designed to let you respond with a matched proposal
25–29	Close	The 90-day plan in 45 seconds (Part 6), then the closing line
29–30	Logistics	Next steps, thank you, out clean

8.2 The two questions to ask

1. **“Which output metric is red right now — and when it went red, how long did it take to know which input caused it?”** (*Shows you think in decomposition; whatever they name, you have the matching family in Part 2 and proposal in Part 4.*)
2. **“Where does the buffer decision actually live today — DS model, policy table, or negotiation — and who owns the speed-versus-accuracy dial?”** (*The answer is almost always ‘it’s complicated,’ which is your opening for the one-dial governance story — Part 5.*)

Reserve: “What does the SSS enrollment funnel lose the most sellers to — fear of cost, loss of carrier control, or distrust of the scorecard?”

8.3 The three sentences that must get said sometime in 30 minutes

1. The **thesis**: the promise is underwriting, and the buffer is priced risk (Part 1.5).
2. The **free lunch**: “risk-proportional buffers buy **32 points** of fast-promise share at the same accuracy commitment — uniform 1.10 days gives you 7.6% fast at 95.8%; $b = 1.645 \times \sigma$ per lane gives you 39.6% fast at 95.0%. One multiplication per lane. I’ve built the model and I’ll show you.” (Part 5.1)
3. The **commitment**: “ninety days: one output metric moved by a mechanism your WBR can audit.” (Part 6)

8.4 If it goes sideways

- **Interviewer arrives late / short on time:** compress to thesis → one story (7.1) → 90-day plan → question 1. Ten minutes, complete.
 - **Interviewer is quiet / lets you drive:** run the clock yourself in the order above; quiet interviewers reward structure.
 - **Interviewer only wants to probe one thing (e.g., supervisory experience):** answer honestly from the Part 9 Section 5 scripts, then bridge: “the honest version is my leadership is cross-functional-pod-shaped — let me show you what that looked like at go-live” → story 7.7.
 - **They challenge the math:** concede the assumptions gracefully (Part 5.3 is pre-written humility), hold the structure: “the parameters are yours to correct — the argument that buffers should be risk-proportional survives any parameter set.”
-

Part 8.5 – The Tuesday dashboard (draw this)

This is the chapter the ex-Amazon hiring manager will force you to *draw*. If you can put this on a whiteboard in three minutes, the rest of the loop is follow-up. Sample numbers are **illustrative WBR series** (Wk 28–33), not Walmart production. Solver numbers are the 12-lane model.

8.5.1 Architecture before pixels

Decision	Answer you say	Why
Audience	Tuesday room: you + three input owners (Adoption, Promise DS, Seller/Carrier). VP sees the <i>same</i> page, not a prettier one.	Two dashboards is zero dashboards.
Cadence	Week = Monday–Sunday . Data freeze Monday AM. Review Tuesday . Action log aged in the meeting.	Amazon WBR grammar. Never a mid-week “flash” that uses a different denominator.
Grain of page 1	Portfolio, order-weighted, with mix-adjusted twins one line below each output.	National averages lie (D4). Twins expose mix.
Dive-deep grain (not page 1)	Seller × lane × speed-program cell.	Bar-raiser dive-deep: stay until the unit is an order, a lane, or a seller — not a US average.
Input vs output	Outputs (O1–O7 + Promise Yield) on top. Inputs (I1–I10) underneath in three owner blocks.	You <i>report</i> outputs. You <i>manage</i> inputs.
North star	Promise Yield = O3 × O1 (fast share × accuracy). Target 50% in 4 quarters (Wk 33 sample: 45.1% × 95.4% = 43.0%).	Un-gameable joint.
Guardrails (hard red regardless of WoW)	$O1 < 95\% \cdot O2 < 90\% \cdot O4 > 5\% \cdot O5 > 2\%$	Policy floors. A “green WoW” that is under a floor is still RED.
RAG	WoW relative % = (this – last) / last. GREEN = favorable > 5% <i>or</i> at / above floor. RED = unfavorable > 5% <i>or</i> floor breach. Else FLAT .	Same LSA rule as the workbook.
What you refuse on page 1	Model MAE histograms, p99 latency charts, seller-level lists, experiment readouts, GMV, take rate, NPS.	Those live in owner tabs / appendix. Page 1 is decisions, not a museum.

8.5.2 Wireframe (copy onto the whiteboard)

SFS PRE-PURCHASE • WBR • Wk 33 (Mon 8/10 – Sun 8/16) • k = ?				
Owner: Sr Mgr Tech Ops (single-threaded) • Challenger: shadow B				
NORTH STAR	Promise Yield	43.0%	WoW +1.7% rel	FLAT → 50%
	= 03 45.1%	× 01 95.4%		
===== OUTPUTS (lagging) =====				
01 Promise accuracy	95.4%	≥95	GREEN	mix-adj twin: ____
02 Seller OTD (30d)	91.3%	≥90	GREEN	(different clock)
03 Fast-promise ≤2d	45.1%	+2q	GREEN	also report PAGE-VIEW
04 Late shipment	4.6%	≤5	GREEN	recovered from 5.2% Wk32
05 Cancellation	1.8%	≤2	GREEN	
06 WISMO / 1k	10.8	-10%	GREEN	
07 Fast-promise CVR Δ	2.4%	A/B	GREEN	
===== INPUTS (this week's levers) =====				
SPEED / ADOPTION (Adoption PM)				
I1 SSS GMV share	45.3%		RollUp	01,02,03
I8 Expedited/same-day elig	56.8%		RollUp	03
I10 SSS activations (30d ret)	305		RollUp	01,03
PROMISE ENGINE (Promise DS + Eng)				
I2 ML-recommended lag	56.5%		RollUp	01,04
I5 Transit MAE (days)	0.72	≤0.8	RollUp	01 [p90 beside]
I6 Buffer in [p50,p90]	70.2%		RollUp	01,03 (sanity, not k)
I9 Engine p99 (ms)	41	≤50	RollUp	03,07
SELLER & CARRIER (Seller Ops / Carrier PM)				
I3 First-scan in window	93.8%	≥93	RollUp	01,02,04
I4 Lane SLA hit	91.5%	≥90	RollUp	01,02
I7 Nodes feed ≥98%	88.9%		RollUp	02,05
===== DECOMP + ACTIONS =====				
01 ≈ I3 × I4 × $\Phi(b/\sigma)$ Wk33 .938×.915×1.1115 = .954				
Red input with green output → contain, do not celebrate (Wk 32 I4).				
ACTIONS: owner • metric that will go green • date • age • RAG				
CALLOUTS (pre-wired; a red discovered in the room is a process miss)				

Owners on the page (never “the team”): Tech Ops (you) owns PY + O1 + the action log. Product owns O3/O7/I8. Seller Perf owns O2. Seller Ops owns O4/O5/I3. CX owns O6. Adoption PM owns I1/I10. Promise DS owns I2/I5/I6. Carrier PM owns I4. Eng owns I7 (with Seller Ops) and I9.

8.5.3 Metric dictionary – grain, window, denominator, failure mode

Say all four every time they pick a line.

Line	Grain	Window	Denominator	Failure mode (how it is gamed)
O1	order (delivered)	order-week Mon–Sun	delivered orders (cancels <i>out</i>)	Pad buffers; drop hard lanes; exclude late-to-scan
O2	order / seller	rolling 30d	scorecard EDD, not printed EDD	Blend with O1; wait out a bad week
O3	order (also page-view)	order-week	SFS orders (or PDP views) with ≤2d <i>displayed</i>	Expand eligibility into cells that cannot execute; mix-shift to metros

Line	Grain	Window	Denominator	Failure mode (how it is gamed)
O4	order	order-week	shipped-after-ship-by / orders	Print labels without first scan
O5	order	order-week	seller-initiated cancel / orders	Push OOS into “customer cancel”
O6	contact / 1k orders	order-week	WISMO contacts	Deflect to email; change taxonomy
O7	experiment	test window	A/B on fast-promise cohort	Peek; no holdout; mix confounded
PY	order	order-week	O3 × O1	Hide which factor moved — always decompose one line below
I1	GMV \$	week	SFS GMV on SSS / SFS GMV	Enroll junk GMV; ignore 30d retention
I2	item	week	items on ML lag / items	Auto-lag that sellers override silently
I3	order	week	first scan in handling window	Manifest time ≠ scan time
I4	lane (not order)	week	lanes meeting published transit	One huge metro masks rural reds — never treat as P(package)
I5	lane, volume-weighted	week	actual – pred days	MAE falls, tail/p90 rises — always pair with p90
I6	lane	week	buffer $\in$ [p50, p90] of error	Band compliance ≠ priced k
I7	node	week	nodes with feed accuracy $\geq$ 98%	Sample only easy nodes
I8	assortment	week	eligible SKU-nodes / catalog	Eligibility without execution
I9	request	week	p99 latency; cache hit	Cache a stale fast promise
I10	seller	week + 30d lag	activations <i>and</i> 30d retained	Vanity enrollments

8.5.4 Causal tree (I-levers → O-metrics) – whiteboard in 60 seconds

```

I1 SSS GMV → 01 accuracy, 02 OTD, 03 fast share
I2 ML lag → 01, 04 late ship
I3 handling / first scan → 01, 02, 04
I4 carrier SLA → 01, 02
I5 MAE + I6 buffer/k →  $\Phi$  term, 03 speed
I7 inventory feed → 05 cancel, 02
I8 eligibility → 03 (gated on I3/I4)
I9 engine p99/cache → 03 display, 07 CVR
I10 SSS activations+ret → I1 (leading), then 01/03
01, 04, tracking → 06 WISMO
03 × elasticity (A/B) → 07
03 × 01 → Promise Yield (north star)

```

} $01 \approx I3 \times I4 \times \Phi(b/\sigma)$

7-day warning before O1 breaks: I3 first-scan (hours, not weeks); I4 7-day lane CUSUM; I5 p90 error (not MAE); I7 feed freshness; I9 cache-hit collapse; I10 30-day retention drop on last wave; Buffer Waste Index inversion (early-days shrinking while σ is rising = you already spent the pad).

8.5.5 RAG policy (memorize)

1. Compute $WoW = (Wk_t - Wk_{t-1}) / Wk_{t-1}$.
2. Hard floor? If $O1 < 95$, $O2 < 90$, $O4 > 5$, $O5 > 2 \rightarrow \text{RED}$, even if WoW is green.
3. Else if $|WoW| > 5\%$ relative $\rightarrow$ GREEN if favorable, RED if unfavorable.
4. Else **FLAT**.
5. **Output green + input red**: do not celebrate. Contain the input this week (Andon / buffer / badge). The output is a delay, not a virtue. Wk 32: I4 89.6% RED, O1 95.0% still above floor because buffer ate it; O4 went 5.2% RED — the seller clock told the truth the customer clock hid.

Good action item: “Recalibrate buffers on 42 SE lanes; owner Promise DS; success = $O4 \leq 5\%$ and $I4 \geq 90\%$ by Wk 33; created 8/5, due 8/12.” (Shipped; O4 5.2% $\rightarrow$ 4.6%.) **Rotten action item:** “Improve accuracy; owner: team; due: ongoing.”

8.5.6 Ninety-second VP tour (say this)

“Page 1 is Promise Yield — 43% last week, 45.1% of orders got a two-day promise and 95.4% of deliveries kept the printed date. Floor on accuracy is 95%; floors on OTD, late ship, and cancel are policy, not vibes. Underneath, ten inputs with named owners. When accuracy moved, we factor it: seller first-scan times carrier SLA times the buffer term. Last month a Southeast hub consolidation made I4 red; we did not wait for O1 to break — we resolved 42 lanes and late-ship recovered under 5%. What I will not put on this page is a latency chart or a seller leaderboard. Those are owner-tab work. What I will bring every Tuesday is: which input is red, whose lever, what date the same number goes green.”

8.5.7 Ex-Amazon HM – the 15 questions most likely in the first 30 minutes

1. Tell me about yourself.
2. Why this role / why Marketplace.
3. Design the WBR from a blank page.
4. How would you improve accuracy *without* slowing the promise?
5. Tell me about a time you disagreed (Have Backbone).
6. Walk me through a time you dove deep into a metric that looked fine.
7. SSS adoption — how do you actually move it?
8. Output green, input red — what do you do?
9. You have no marketplace on your resume.
10. This needs two years supervisory.
11. Show me how you’d set buffers (k / Solver).
12. What’s the north star vs the guardrails?
13. Peak week, accuracy down four points.
14. Why you vs an Amazon SFP transfer?
15. What questions do you have for me?

Spoken answers: Q-bank §1–2, §11–13. Draw: this page. Math: Part 5 (A 7.6% fast $\rightarrow$ B 39.6% at 95%; D 97.8%).

(Part 9 — the full question bank — follows. Sections 11–16 are the ex-Amazon loop: LPs, dashboard, bar-raiser, cases, Fermi, Amazon-trained vocabulary.)

Walmart Marketplace – Hiring Manager Interview Q&A Bank

Role: Senior Manager, Technology Operations — Pre-Purchase Delivery Experience (req R-2596147), Seller Fulfillment Solutions **Candidate:** Rajesh Rishiyur Chandrasekaran · **Interview:** Sept 9, 2026 · 30 minutes hiring manager; this bank covers the **full ex-Amazon loop** (HM + bar-raiser follow-ups + metric deep-dive + ops case) **Core math carried into the room:** $\text{promised_EDD} = \text{order time} + \text{handling} + \text{predicted transit} + \text{buffer}$ · $\text{promise kept} \Leftrightarrow \text{handling error} + \text{transit error} \leq \text{buffer}$ · $\text{promise accuracy} \approx P(\text{seller ships in window}) \times P(\text{carrier meets transit}) \times \Phi(\text{buffer}/\sigma)$ — in shorthand, $\mathbf{O1} \approx \mathbf{I3} \times \mathbf{I4} \times \Phi(\mathbf{b}/\sigma)$. **Promise Yield** = fast-promise penetration $\times$ promise accuracy. Φ bookmarks: $\Phi(1.00) = 84.1\%$ · $\Phi(1.645) = 95.0\%$ · $\Phi(2.00) = 97.7\%$ · $\Phi(2.33) = 99.0\%$. Canonical policy pair: **A** uniform $b = 1.10d \rightarrow 95.8\%$ accuracy / 7.6% fast · **B** $b = 1.645\sigma \rightarrow 95.0\%$ / 39.6% fast (`model_reference.py`). **How to use:** Sections 1–10 are the original HM bank (kept). Sections 11–16 are the Amazon-trained overlay — Leadership Principles, dashboard/WBR design, bar-raiser stress, full cases, Fermi, and “they ask this because they are ex-Amazon.” Spoken English, first person. Resume numbers are canonical; marketplace arithmetic is labeled **illustrative**. $>$ lines are say-this. **Math is inline, on purpose.** Every section from 1 to 16 carries its own arithmetic where the metric is discussed — definition $\rightarrow$ formula $\rightarrow$ plugged numbers $\rightarrow$ what it means $\rightarrow$ what you would do. You should never have to flip to another chapter mid-answer. Canonical `model_reference.py` figures are exact; anything toy is labeled **ILLUSTRATIVE**.

Section 1 – The opener

The spine: one equation, three chapters

Everything in this bank hangs off the equation the team already runs on. Say it early, then attribute your career to its terms.

$$\text{promised_EDD} = \text{t_order} + \text{handling_config} + \text{transit_predicted} + \text{buffer}$$

Term	Plain English	Who owns it here	
t_order	when the customer taps Buy	the customer	—
handling_config	hours the box sits at the seller before the first carrier scan	seller declares it; Walmart measures it (I3)	Flexport shipment visibility — did the shipper actually ship inside the window, and did the first milestones arrive clean
transit_predicted	the model's expected days inside the carrier network	Promise DS + Carrier PM (I4, I5)	Elementum — 200 lanes of measured transit distributions, promise at the tail
buffer	the pad, because handling and transit will both be wrong sometimes	this role (I6 / k)	Manhattan DOM — estimated lead time / promise days out of node sourcing through allocation

Kept iff the combined execution error fits the pad:

$$\epsilon = (\text{handling_actual} - \text{handling_config}) + (\text{transit_actual} - \text{transit_predicted})$$

promise kept $\Leftrightarrow \epsilon \leq b$

The whole math kit, on one card — so nothing in Section 1 or 2 requires flipping to Part 5.

Worked promise, round numbers. Order Tue 3:00 PM PT · handling 1.0 d · predicted transit 2.0 d · buffer 1.0 d:

$$1.0 + 2.0 + 1.0 = 4.0 \text{ days} \rightarrow \text{printed EDD} = \text{Saturday 3:00 PM}$$

Three worlds, same Saturday:

World	Seller error	Carrier error	ϵ	$\epsilon \leq 1.0?$	Outcome
Clean	0	0	0	yes	kept
Buffer earns its keep	+8 h = 8/24 = 0.33 d	0	0.33	yes	kept — customer never knows
Buffer too thin	0	+2.0 d	2.00	no	broken by ~1 day

Coverage lookup. For one lane, model $\epsilon \sim \text{Normal}(0, \sigma^2)$, so $P(\text{pad covers the noise}) = \Phi(b/\sigma)$. Bookmarks worth memorizing:

$z = b/\sigma$	$\Phi(z)$	Say it as
1.00	84.1%	one sigma of pad — one order in six breaches on noise alone
1.645	95.0%	the 95%-floor dial — this is where k comes from
2.00	97.7%	two sigmas
2.33	99.0%	what Legal asks for when it says “be safe”

Why uniform pads fail — the canonical pair (`model_reference.py` , exact):

Policy	Buffer rule	Accuracy	Fast share ≤ 2 d
A	1.10 days on every lane	95.8%	7.6%
B	$b = 1.645 \times \sigma$ per lane	95.0%	39.6%

Same protection, re-allocated by measured risk: **+32 points of fast share at the same ~95% commitment**. The arithmetic underneath is one multiplication per lane. σ spans **0.22 d** (California Intra) to **1.60 d** (Mountain Rural) — a 7× spread:

$$b_{CA} = 1.645 \times 0.22 = 0.36 \text{ days} \quad b_{MTN} = 1.645 \times 1.60 = 2.63 \text{ days}$$

Uniform 1.10 gives California three times the pad it needs ($\Phi(1.10/0.22) = \Phi(5.0) \approx 100\%$) and starves the mountain ($\Phi(1.10/1.60) = \Phi(0.69) = 75.4\%$ — one typical surprise in four already blows the promise).

The decomposition, with the sample week. $O1 \approx I3 \times I4 \times \Phi(b/\sigma)$. Wk 33 (illustrative WBR series): $I3 = 93.8\%$, $I4 = 91.5\%$, $I5$ (transit MAE) = 0.72 d, $I6 = 70.2\%$ of lanes in band, $O1 = 95.4\%$.

$$\begin{aligned} I3 \times I4 &= 0.938 \times 0.915 = 0.85827 \rightarrow 85.8\% \\ \text{implied } \Phi \text{ term} &= 0.954 / 0.85827 = 1.1115 \rightarrow \text{above 1 on purpose (see §2)} \end{aligned}$$

Sensitivity you can do out loud. Hold $I4$ and the residual, drop $I3$ by 3 points:

$$O1' = 0.908 \times 0.915 \times 1.1115 = 0.9235 \rightarrow 92.3\%$$

What it means: a 3-point seller-handling slip costs 3.1 points of customer promise accuracy and breaks the 95% floor with no change in buffer policy. The owner is Seller Ops, named in one line.

Drift, the failure that hides. A lane sits at $\sigma = 1.2$ with a frozen pad $b = 1.65 \times 1.2 = 1.98$ days. Carrier consolidates; σ widens to 1.9:

$$\begin{aligned} \text{before: } & \Phi(1.98 / 1.2) = \Phi(1.65) = 95.0\% \\ \text{after: } & \Phi(1.98 / 1.9) = \Phi(1.04) = 85.1\% \\ \text{re-solve: } & b_{\text{new}} = 1.65 \times 1.9 = 3.14 \text{ days} \quad (\text{at the exact } k = 1.6449 \rightarrow 3.13; \text{ same story}) \end{aligned}$$

What it means: nobody changed the buffer and the lane lost ~10 points of coverage. That is why σ carries a drift flag and k gets re-solved, not frozen.

Tell me about yourself

Tell me about yourself. (90-second version — the default)

I'm a supply chain technology operator — fifteen-plus years on the systems that decide what a customer gets promised, and whether that promise is kept. The fastest way to tell you why I fit this seat is your own equation: promised date equals order time, plus seller handling, plus predicted transit, plus buffer. I've owned three of those terms — separately.

Buffer and promise days: Manhattan Associates, distribution order management — order allocation and promising across three mega-DCs and a hundred stores. The estimated lead time we could commit to came out of sourcing each order from the best node through allocation; choose the node, and you've chosen the promise. I held a 99% order fill rate through peak and protected roughly \$8 million in sales on one Cyber Monday, with structural stockouts down 20%.

Predicted transit: Elementum. Our plan ran on carrier-brochure lead times — single optimistic numbers nobody had measured. I measured real transit distributions across 200 lanes and set the promise at the tail, the 99th percentile, not the brochure. Ninety-nine percent lead-time accuracy, and because measured spread came in tighter than assumed, 580,000 units and about \$1.5 million of safety stock came free — with a drift flag per lane, so a widening spread triggered review instead of a silent re-plan.

Handling — whether the shipper actually ships inside the window: the Flexport shipment-visibility program. Seven pilots to 68 enterprise customers behind published data-quality and milestone-accuracy gates, cost-to-serve from 66 cents to 3 cents a container, and four systems consolidated into Snowflake taking data accuracy from 60 to 90%. First-scan truth is a measurement problem before it's a seller problem.

Three terms, three chapters: promise days at Manhattan, the transit tail at Elementum, handling and first-scan telemetry at Flexport. This role is the first seat where all three sit on one page — and that page prints a date a customer believes.

Tell me about yourself. (2-minute version — if they want more, add this; don't restart)

A quick tour of how I got here, in the same order. Manhattan Associates, distribution OMS: allocation and promising across three mega-DCs and a hundred stores — and the promise days were an *output* of allocation, not a setting. That's where I learned what a broken promise costs: a 4 AM Cyber Monday allocation incident where I triaged by promise-at-risk first, held 99% fill rate and protected roughly \$8 million. At MetricStream I owned SOWs and ran onshore-offshore delivery teams across concurrent client programs. Elementum was the transit term: 200 lanes of measured distributions, promise at the p99, and a thousand multimodal routes onboarded over X12 EDI and XML with shipment-data completeness going 60 to 95% — lane transit modeling and carrier scan events are home turf. Flexport was the handling-and-visibility term plus the adoption machine and the input-versus-output WBR discipline. Sephora is the longest chapter and the execution one: the dual-DC JDA migration in eight months on time and on budget, a 20-person agile conversion, and the unified omni DC with Locus robotics at +15% processing efficiency. The thread never changes: I take a broken operational number, decompose it into the inputs that drive it, and run the program that closes it. I can sit with a seller-ops analyst in the morning, pull the SQL in the afternoon, and walk the trade-off into a VP review the same week.

If he asks "so which term is hardest?" — have the arithmetic ready.

Rank the three terms by what a point of improvement is worth, using the Wk 33 decomposition:

Term	Lever	One-point move	O1 effect (holding the rest)
handling	I3 93.8% → 94.8%	+1.0 pt	$0.948 \times 0.915 \times 1.1115 = \mathbf{96.4\%}$ (+1.0 pt)
transit	I4 91.5% → 92.5%	+1.0 pt	$0.938 \times 0.925 \times 1.1115 = \mathbf{96.4\%}$ (+1.0 pt)
buffer	b unchanged, σ -10% on a 1.2- σ lane	$\Phi(1.98/1.08) = \Phi(1.83)$	96.7% vs 95.0% coverage: +1.7 pts of the Φ term

What it means: the two execution terms pay linearly and belong to other orgs; the buffer term pays through Φ , which is non-linear and belongs to me. So I push σ and k first — it is the term where I own both the analysis and the decision.

Walk me through your resume.

Walk me through your resume.

Chronologically, five chapters, one thread. Manhattan Associates — Distribution OMS consultant: order allocation and promising across 3 mega-DCs and 100 stores, where the estimated lead time came out of node sourcing through allocation; 99% fill rate through peak, ~\$8M protected, stockouts down 20%. MetricStream — solutions manager: owned SOWs, ran onshore-offshore delivery teams. Elementum — delivery lead: 200 lanes to 99% lead-time accuracy, safety stock down 580K units and \$1.5M, a thousand routes onboarded via EDI and XML with completeness going 60 to 95%. Flexport — TPM: shipment-visibility platform 7 to 68 customers, cost-to-serve down 95%, four systems consolidated into Snowflake with accuracy 60 to 90%, and a roadmap re-sequenced for 66% less engineering effort and \$5M avoided. Sephora — TSCPM: the dual-DC JDA migration in eight months, the S/4HANA inventory operating-model redesign, and the UDC omni DC with robotics, up 15% on processing efficiency. The thread: every chapter is the delivery promise from a different seat — predicting it, promising it, measuring it, or proving it.

The arithmetic behind two of those claims (say one, not both, and only if pushed):

cost-to-serve cut: $(\$0.66 - \$0.03) / \$0.66 = 0.955 \rightarrow 95\%$ reduction
customer growth: $68 / 7 = 9.7\times$ base growth, while ticket response went 72% → 90%

What it means: the book grew ~10× and the unit cost fell 95% *at the same time* — that only happens if the adoption gates keep bad data out. That is the same reason SSS enrollments without 30-day retention are not adoption.

Why leaving / why here

Why are you leaving Sephora?

I've delivered what I went there to do. The WMS migration is live, the omni DC is live and 15% more efficient, and the promising governance is the standing reference. I'm not running from anything — I'm running toward scale. At Sephora, a point of promise accuracy is thousands of orders; at Walmart Marketplace it's millions. I want to operate where a basis point of accuracy or a day of promised speed moves a number the size of a business.

Why Walmart?

Because Walmart is where the hardest version of my problem lives at the largest scale that isn't Amazon — and it's the challenger seat, which is the better operating job. Walmart has the traffic, the assortment is coming from Marketplace, and the delivery experience is the piece that converts traffic into trust. I've spent fifteen years on exactly that piece. I also respect how this team runs — a weekly business review that separates input levers from output metrics is the operating rhythm I've run for years, so I'd be productive in week one, not month three.

Why Marketplace specifically?

Marketplace is the growth engine, and seller-fulfilled is its hard problem: Walmart owns the customer-facing delivery experience for inventory it never touches. That means the job is influence at scale — moving thousands of independent sellers through tools, incentives, and scorecards rather than direct control. That's the exact motion I've run: 68 enterprise customers adopted a platform because I standardized the offer and gated the waves, and a thousand freight lanes onboarded because the data-quality toll made adoption trustworthy. Control is easy; influence with evidence is the craft, and Marketplace is where it pays.

What "scale" is worth, arithmetically (ILLUSTRATIVE, on the 12-lane model's ~1M orders/week book — I would replace the denominator with the WBR's own on day one):

1 basis point of 01 = $1,000,000 \times 0.0001 = 100$ orders/week that keep their date
 priced at the model's \$12 breach cost: $100 \times \$12 = \$1,200/\text{week} \approx \$62\text{K}/\text{year}$ — per bp
 a 4-point peak miss: $1,000,000 \times 0.04 = 40,000$ breaches $\times \$12 = \480K in one week

What it means: at this denominator, accuracy is a P&L line, not a quality slogan — and it is why I would rather Andon 40,000 orders' worth of specific cells than buy a national pad that also spends fast-share on every customer who was never at risk.

Why this role — Senior Manager, Tech Ops, Pre-Purchase Delivery Experience?

Because I have owned every term of your promise equation — one term at a time, in three different seats — and this is the first job description where owning all three at once *is* the job.

`promised_EDD = t_order + handling_config + transit_predicted + buffer`

Term one — buffer, the promise days themselves. Manhattan Associates. I ran distribution order management: order allocation and promising across three mega-DCs and a hundred stores. The estimated lead time we could put in front of a customer was an *output* of sourcing the order from the best node through allocation — pick the node, you've picked the days you may promise. That is precisely this role's buffer-and-promise problem, in a closed network where I could see both ends. Peak proved it: 99% fill rate held, roughly \$8 million protected, stockouts down 20%.

Term two — predicted transit. Elementum. I replaced carrier-brochure lead times with measured distributions across 200 lanes and moved the promise to the tail: 99% lead-time accuracy, 580,000 units and about \$1.5 million of safety stock released because measured spread was tighter than assumed, and a drift flag per lane so a widening spread got reviewed, never silently re-planned. That is σ estimation and buffer calibration, denominated in inventory instead of days.

Term three — handling, the seller-processing question. The Flexport shipment-visibility program. The product's entire job was to answer whether the origin party actually tendered the freight and whether the first milestone events arrived clean — the same question as I3, first carrier scan inside the configured handling

window. I scaled it 7 pilots to 68 enterprise customers behind published gates (data quality $\geq 90\%$ against source of truth, milestone/ETA accuracy $\geq 80\%$, verification SLA, engagement, stability, CSAT), took cost-to-serve from 66 cents to 3 cents a container, and consolidated four systems into Snowflake at 60 $\rightarrow$ 90% data accuracy.

Three seats, three terms. Pre-Purchase Delivery Experience is where they converge: the buffer policy, the transit model's tail, and the seller-handling telemetry — all priced against conversion, all on one page, with one owner.

And the convergence is worth arithmetic, not adjectives. The reason a company needs one person holding all three is that they interact multiplicatively, not additively:

$$O1 \approx I3 \times I4 \times \Phi(b/\sigma)$$

Wk 33: $0.938 \times 0.915 \times 1.1115 = 0.954 \rightarrow 95.4\%$

Owning only handling gets you I3. Owning only the model gets you σ . Owning only the pad gets you b. **The product is where the customer lives** — and the same structure says who to call when it moves:

If the mover is...	Term	Named owner	My move this week
seller shipped late	I3	Seller Ops	observed handling (I2), then surgical badge pull on the cell
carrier missed the lane	I4	Carrier PM	contain: widen b or demote the lane, then root-cause
noise outgrew the pad	$\Phi(b/\sigma)$	me	re-solve k; σ drift flag; shadow four weeks before display

*This role needs someone who can hold promise accuracy and promise speed at the same time — and I've spent my career proving those aren't a trade-off if you do the math at the lane level. The proof is one multiplication per lane: uniform 1.10 days of pad buys 7.6% fast share at 95.8% accuracy; $b = 1.645\sigma$ buys **39.6% fast share at 95.0%**. Same protection, priced by measured risk.*

Section 2 – Role-fit and domain depth

The promise itself

How would you improve promise accuracy without slowing the promise?

The lazy fix is buffer — and buffer spends speed to buy accuracy, which games one metric against the other. The factorization says there are three ways to gain accuracy without adding a single buffer-day: raise the probability the seller ships in window — handling adherence, measured on first scan; raise the probability the carrier hits transit — lane-level SLA management; and shrink σ itself — a tighter transit model means the same buffer covers more of the distribution. At Elementum I lived the proof: replacing brochure lead times with measured distributions across 200 lanes took accuracy to 99% while *releasing* 580K units of buffer stock. Accuracy and speed both improved because uncertainty fell.

The arithmetic, three doors, zero buffer-days. Start from the sample week: $01 \approx I3 \times I4 \times \Phi(b/\sigma) \rightarrow 0.938 \times 0.915 \times 1.1115 = 0.954$.

Door	The move	The arithmetic	New coverage	Promise-day cost
Seller ships in window (I3)	93.8% $\rightarrow$ 95.8%	$0.958 \times 0.915 \times 1.1115$	97.4% O1	0
Carrier hits transit (I4)	91.5% $\rightarrow$ 93.5%	$0.938 \times 0.935 \times 1.1115$	97.5% O1	0
Shrink σ (the model)	σ 1.2 $\rightarrow$ 1.0 at the same $b = 1.98$	$\Phi(1.98/1.00) = \Phi(1.98)$	97.6% vs $\Phi(1.65) = 95.0\%$	0
<i>The lazy fix, for contrast</i>	pad b 1.98 $\rightarrow$ 2.30 on $\sigma = 1.2$	$\Phi(2.30/1.20) = \Phi(1.92)$	97.2%	+0.32 days on every order in the lane

What it means: the pad is the only door that charges every customer — including the 95% who were never at risk — and it buys the *least* coverage per unit of effort here. **What I'd do:** rank the three cheap doors by volume $\times$ gap, and put the pad last.

Buffer is the tax you pay on uncertainty you haven't reduced yet. I'd rather cut the uncertainty than raise the tax.

What actually drives fast-promise penetration?

Penetration is the share of orders promised in two days or less, and it's a chain: eligibility coverage times the promise math clearing the two-day bar. The levers, ranked by cost. Cheapest: buffer calibration — a lane over-buffered relative to its σ is speed you already paid for and aren't shipping; in the illustrative model I built, σ -proportional buffering buys about 32 points of penetration versus a naive uniform 1.1-day buffer at the same accuracy. Next: handling-time truth — sellers pad handling defensively; ML-recommended handling times reclaim those days. Then: transit-model MAE — every tenth of a day of model error is buffer you must hold. Structural and slowest: inventory placement and speed-eligibility coverage. I'd work that list top down because the top is math and the bottom is capex.

Where the 32 points come from — one multiplication per lane. Fast $\Leftrightarrow h + \hat{t} + b \leq 2.0$. Canonical 12-lane cohorts (`model_reference.py`), top four by session volume:

Lane	σ	$h + f$	+1.10 uniform	Fast?	$b = 1.645\sigma$	Promise	Fast?
West Metro / Ground	0.30	1.45	2.55	no	$1.645 \times 0.30 = 0.49$	1.94	YES
Southeast Metro / Ground	0.42	1.60	2.70	no	$1.645 \times 0.42 = 0.69$	2.29	no
Northeast Metro / 2-Day	0.28	1.30	2.40	no	$1.645 \times 0.28 = 0.46$	1.76	YES
California Intra / Same-Day	0.22	0.85	1.95	YES	$1.645 \times 0.22 = 0.36$	1.21	YES

Under A exactly **one** of twelve lanes clears the badge; under B, **three** do — and they are three of the four biggest session pools ($2.55M + 2.10M + 1.10M = 5.75M$ of $18.51M$ weekly sessions = **31.1% session-weighted**). Order-weighted it prints **39.6%**, because a faster promise also converts better, so the fast lanes carry more than their session share.

A: 7.6% fast at 95.8% accuracy

B: 39.6% fast at 95.0% accuracy

$\Delta = +32.0$ points of fast share

What it means: penetration was never blocked by physics on those lanes — it was blocked by spending 1.10 days of pad on a $0.30\text{-}\sigma$ lane that needed 0.49. **What I'd do:** shadow B on those three lanes first; they are the largest, the tightest, and the ones where the 2-day cliff actually flips.

How would you set a lane's buffer?

With a policy, not a negotiation — published before the data, the way I published a 0.5% reconciliation tolerance before mock cutover one at Sephora. The policy: buffer = $k \cdot \sigma$ per lane, with one portfolio dial. Set k at 1.645 and Φ gives you a 95% accuracy floor across the portfolio exactly. What that buys is honesty per lane: tight lanes get small buffers and fast promises; volatile lanes get more absolute days. The equimarginal check comes next: at the optimum, the last buffer-day on every lane buys the same accuracy — so a lane whose σ is so wide it needs buffering out of the fast band shouldn't get buffer, it should get demoted. And every lane carries a drift flag, the same hygiene I ran at Elementum: a widening spread triggers a review, never a silent re-plan.

The dial, with the lookup table on the same page. $b = k \cdot \sigma$, and k comes straight off the normal CDF:

Accuracy floor you publish	$k = \Phi^{-1}(\text{floor})$	b on $\sigma = 0.22$ (CA Intra)	b on $\sigma = 1.60$ (Mtn Rural)
84.1%	1.000	0.22 d	1.60 d
95.0%	1.645	0.36 d	2.63 d
97.7%	2.000	0.44 d	3.20 d
99.0%	2.326	0.51 d	3.72 d

Check the two that matter: $1.645 \times 0.22 = 0.36$ days (about 8.7 hours of pad) and $1.645 \times 1.60 = 2.63$ days. One dial, $7\times$ different absolute protection, identical 95% commitment on both.

And the drift discipline, because k is only right while σ is. A lane at $\sigma = 1.2$ with a frozen pad $b = 1.65 \times 1.2 = 1.98$:

today: $\Phi(1.98 / 1.2) = \Phi(1.65) = 95.0\%$ ✓ on the floor
 carrier consolidates, $\sigma \rightarrow 1.9$
 same frozen pad: $\Phi(1.98 / 1.9) = \Phi(1.04) = 85.1\%$ ✗ ten points gone, silently
 re-solved pad: $b_{\text{new}} = 1.65 \times 1.9 = 3.14$ days (exact $k = 1.6449 \rightarrow 3.13$)

What it means: nobody edited a buffer and the lane still lost ten points of coverage — the failure is *not re-measuring* σ . **What I'd do:** a drift flag per lane (the Elementum hygiene), k re-solved weekly and printed in the WBR header, and a demotion rule so a lane whose honest pad pushes it past 2 days leaves the fast band instead of pretending.

One dial, σ -proportional, floor published before the quarter starts. A buffer policy you negotiate lane by lane isn't a policy — it's a lobby.

Promise accuracy and seller on-time delivery both exist as metrics. Why both, and what's the difference?

They answer different questions to different audiences. Promise accuracy is the customer's metric: did the package arrive on or before the EDD Walmart printed on the page? Seller OTD is the scorecard metric: did the seller execute their commitment? They diverge by design — the buffer can keep a promise even when a seller ships late, and a perfect seller can lose a promise to a carrier meltdown. You need both because accountability is split. Run only promise accuracy and you'll solve seller problems with buffer — spending customer-facing speed to hide an input failure. Run only OTD and you'll punish sellers for carrier variance they don't control. Same discipline as Sephora's inventory model: fill rate and promise accuracy are computed on different definitions, and blending them makes both fiction.

The divergence, counted. ILLUSTRATIVE — 100 orders, one week, buffer = 1.0 day.

What physically happened	Orders	O1 (customer's printed date)	O2 (seller scorecard)
Seller on time, carrier on time	88	kept	on-time
Seller 8 h late ($\epsilon = 8/24 = 0.33 \leq 1.0$), buffer absorbs it	5	kept	miss (handling window)
Seller perfect, carrier +2 d ($\epsilon = 2.0 > 1.0$), no excusal	3	miss	miss
Seller perfect, carrier +2 d, SSS excuses carrier delay	4	miss	on-time (excused)
Total	100	$88 + 5 = 93 \rightarrow \mathbf{O1 = 93\%}$	$88 + 4 = 92 \rightarrow \mathbf{O2 = 92\%}$

Nine of a hundred orders disagree between the two clocks — and they disagree in *both directions*. Wk 33's real gap is the same fingerprint: O1 95.4% vs O2 91.3% = **4.1 points**, from (a) the 30-day smear on O2, (b) buffer-recovered seller misses, (c) SSS excusal.

What it means: you cannot reconcile them into one number, and any attempt to do so hides the two cells that matter — the five where buffer paid for a seller's miss, and the four where a hurricane was charged to nobody. **What I'd do:** print both, always, with the clock named on the line, and ask which clock the person in front of me is holding before I answer.

How would you drive Simplified Shipping Settings adoption?

SSS is an accountability transfer — sellers give up shipping-config control, Walmart absorbs carrier-delay accountability — so adoption is a fear-removal problem, not a feature problem. My playbook is the one that

took Flexport's platform from 7 to 68 customers. First, standardize the offer so the value is legible: what the seller gains, in their numbers. Second, wave the rollout with an entry bar — the analog of my 50-container calibration toll is shadow-running a seller's orders under SSS settings before flipping, so day one is verified, not hoped. Third, publish graduation criteria per wave and hold them — a wave that fails a criterion doesn't board the next cohort. Fourth, measure the honest pair: onboarding throughput *and* 30-day retention. Enrollments without retention is churn wearing a growth costume.

The adoption arithmetic I'd put on the page. Sample series: I1 (SSS share of SFS GMV) 38.5% → **45.3%** over six weeks; I10 = **305** activations retained at 30 days.

```
slope           = (45.3 - 38.5) / 6 weeks = 1.13 points/week
weeks to 50% = (50.0 - 45.3) / 1.13  ≈ 4.2 weeks  → majority-of-book in ~a month, IF slope
holds
retention math: to print 305 retained at 80% retention → 305 / 0.80 = 381 starts/week
                  at 50% retention                      → 305 / 0.50 = 610 starts/week
```

What it means: the same headline number — 305 — needs 381 or 610 starts depending on a figure nobody puts on the slide. That is why I refuse to report I10 without its retention denominator. **What I'd do:** print activations and 30-day retention as one paired line, and gate the next wave on retention, not on enrollments.

A seller is consistently missing their stated handling time. Walk me through what you do.

Decompose before you punish. First, is the measurement right — is first scan in window actually late, or is a label printed early and scanned late by the carrier's induction? At Elementum, a set of "late" lanes turned out to be a timestamp-mapping problem at the forwarder, not late freight — fix measurement before consequences. Second, which failure is it: a setting problem — their stated handling doesn't match their real operation, fixable with an ML-recommended handling time — or an execution problem, real capacity or cutoff issues. Third, the ladder: correct the setting, coach with the scorecard evidence, and only then pull fast-promise eligibility for that seller or SKU set. The equimarginal principle applies to people too: never buffer a whole lane — taxing every customer — to absorb one seller's miss.

Why the size of the miss decides the response. ILLUSTRATIVE — a seller declaring 1-day handling who hits it 85% of the time, on 100 orders, buffer = 1.0 day.

If the 15 misses average...	ϵ from handling	$\epsilon \leq 1.0?$	O1 on those 15	Portfolio O1	Seller's O4
8 hours late	0.33 d	yes	all kept	100%	15% — hard red
20 hours late	0.83 d	yes	all kept	100%	15% — hard red
1.5 days late	1.50 d	no	all broken	85%	15% — hard red

What it means: the scorecard sees one number (15% late) for three completely different customer outcomes. The distribution of *how late*, not the count of late, decides whether this is a coaching conversation or an Andon. **What I'd do:** pull the lateness distribution for that seller before I pull their badge — and if the mass sits under the pad, fix the declared handling with observed handling (I2) instead of spending enforcement on a seller whose customers are fine.

Promise accuracy dips. How do you tell a carrier problem from a seller problem from a model problem?

The promise-kept condition is the diagnostic: handling error plus transit error must fit inside the buffer, so every breach decomposes on scan events. First scan after the handling window — seller problem, and I trend it as I3, seller adherence. First scan on time but delivery beyond predicted transit — now split on history: if actuals moved versus that lane's own distribution, it's carrier; if actuals match history and the model predicted wrong, it's model. At Flexport I ran exactly this triage discipline: a week that printed 78% manual looked like operators slipping — the root cause was a carrier's format change. The WBR should attribute a red to seller, carrier, or model within minutes, because the correction owner differs completely for each.

The five-minute version, with the numbers in it. $O1 \approx I3 \times I4 \times \Phi(b/\sigma)$.

Wk 33 baseline: $0.938 \times 0.915 \times 1.1115 = 0.954$ ($I3 \times I4 = 0.85827$; residual $\Phi = 0.954/0.85827 = 1.1115$)

Now hold two terms and move one — the biggest single-term mover names the family:

Scenario	Arithmetic	O1	Δ	Family / owner
Seller slips 3 pts	$0.908 \times 0.915 \times 1.1115$	92.3%	−3.1	B-family · Seller Ops
Carrier slips 3 pts	$0.938 \times 0.885 \times 1.1115$	92.3%	−3.1	C-family · Carrier PM
σ widens 1.2 → 1.9 at frozen $b = 1.98$	Φ term 95.0% → 85.1%	≈ 85% on that lane	−10 on the lane	A-family · me + Promise DS

Two honest cautions I say out loud. The residual **1.1115 exceeds 1** on purpose — I4 is measured at *lane* grain (“91.5% of lanes met SLA”), not order grain, so multiplying it as a package survival rate double-counts geography; and buffer *recovers* misses, which is exactly why O1 (95.4%) sits above I3 (93.8%). The identity is a flashlight for which family to open, not a conservation law.

What it means: three different owners produce the same 3-point red, and the arithmetic tells them apart in one line each. **What I'd do:** compute all three deltas before the meeting, so the red arrives with a name on it.

It's your first week. What do you measure?

Five numbers, plus one meta-question. The trend of promise accuracy cohorted by seller segment, lane, and speed program — not the blended average. The breach decomposition — what share of misses is seller handling, carrier transit, model error. The Buffer Waste Index — mean days early on kept promises, ranked by lane, which is the speed-for-free backlog. The SSS funnel — throughput and 30-day retention. And inventory-feed accuracy at the top ship-from nodes, because feeds drive cancellations. The meta-question, which I ask at every company: which of these numbers does the room trust least? That number was the starting point at Flexport, at Elementum, and at Sephora, and I'd bet it's the starting point here.

Each of the five comes with the arithmetic I expect to run on it in week one:

#	Number	The calculation	What a bad answer looks like
1	O1 by cohort	O1 raw vs mix-adjusted at last quarter's cohort weights; the delta is the mix effect	raw 94.2% → 94.0% while every cohort worsened (Simpson)
2	Breach decomposition	share of misses where first scan was late / lane missed SLA / neither → residual	the residual is the biggest bucket, i.e. nobody knows

#	Number	The calculation	What a bad answer looks like
3	Buffer Waste Index	mean days early on kept promises $\times$ lane volume, ranked	0.61 days of waste on the biggest lane (see below)
4	SSS funnel	activations $\div$ retained: $305 / 0.80 = 381$ starts	retention unmeasured
5	Feed accuracy	$100 - 88.9 = 11.1\%$ of nodes below the 98% bar	cancels quietly out of every denominator

What it means: all five are queries, not programs — I can have them by Friday of week one without asking anyone to build anything. **What I'd do:** publish them with grain and date on each, then let the room tell me which one they don't believe.

What's wrong with a 30-day rolling scorecard?

Two things: it's slow to see and slow to forget. A step change — a carrier meltdown, a model regression — gets averaged across thirty days, so you detect it late, and it haunts the score for weeks after it's fixed. Mix shift compounds it: the denominator's composition moves, so the score changes when nothing operational did. My fix isn't to kill it — a 30-day window is fair for seller-facing consequences, where stability matters — it's to never *operate* on it. The WBR runs on weekly cohorts by order date, with the decomposition attached, so a step change shows up as a step. Grain and date on every number: that's the discipline I held at Flexport, where I discarded a flattering truncated week rather than print it.

The smear, in numbers. ILLUSTRATIVE — a seller running 95% on-time who has one catastrophic week at 75%.

one week is $7/30$ of the window
 $30\text{-day OTD} = 0.95 - (7/30) \times (0.95 - 0.75) = 0.95 - 0.0467 = 0.9033 \rightarrow 90.3\%$
 floor is 90%. The scorecard says GREEN.

A 20-point collapse for a week moves the 30-day number **4.7 points** — and then, after the fix ships, the ghost sits in the denominator for **23 more days**.

Compare the same week on a 7-day panel: **75%**, 15 points under the floor, on the Tuesday it happened.

What it means: the rolling window is fair to sellers and useless to operators — it is a low-pass filter on exactly the step changes we exist to catch. **What I'd do:** keep O2 as the policy floor for consequences, add a 7-day cohort with a CUSUM tripwire for action ($n \geq 30$, sustained $2\times$ baseline), and never let a 30-day number set this week's decision.

How do you get the promise program through peak season?

Peak is a distribution shift — σ widens everywhere at once, carriers breach lanes they've held all year, seller handling slips under volume. Four moves, all before peak. Publish the peak buffer policy in advance — wider k on named lane classes, agreed like my 0.5% mock-cutover tolerance, before anyone's under pressure. Freeze non-essential change — at Sephora the UDC program held a hard October-to-December freeze; the building's peak outranked the program's elegance. Pre-wire demotion rules — drift flags that pull lanes from the fast band automatically when transit spreads blow out. And run a war-room cadence triaged by promise-at-risk — the discipline from my Manhattan years, where a 4 AM Cyber Monday triage held fill rate at 99% and protected roughly \$8 million. Silence became the alarm after that night; I'd install the same heartbeat here.

The regime break, quantified. The model's peak assumption is $\sigma \times 1.45$. k is measured *in sigmas*, so the correct response is that **k does not change — the days do**:

summer: $b = 1.645 \times 0.60 = 0.99$ days on a $\sigma = 0.60$ lane $\Phi(1.645) = 95.0\%$
 peak: $\sigma \rightarrow 0.60 \times 1.45 = 0.87$; correct $b = 1.645 \times 0.87 = 1.43$ days (+0.44 days)
 frozen: keep $b = 0.99$ while $\sigma = 0.87 \rightarrow \Phi(0.99/0.87) = \Phi(1.135) = 87.2\%$

7.8 points of coverage lost on that lane by changing nothing. The workbook's Q4 tab runs the same failure at portfolio scale and prints **91.9%** accuracy against a 95% floor.

What it means: the November firestorm is caused in July, by freezing days instead of re-estimating σ . **What I'd do:** publish in October — re-solved k , the named lane classes, the expected fast-share cost (+0.44 days is a real penetration hit), Andon rules, and a change freeze. Success metric: zero emergency national pads between Thanksgiving and Christmas.

How would you approach same-day delivery for seller-fulfilled inventory?

Why “no buffer to hide behind” is literal. $\Phi(0/\sigma) = \Phi(0) = 50\%$. A same-day promise with zero pad is a coin flip on the model-noise term alone, before a seller or carrier has even missed anything. To hold 95% on the tightest lane in the model:

$b = 1.645 \times 0.22 = 0.36$ days = $0.36 \times 24 = 8.7$ hours of pad

What it means: 8.7 hours of protection does not exist inside a same-day window — so same-day cannot be underwritten by buffer at all. Accuracy has to come from I3 measured in hours, near-perfect feeds, and a courier window that exists on the calendar. That is why eligibility *is* the promise here, and why a graceful demotion to next-day is the only honest fallback.

Respectfully — it's the hardest promise in the book, because at same-day speed there is no buffer to hide behind; the eligibility decision *is* the promise. I'd scope it as gates before geography. Eligibility per seller-node: near-perfect inventory-feed accuracy, handling measured in hours, courier integration proven. Pilot metros chosen for density, with a published gate that can stop the rollout — the analog of my Locus robotics gate at Sephora: 98% pick-accurate throughput before the bots earned volume, and when week two measured 94% we throttled 30% of volume to manual and fixed the zone logic. Same-day earns volume the same way. The fallback is graceful demotion to next-day, never a broken promise — and expansion rides wave criteria, not enthusiasm.

The model, the engine, the data

What's ML's role in promise generation — and where does it fail?

Three jobs: predicted transit per lane — ideally a distribution, not a point, because the tail keeps promises, not the mean; recommended handling time per seller-item; and buffer calibration on top of both. Where it fails, from having operated next to these systems: distribution shift — models trained on history lag reality exactly when it matters, at peak and carrier changes; cold-start on sparse lanes; feedback loops — the promise changes buyer behavior, which censors the data the model retrains on; and silent degradation, the worst class. My guardrails are operational, not algorithmic: lane-grain MAE on the WBR every week, drift flags per lane, shadow mode before any rollout, and an eval-set discipline — I built an LLM triage agent on exactly that rule: every change must keep the known past passing before it touches the live present.

Why “a distribution, not a point” is an arithmetic claim, not a preference. One real-shaped lane — right-skewed, as freight always is (**ILLUSTRATIVE** 100 shipments, the shape I measured 200 times at Elementum):

mean = 12.50 days σ = 2.64 days p50 = 12 p90 = 15 p99 = 22

Promise you set	Days	Share of shipments that beat it
the mean	12.5	~61% — two in five arrive late
p50 (median)	12	61%
p90	15	91%
p99 (what I shipped)	22	100%

What it means: a promise set at the mean is close to a coin flip, and the gap between p90 (15) and p99 (22) is **seven days** — all of it in the tail a point estimate cannot see. A model can be excellent at the middle and useless for promising. **What I’d do:** ask DS for calibrated quantiles, not a point — and audit calibration by asking whether actuals land inside the predicted p90 about 90% of the time, because that is the number $\Phi(b/\sigma)$ silently assumes.

How would you work with the Promise Optimization data science team?

On a contract: they own the model, I own the outcome, and we share one instrument. The instrument is the WBR: lane-grain MAE, calibration curves, and the breach decomposition, same definitions every week — I froze seven metric definitions before automating Flexport’s review into Snowflake precisely so the pipeline automated trust instead of starting a re-derivation fight. What I bring them is a ranked backlog, not vibes: the Buffer Waste Index says which lanes’ model error is most expensive in penetration terms, so their next model improvement lands where the business feels it. And rollouts run my gates — shadow mode, cohort calibration, a rollback — because a model change is a cutover, and cutovers get mock runs.

The promotion bar, priced. Champion-challenger in shadow; the challenger must beat the champion on the frontier for **four consecutive weeks**.

a coin-flip model clears four straight by luck: $0.5^4 = 0.0625 \rightarrow 6.25\%$

What it means: I am explicitly accepting a ~6% false-promotion rate on a pure coin flip, and buying that down with weekly n large enough that a week is not a coin flip — hundreds of thousands of orders per lane cohort, not thirty. **What I’d do:** state the bar and its false-positive budget up front, so a promoted model is a measured decision rather than an enthusiasm, and so DS knows exactly what “done” means before they start.

How do you trade conversion lift against the cost of a late delivery?

With the breach economics made explicit: a broken promise costs a WISMO contact, refund and return risk, and trust decay that suppresses the customer’s next purchase — while a faster promise buys conversion now. The equimarginal question at lane grain: does one more day of promised speed on this lane buy more conversion than the expected breach cost it adds?

The trade, worked end to end. Model parameters (`model_reference.py`): speed elasticity $\beta = 0.16$ per promise-day, contribution margin **\$6.50**, breach cost **\$12**. **ILLUSTRATIVE** base: 1,000 sessions on one lane at

7.0% conversion.

one day faster is worth $e^{0.16} = 1.1735 \rightarrow +17.4\%$ relative conversion
 2-day promise: $1,000 \times 7.00\% = 70.0$ orders
 1-day promise: $1,000 \times 7.00\% \times 1.1735 = 82.2$ orders (+12.2 orders)
 margin gained: $12.2 \times \$6.50 = +\79

Now the two ways to buy that day:

Where the day comes from	Accuracy effect	Expected breach cost	Net
Out of the pad (one sigma removed: 95.0% $\rightarrow$ 84.1%)	-10.9 pts	$82.2 \times 0.159 \times \$12 = \$157$ vs $70.0 \times 0.05 \times \$12 = \$42 \rightarrow +\115	-\$36. Do not do it.
Out of σ, or a later cutoff (accuracy unchanged)	0	unchanged	+\$79. Do it.

What it means: the same day of speed is worth +\$79 or -\$36 depending entirely on *which term you took it from* — and the naive “cut the buffer, get speed” move is the one that loses money. **What I’d do:** never argue speed-versus-accuracy in the abstract; put the four numbers (elasticity, margin, breach cost, current Φ) on the board and let the sign of the net decide. If Finance disputes the \$12, that is a governance conversation with a number in it, which is the point. Promise Yield — penetration times accuracy — is the single number that keeps the trade honest, because optimizing either factor alone games the other. And there’s a governance subtlety I’d surface rather than bury: a profit-optimal policy may hold *more* accuracy than the 95% floor. If the economics and the floor disagree, that’s a finding for the room, not a silent override.

What’s the difference between WFS and SFS, and how do they interact?

WFS is control: Walmart holds the inventory, so handling and transit are Walmart’s own execution — promise generation is easy because the inputs are known. SFS is influence: sellers hold the inventory, Walmart owns the customer-facing promise on execution it doesn’t control — which makes it the harder and more interesting operating problem.

What control is worth in days. ILLUSTRATIVE — the same physical lane, two governance regimes:

WFS (handling observed inside Walmart's own building) $\sigma = 0.30 \rightarrow b = 1.645 \times 0.30 = 0.49$ days
 SFS (handling declared by a seller, unverified) $\sigma = 0.60 \rightarrow b = 1.645 \times 0.60 = 0.99$ days
 difference = 0.49 days

What it means: half a day of promise, on identical trucks, purely from whether the handling number is measured or asserted. That half-day is the price of the information gap — and it is exactly what SSS plus observed handling time is buying back. **What I’d do:** frame SSS internally not as a settings feature but as a σ -reduction program, because that is what it does to the promise: 0.99 days $\rightarrow$ 0.49 days is the same as a free half-day of speed at constant accuracy. They interact through the customer’s expectations: WFS sets the delivery bar customers assume site-wide, and SFS tools — SSS, ML handling times, the scorecard — are the bridge that lets independent sellers meet a bar they didn’t set. My career is the influence version: 68

customers, a thousand lanes, twenty-five sites, all moved by evidence and gates rather than org-chart authority. That's the SFS skillset.

What changes for cross-border sellers?

The arithmetic of a fat tail. Take the cross-border lane shape from above — $\sigma = 2.6$ days, handling + transit = 5.0 days:

uniform 1.10-day pad:	$\Phi(1.10 / 2.6) = \Phi(0.42) = 66.4\%$	→ one order in three breaks
honest pad at 95%:	$b = 1.645 \times 2.6 = 4.28$ days	
resulting promise:	$5.0 + 4.28 = 9.28$ days	→ structurally never "fast"

What it means: on this lane the choice is not speed-versus-accuracy, it is *honest-slow versus dishonest-fast* — and a uniform US buffer is dishonest-fast by 29 points. **What I'd do:** promise it at 9-plus days and keep the accuracy, put it outside the fast band by policy, and route the cells where that costs real demand to the WFS conversion funnel rather than heroic padding.

The distribution changes shape: longer transit, fatter tails, and customs as a variance spike no mean-based model sees. Three consequences. Measurement first — completeness on cross-border scan events is usually the gap; at Elementum I onboarded a thousand multimodal routes over X12 EDI and XML and took shipment-data completeness from 60 to 95%, and nothing about a cross-border promise works until that pipe is honest. Second, the equimarginal principle bites hardest here: many cross-border lanes need so much buffer to hit the floor that they mathematically can't live in the fast band — promise them honestly slower and keep the accuracy. Third, inventory-feed trust at foreign nodes needs its own SLA, because a cancellation is the worst outcome of all — worse than a slow promise.

The concepts, named

If you could only steer this team on one number, what would it be?

Promise Yield — fast-promise penetration times promise accuracy. Each factor alone is gameable: chase accuracy and you pad buffers until every promise is slow and safe; chase penetration and you shave buffers until the promises are fast and broken. The product punishes both games. It's the same reason inventory people never report fill rate without inventory turns — single-sided metrics buy their wins from the other side of the ledger. Underneath it, the WBR still decomposes into the input levers — handling adherence, carrier SLA, MAE, buffer calibration — because you steer on inputs. But the headline the room aligns on should be the joint number.

PY, computed, plus the honest limit of it:

Wk 33 baseline:	03 45.1% × 01 95.4% = 43.0%	4-quarter target: 50%
Policy A:	7.6% × 95.8% = 7.3%	
Policy B:	39.6% × 95.0% = 37.6%	(the model prints 37.7%)

Strategy someone might try	PY arithmetic	PY	Verdict
Pad everything for safety	20% × 99%	19.8%	PY collapses — the padding game fails on its face
Real fix (risk-proportional buffers)	39.6% × 95.0%	37.6%	5× Policy A's yield, no new models

Strategy someone might try	PY arithmetic	PY	Verdict
Cliff-chase: badge everything, eat breaches	$60\% \times 90\%$	54.0%	PY rises — and O1 90% breaches the 95% floor → hard RED

What it means: PY kills the padding game outright, but it does **not** kill the fast-and-broken game — which is why it is a north star *with guardrails* ($O1 \geq 95\%$, $O2 \geq 90\%$, $O4 \leq 5\%$, $O5 \leq 2\%$), never a north star alone. I would rather say that out loud than be caught by it. **What I'd do:** report PY on top, decompose one line below, and let a floor breach print RED regardless of how good the joint looks.

What is buffer waste, and what would you do about it?

Buffer waste is the speed you bought and didn't ship: on kept promises, the mean number of days the package landed *early*. Early delivery feels like winning, but systematically early means the buffer was bigger than the uncertainty — you paid for insurance you didn't need, in the currency of a slower promise on the item page. Ranked by lane, the Buffer Waste Index is a free-speed backlog: shrink buffers where the model error is provably low and penetration rises with zero accuracy cost. It's the exact math I ran at Elementum in inventory currency — measured spreads came in tighter than assumed spreads, and 580K units and \$1.5 million of buffer stock came out. Days and dollars are the same waste wearing different units.

BWI on the biggest lane in the model — and why it is worth exactly one badge. West Metro / Ground: $\sigma = 0.30$, $h + \hat{t} = 1.45$, the largest session pool at 2.55M/week.

BWI $\approx b - E[\varepsilon \mid \text{kept}] \approx b - 0 =$ the pad itself, when σ is small relative to b
 uniform policy: $b = 1.10 \rightarrow$ promise $1.45 + 1.10 = 2.55$ days NOT fast
 honest pad: $b = 1.645 \times 0.30 = 0.49 \rightarrow$ promise 1.94 days FAST
 harvest: $1.10 - 0.49 = 0.61$ days of pure waste per order

What it means: 0.61 days of insurance nobody needed was the only thing keeping Walmart's largest metro lane out of the two-day badge. No model change, no carrier negotiation, no seller behavior — arithmetic.

What I'd do: rank every lane by BWI $\times$ volume, and work the list top-down; the ranking *is* the roadmap, and it costs one query.

And the same idea in inventory units, which is where I first ran it. ILLUSTRATIVE per SKU-lane cell:

safety stock $\approx z \times \text{daily demand} \times \sigma_{\text{leadtime}}$
 brochure's assumed spread $\sigma = 4.0$ d: $2.33 \times 100 \times 4.0 = 932$ units
 measured spread $\sigma = 2.6$ d: $2.33 \times 100 \times 2.6 = 606$ units
 released per cell: $932 - 606 = 326$ units

Across the book that summed to the real numbers on my resume: **580,000 units and about \$1.5 million** of carrying cost — roughly 1,800 SKU-lane cells' worth at that per-cell release. The inventory never moved; the uncertainty did.

Walk me through the promise math this team runs on.

The promise is a sum: order time plus handling plus predicted transit plus buffer equals the EDD on the page. The promise is kept when the errors fit the buffer: handling error plus transit error at or under it. So accuracy factors into three probabilities — seller ships in window, carrier meets transit, and Φ of buffer over σ , the share of residual variance the buffer absorbs. What I like about the factorization is that it's an org chart: each

factor has an owner — seller ops owns the first, carrier programs own the second, the model and buffer policy own the third — and the WBR is just this equation read left to right with a red or green on each term. When O1 goes red, the first question is which factor moved. That's a five-minute diagnosis instead of a week of opinion.

All of it, on one napkin, with numbers — this is the version I would actually write on his whiteboard:

- 1) $\text{promised_EDD} = t_{\text{order}} + \text{handling_config} + \text{transit_predicted} + \text{buffer}$
 Tue 3pm + 1.0 d + 2.0 d + 1.0 d = Saturday 3pm
- 2) $\varepsilon = (\text{handling_actual} - \text{handling_config}) + (\text{transit_actual} - \text{transit_predicted})$
 kept $\Leftrightarrow \varepsilon \leq b$
 seller 8h late, carrier fine: $8/24 + 0 = 0.33 \leq 1.0 \rightarrow \text{KEPT}$
 seller fine, carrier +2 days: $0 + 2.00 = 2.00 > 1.0 \rightarrow \text{BROKEN}$
- 3) per lane, $\varepsilon \sim \text{Normal}(0, \sigma^2) \Rightarrow P(\text{pad covers the noise}) = \Phi(b/\sigma)$
 $\Phi(1.00) = 84.1\%$ $\Phi(1.645) = 95.0\%$ $\Phi(2.00) = 97.7\%$ $\Phi(2.33) = 99.0\%$
- 4) $O1 \approx I3 \times I4 \times \Phi(b/\sigma)$
 Wk 33: $0.938 \times 0.915 \times 1.1115 = 0.954$ ($I3 \times I4 = 0.858$; residual = 1.1115)
- 5) the dial: $b = k \cdot \sigma$, $k = \Phi^{-1}(\text{floor})$
 95% floor $\rightarrow k = 1.645 \rightarrow b = 0.36 \text{ d on } \sigma = 0.22 \cdot 2.63 \text{ d on } \sigma = 1.60$
- 6) the KPI: Promise Yield = $O3 \times O1 = 0.451 \times 0.954 = 43.0\%$

What it means: six lines take you from a physical box to a governed policy dial to a single KPI, and every line has an owner attached. **What I'd do:** teach the pod to derive it rather than memorize it — a team that can re-derive line 4 will never mistake I4 for a package survival rate, and a team that can re-derive line 5 will never negotiate buffers lane by lane.

Why does inventory-feed accuracy at ship-from nodes matter to a *delivery promise* team?

Because the worst promise isn't slow — it's cancelled. A stale feed sells inventory that doesn't exist, and the cancellation-rate metric is where it lands. I've owned this problem from the inside: the S/4HANA operating model at Sephora was precisely the definition of "sellable" — available-to-sell rules, out-of-bounds governance, one signed table across store, DC, dotcom, and comingle nodes, because most overselling bugs are vocabulary bugs. For seller nodes I'd run feeds like I ran lane onboarding at Elementum: a data-quality toll before a node's inventory backs a fast promise — freshness SLA, staleness alerts, accuracy sampled against fulfillment outcomes. A feed is onboarded when it passes the toll, not when it connects.

Where the cancellations actually live. $I7 = 88.9\%$ of nodes at $\geq 98\%$ feed accuracy, so $100 - 88.9 = 11.1\%$ of nodes are below the bar. **ILLUSTRATIVE** — say those dirty nodes carry 15% of orders and cancel at 4% while clean nodes cancel at 1.2%:

portfolio 05 = $0.15 \times 4.0\% + 0.85 \times 1.2\% = 0.60 + 1.02 = 1.62\%$ (under the 2% floor — GREEN)
 dirty nodes' share of all cancels = $0.60 / 1.62 = 37\%$

What it means: a green portfolio number is hiding a slice that produces **37% of every cancelled order** — and cancels are worse than lateness, because there is no tracking page to apologise with and no buffer that can cover a unit that was never there. **What I'd do:** rank nodes by (orders $\times$ cancel rate), onboard the top of that

list to a near-real-time feed, and refuse fast-promise eligibility to any node that has not passed the freshness toll. You cannot buffer your way out of a phantom unit.

Why are promise-engine p99 latency and cache hit rate on the input-lever list?

Because the promise renders on the item page, in the conversion path. Slow p99 either degrades the page or forces a cache fallback — and a stale cache serves yesterday's promise, which is an accuracy breach manufactured by infrastructure. So the engine carries an ops contract: a latency budget, a cache-hit floor, and a kill switch that degrades to a conservative default promise when the engine can't answer honestly — never a fast promise you can't verify. That's a principle I ran at Flexport, where every data source sat behind its own kill switch: no data is better than bad data. A promise you can't stand behind is bad data on the most expensive page in the company.

The fallback tax, priced. ILLUSTRATIVE — I9 = 41 ms p99 against a 50 ms budget today; suppose 2% of PDP views miss the budget and fall back to a padded default promise that converts at 5.5% instead of 7.0%:

per 1,000 views:	20 fall back × (7.0% − 5.5%) = 20 × 0.015 = 0.3 orders lost
per 1,000,000 views/week:	300 orders lost
priced at \$6.50 margin:	300 × \$6.50 = \$1,950/week ≈ \$101K/year

What it means: a latency chart nobody reads is a six-figure annual revenue line, and it files zero tickets — the exact silent-failure shape as buffer padding. **What I'd do:** carry I9 with a *revenue* annotation, not a milliseconds annotation, and make the fallback promise conservative-and-logged so I can measure how often infrastructure is quietly choosing the slow date.

How would you improve seller handling adherence — first scan in the window?

First, trust the measurement: scan-based, never manifest-based — a printed label isn't a shipped package. Then split the population. Setting problems: the seller's stated handling doesn't match their real operation — weekend calendars, cutoff times, seasonal capacity — and the ML-recommended handling time fixes the number to the truth; the metric here is the share of items on ML-recommended handling, which is on this team's input list for good reason. Execution problems: real slips, coached with scorecard evidence, then eligibility consequences. And run the carrier-ruler discipline I built at Elementum on sellers: allocate the benefit — fast-promise eligibility — by measured performance. Adherence improves fastest when the fast badge is visibly earned and losable.

Two arithmetic checks I run before touching a seller. First, the clock gap: I3 = 93.8% in-window means **100 − 93.8 = 6.2%** of orders missed the scan window, while O4 says only **4.6%** shipped after ship-by — a **1.6-point** gap that is definitional, not operational (a label can satisfy ship-by without a carrier scan; a scan at 3:40 PM misses a 3:00 PM window while the label was on time). Reconcile before you punish.

Second, size the I2 (observed-handling) lever. **ILLUSTRATIVE** mix: I2 = 56.5% of items on ML-recommended lag, and suppose declared-handling items miss at twice the rate of ML-lag items. Solve for the two rates from the observed 6.2%:

0.565x + 0.435(2x) = 0.062	→ 1.435x = 0.062	→ x = 4.3% (ML-lag items), 2x = 8.6% (declared)
push I2 from 56.5% to 80%:	0.80 × 4.3% + 0.20 × 8.6% = 5.2% out of window	
	→ I3 rises 93.8% → 94.8%	
01 effect:	0.948 × 0.915 × 1.1115 = 96.4%	(+1.0 point)

What it means: moving one input metric 23 points buys one point of customer promise accuracy — real, but not a miracle, and worth knowing *before* I promise it to a VP. **What I'd do:** ship observed handling as the default (Amazon's AHT equivalent) rather than an opt-in, because the 43.5% still on declared handling is where the misses concentrate.

A lane's carrier SLA adherence collapses this week. What do you do?

First, size the damage in one line. A lane's SLA collapse is usually a *mean shift*, not a spread change — the trucks got slower, not noisier. On a $\sigma = 1.2$ lane carrying a k-solved pad $b = 1.98$, a +1.0-day shift in actual transit:

before: $\Phi(1.98 / 1.2) = \Phi(1.65) = 95.0\%$
 after: $\Phi((1.98 - 1.0) / 1.2) = \Phi(0.82) = 79.3\% \rightarrow 15.7$ points of coverage gone
 to hold 95% under the shift: $b = 1.00 + 1.645 \times 1.2 = 2.97$ days (+0.99 days of promise)

What it means: the honest choices are “+1 day of promise on this lane” or “this lane leaves the fast band” — and I would rather demote it than print a date I have already computed to be a coin-flip-plus. There is no third option where the customer doesn't pay.

Contain, then cure. Same day: the lane's promises stop lying — widen the buffer or demote the lane from the fast band immediately, because sending a promise you already know is stale is the one unforgivable move. This is the void-honestly rule: a missed window gets an honest note, never a stale send. Then root-cause with the carrier team: capacity, weather, a network change — and check whether it's actually a measurement artifact before escalating. Recovery is tracked on the lane's transit MAE, and re-promotion is earned by the number, not the apology — the routing-guide muscle from Elementum, where load allocation followed measured performance and the CFO could audit every claim. Structurally: lanes big enough to matter deserve a second carrier.

How would you use lane transit-model MAE, and what would you push the DS team on?

MAE is the compound-interest metric: model error feeds σ , σ feeds the buffer, the buffer feeds penetration — so a tenth of a day of MAE improvement cascades into fast-promise share at constant accuracy. Two pushes. Grain: MAE at network level is decorative — an 85% aggregate can hide a 50% tail, the lesson I learned measuring forecast accuracy at the planning grain at Elementum — so it's lane-grain MAE, weighted by order volume, on the WBR. Calibration: a point estimate isn't enough, because promises are kept at the tail, not the mean; I'd ask for calibrated distribution forecasts — does the actual land inside the predicted P90 90% of the time? — since that's the number $\Phi(b/\sigma)$ silently assumes.

“Compound interest” is a conversion factor, and here it is. For unbiased roughly-normal errors, $MAE = \sigma \cdot \sqrt{2/\pi} = 0.798\sigma$, so $\sigma = 1.253 \times MAE$. With the dial at $k = 1.645$:

I5 = 0.72 days MAE $\rightarrow$ implied $\sigma \approx 1.253 \times 0.72 = 0.90$ days
 one tenth of a day of MAE improvement: $\Delta\sigma = 1.253 \times 0.10 = 0.125$ days
 $\Delta b = 1.645 \times 0.125 = 0.21$ days of pad released

What it means: every 0.1 day DS takes out of MAE hands me **0.21 days of promise speed** at unchanged accuracy — that is why I fund model work out of the penetration budget, not the science budget.

Why MAE alone gets a promise team fired. ILLUSTRATIVE — ten Dallas → Sacramento transits against a 2.1-day prediction, days: 1.8, 1.9, 2.0, 2.0, 2.1, 2.1, 2.2, 2.4, 2.8, 4.0.

absolute errors: 0.3, 0.2, 0.1, 0.1, 0.0, 0.0, 0.1, 0.3, 0.7, 1.9
 MAE = $3.7 / 10 = 0.37$ days p90 error = 0.70 days max error = 1.90 days

A 1.0-day buffer covers the p90 comfortably and loses to the 4.0-day trip by 0.9 days. **What it means:** a model that fits the 1.8–2.2 cluster harder improves MAE and changes nothing about the only trip that broke a promise. **What I'd do:** put p90 error and buffer-exceedance rate on the WBR beside I5 this week, and refuse to let MAE be the sole model-health line.

And the tail metric done correctly — the p99 definition people get wrong. p99 is **cumulative**, not a per-day density: it is the smallest number of days d such that $P(L \leq d) \geq 0.99$. Not “the day with 1% of shipments.” Same **ILLUSTRATIVE** 100-shipment lane as above:

Transit days d	shipments	cumulative	$P(L \leq d)$
9	5	5	0.05
10	14	19	0.19
11	21	40	0.40
12	21	61	0.61 ← p50
13	15	76	0.76
14	9	85	0.85
15	6	91	0.91 ← p90
16	3	94	0.94
19	3	97	0.97
22	3	100	1.00 ← p99

mean = 12.50 days $\sigma = 2.64$ days p50 = 12 p90 = 15 p99 = 22
 normal approximation: mean + $2.326\sigma = 12.50 + 6.14 = 18.6$ days
 empirical p99: 22 days

What it means: on right-skewed freight the normal shortcut understates the real tail by **3.4 days** — which is exactly why production sets buffers off the empirical quantile per lane and keeps the normal for the whiteboard. This is the Elementum promise: 200 lanes, promise at the p99, 99% lead-time accuracy held — because the promise was built from the cumulative tail, not from a brochure number or a mean.

When should a lane be pulled from the fast band instead of buffered into compliance?

The arithmetic that settles it. The marginal accuracy bought by the last buffer-day on a lane is $\phi(z)/\sigma$. At $k = 1.645$, $\phi(1.645) = 0.1031$, so:

California Intra (σ 0.22): $0.1031 / 0.22 = 0.469$ accuracy per buffer-day
 Mountain Rural (σ 1.60): $0.1031 / 1.60 = 0.064$ accuracy per buffer-day
 ratio: $0.469 / 0.064 = 7.3\times$

And the structural version, which is even blunter — Mountain Rural's handling + transit is **4.60 days before any buffer at all**:

$4.60 > 2.00 \rightarrow$ the lane cannot be in the fast band at ANY pad, honest or not

What it means: a buffer-day spent on the mountain buys one-seventh the accuracy of the same day spent in California, and on that particular lane no amount of buffer can produce a two-day promise — so “buffer it into compliance” is not a slower fix, it is an impossible one. **What I'd do:** demote it, promise it honestly at ~7 days, and hand the cell to the WFS conversion funnel where demand justifies it.

The equimarginal principle answers it cleanly: at the optimum, the last buffer-day on every lane buys the same accuracy — $\phi(b/\sigma)/\sigma$ constant — which means high- σ lanes rationally get *fewer* sigmas of protection, not more. Follow that logic to its edge: if the buffer a lane needs to hit the floor pushes its promised EDD past two days, the lane can't be in the fast band by definition — you're not protecting a fast promise anymore, you're decorating a slow one. Demote it honestly, fix the σ — carrier, model, or seller — and let it earn the band back. A gate that can't stop the program is decoration; I throttled my own robotics rollout at 94% against a 98% bar, and the launch was stronger for it.

You're not from e-commerce. What's your model of how the promise touches conversion?

The promise is the last thing a customer reads before deciding, and it moves conversion twice: the speed shown — a two-day promise converts better than a five-day one, which is why penetration carries a conversion-lift metric — and the confidence earned, which is slower and compounding: a customer burned by a broken EDD discounts the next promise they see, and WISMO contacts are the leading indicator of that decay. That's why Promise Yield is the right joint frame: the speed term is the immediate conversion lever, the accuracy term protects the discount rate customers apply to every future promise. I watched the retailer version at Sephora — promising governance existed because an oversell costs more trust than a slower honest promise ever costs conversion.

Both halves, with arithmetic. *Speed*, measured by the A/B line (O7 = 2.4% on the sample series; I read that conservatively as a **relative** lift, and I say so, because the workbook does not label “pp” vs “relative”):

control CVR = 6.000%
 treatment CVR = $6.000\% \times 1.024 = 6.144\%$ extra = 0.144 percentage points = 0.00144
 per 1,000,000 sessions at \$35 AOV: $1,000,000 \times 0.00144 \times \$35 = \$50,400/\text{week}$

Confidence, the slower half — the same \$12 breach cost is the placeholder for it, and the reason 15 orders that never got placed cost more than they look:

1,000 PDP views, 2-day promise at 7.0% CVR $\rightarrow$ 70 orders
 4-day promise at 5.5% CVR $\rightarrow$ 55 orders
 the silent tax: 15 orders that never existed — zero tickets, zero reds, zero WISMO

What it means: the speed half is measurable weekly and worth tens of thousands per million sessions; the confidence half never files a ticket, which is precisely why padding wins by default in any review built from tickets and reds. **What I'd do:** insist that every buffer or display change ships behind an A/B that reads *conversion and breach cost together*, so the invisible half of the ledger has a number on the page.

Section 3 – Program & operating cadence

The weekly business review

How do you run a weekly business review?

Same architecture every week: outputs on top, inputs underneath, every red pre-wired before the room — cause, owner, date. Same day, same definitions, and every number carries its grain and its date, because two dashboards is zero dashboards. I ran Flexport's Monday review this way for over a year, by hand at first — reconciling four systems weekly — then automated into Snowflake behind a parallel-run gate so the numbers never changed meaning mid-flight. The test of a good WBR is speed of attribution: when promise accuracy goes red, the room should know within minutes whether it was seller handling, carrier transit, or model error. The meeting decides; it never discovers.

What “pre-wired” means arithmetically — the page I would walk in with:

north star	Promise Yield = $03 \times 01 = 0.451 \times 0.954 = 43.0\%$	target 50%
RAG rule	WoW = $(\text{this} - \text{last}) / \text{last}$; $ \text{WoW} > 5\%$ relative → GREEN/RED, else FLAT	
	e.g. 06 WISMO 11.5 → 10.8: $(11.5 - 10.8) / 11.5 = 6.1\%$ favorable → GREEN	
hard floors	$01 < 95 \cdot 02 < 90 \cdot 04 > 5 \cdot 05 > 2 \rightarrow$ RED regardless of WoW	
attribution	$01 \approx I3 \times I4 \times \Phi(b/\sigma)$: $0.938 \times 0.915 = 0.858$, residual 1.1115	

What it means: every red on that page already carries its factor decomposition, so the meeting spends its time on the *ask*, not on the arithmetic. **What I'd do:** freeze the definitions first (I froze seven at Flexport before automating them), because a WBR whose numbers can be re-derived mid-meeting is a debate club.

A WBR where the reds arrive without a cause, an owner, and a date isn't a review — it's a viewing.

Explain input levers versus output metrics, and why the separation matters.

Outputs are what the business feels — promise accuracy, penetration, late shipment, WISMO — and they're lagging: by the time they move, the cause is weeks old. Inputs are what a team can actually turn this week — buffer calibration, SSS enrollment, handling adherence, feed accuracy — and they're leading. The separation matters because you manage inputs and report outputs; teams that stare at outputs end up explaining, not correcting. The discipline test I apply: every output should decompose into named inputs with owners, and if an output goes red while every input shows green, your input set is incomplete — that's a finding, not an excuse. That's the operating rhythm I ran at Flexport, and it's the rhythm this team already runs, which is half of why I want the seat.

An output metric goes red in your WBR. Walk me through your response.

First hour: verify the number before believing it — cohort definition, denominator mix, pipeline health; I've seen more red weeks caused by measurement than by operations. Then decompose along the factorization — seller, carrier, model, buffer — and segment: one lane or everywhere, one cohort or all. Then first-principles triage, because the noise is never the cause: at Flexport a 78%-manual week looked like operators slipping and was actually a carrier format change, fixed at the source with a dated ask. The red leaves the room with a root

cause, an owner, and a date, and the fix is verified by the same number going green in the same cadence. The program never stops moving while you analyze.

How do you set targets?

Baseline first — nothing moves until the starting number is on the wall, because a program without a baseline can't prove it did anything. Then targets by evidence, not fiat: the walk from 40% automation to a 70% gate to a 90% end-state at Flexport was staged on what the scorecard showed each quarter, not on a round number someone liked. Two more rules. Publish the bar before the pressure: I set a 0.5% cutover tolerance with finance before mock one, because a line negotiated after results arrive isn't a line. And separate floors from stretch: my robotics gate was 98% with a 99% stretch — the floor can stop the program, the stretch just aims it.

The staged walk, with its arithmetic — this is what “by evidence” looks like:

```
automation: 40% baseline → 70% no-touch gate → 90% end-state
             remaining manual work: 60% → 30% → 10% (each stage halves-plus the residual)
cost-to-serve: $1.00 hand-carried → $0.66 steady manual → $0.03 automated
              ($0.66 - $0.03) / $0.66 = 95.5% reduction
cross-check: the platform priced one eliminated interaction at $0.18
              $0.66 / $0.18 ≈ 3.7 residual human interactions per container
              $0.03 / $0.18 ≈ 0.17 – i.e. one interaction per six containers
```

What it means: the target wasn't “90% because it sounds good” — it was the residual-interaction count the scorecard said was reachable, and the \$0.18 cross-walk kept the dollars honest at a rate Finance owned, not one I invented. **What I'd do here:** set the Promise Yield walk the same way — 43.0% today, staged against what BWI harvesting and the SSS slope can actually deliver, not against a round 50%.

How do you keep action items from dying between meetings?

One log, one grammar — the discipline from my UDC issues log: 836 rows over fourteen months, every row carrying status, priority, owner, and date, no side lists, and 85.4% closed by program end.

The log's own arithmetic, which is how I knew it was alive:

```
836 rows × 85.4% closed = 714 closed, 122 open at program end
836 rows / ~61 weeks    = ~14 new rows/week – the honest intake rate
go-live spike: 271 rows, then 65 → 21 → 14 → 4 over following months
decay: 65/271 = 24% of the spike in month one, then 32%, 67%, 29% of the prior month
```

What it means: a spike with fast decay is a healthy cutover; a spike that plateaus is a design problem — and the 14-rows-per-week intake is what told me whether the review cadence could actually absorb the flow.

What I'd do: age every action item in the meeting, and treat “no movement in two weeks” as a red in its own right. Three mechanics keep it alive. The weekly review reads blockers and criticals first, so the important rows get the room's attention. Ageing is visible — an item that hasn't moved in two weeks is itself a red. And deferral is explicit: a parking-lot status means “decided not now,” in writing, with an owner — never silently deleted, because silent deletions teach the team the log lies. Any VP could open my log any Monday and see the truth the team saw.

Authority, gates, and hard calls

This role leads without direct authority over Product, DS, or Seller Ops. How do you do that?

Structurally, not charismatically. Three mechanisms from my track record. Pre-wire: at Sephora’s architecture council I put a per-system change table in front of every owner before the meeting — roughly eighty named participants across fifteen functions — and asked the room for exactly three decisions; sign-off landed because nobody got a first read in public. Evidence over rank: the Flexport roadmap flip happened because my manager pressure-tested the cost-to-serve math privately for two days before the room ratified it — rank, don’t argue. And let them author: the 20-person agile conversion stuck because the team designed its own ceremonies while I published the evidence weekly. Influence is making the right decision easy to make.

How do you use stage gates?

With numeric entrance and exit criteria, published in advance — and with real teeth, because a gate that can’t stop the program is decoration. The JDA migration ran the full lifecycle as gates: discovery closed on a written exit, UAT opened on proof — function testing done, conversion built, environments ready — go-live read three mock cutovers against a pre-published tolerance, and hypercare exited on under two blockers a day, hit in week three of four.

The gate’s arithmetic, and why the third mock was the only one that mattered:

```
tolerance published BEFORE mock one: 0.5% reconciliation miss
mock 1: 7.0% = 14× the line → NO-GO
mock 2: 1.8% = 3.6× the line → NO-GO
mock 3: 0.3% = 0.6× the line → GO, with 40% margin under the line
total improvement: 7.0 → 0.3 = a 23× reduction, measured on one frozen definition
```

What it means: the decision was arithmetic, not confidence — and because the line was published first, mock one’s 7% went upstairs with my name on it instead of becoming a negotiation. **What I’d do here:** the same shape for any promise-policy change — a published shadow-mode bar (four consecutive weeks on the frontier) so promotion is a reading, not a vote. The proof the gates were real: my own robotics rollout got throttled by its own gate at 94% against a 98% bar. Gates got a program I’d never run before — a WMS migration — delivered in eight months, on time, three percent under budget.

A program you own is slipping. What do you do?

Make the slip visible before it’s unmissable. The instrument catches it early — a burn-up against a flat planned line caught in-sprint scope growth at 56% completion on my Flexport launch, and the tripwire went on: new work estimated and swapped, never absorbed. Then re-plan in the open: when Locus retail testing slipped three weeks at Sephora, the revised build plan re-slotted the E2E chain in its own revision sheet — never absorbed into weekends, because a plan that hides slips teaches the team to stop reading it. Then the escalation, same-day, framed as a priced decision — this scope or that date, A costs this, B costs that — never as a complaint. Executives resolve trade-offs fast when the trade-off is priced.

How do you say no?

With a ranking, so the no is arithmetic instead of opinion. At Flexport I scored the backlog on benefit times impact over dev-weeks and re-sequenced an eighteen-month roadmap — the no to operator-first wasn’t “I disagree,” it was “partner-first moves twice the cost reduction, here are the shared axes, audit any input.” Rank, don’t argue. The second kind of no is the fence: at the architecture council I presented what we were *not* changing as its own slide — the comingle reserve math stays untouched — and that written out-of-scope line is what held when well-meaning teams arrived mid-build with improvements. A no that lives in a document survives; a no that lives in a meeting gets re-litigated weekly.

How do you prioritize across a portfolio of programs?

Floors first, then value per effort, then sequencing logic. Floors first: nothing that breaches the accuracy floor or the safety envelope gets ranked at all — a score bonus never admits a below-floor item. Then benefit-impact per engineering-week on shared axes, the discipline that cut 66% of effort and \$5 million from the Flexport plan. Then sequencing: build the chassis once — we launched the partner persona first because its backend was what operators would ride later, so value landed in six months instead of eighteen. And the portfolio stays honest through a weekly read of planned versus actual, because a priority list nobody burns down against is a wish list.

Cadence mechanics

How do you choose which metrics go on the WBR?

Ruthlessly. One output family the room aligns on — here, Promise Yield with its two factors and the guardrails: late shipment, cancellation, WISMO. Then six to eight inputs, each provably upstream of an output and each with a named owner — a metric nobody owns or acts on comes off the board. Definitions frozen and written: when I automated Flexport's review into Snowflake, I froze the seven metric definitions exactly as the team used them, so the pipeline automated trust instead of reopening a re-derivation fight. And every number carries grain and date on its face — the confusion cost of quoting an event-grain number next to a weekly-stock number in the same breath is a tax I've paid once and won't pay again.

How do you know when a program is done — and how do you run hypercare?

Done means the metrics say it's done, against exit criteria written before go-live. My JDA hypercare was a four-week schedule with a written exit — under two blocker tickets a day — and we hit it in week three. Two disciplines matter. The exit is a number, not a feeling, because hypercare without an exit criterion becomes a permanent support channel. And the post-cutover spike is read by its shape: on UDC, the 271-row issue spike at the combined go-live burned down 65, 21, 14, 4 over the following months — a spike with fast decay is a healthy cutover; a spike that plateaus is a design problem. I'd run any promise-engine or SSS launch here on the same arc.

Two teams show up with different numbers for the same metric. What do you do?

Reconcile before compare — the one-canonical-source rule, and it once saved a product at Flexport. Leadership was staring at our 25% schedule reliability against a competitor's 63% and heading toward a write-off; the comparison ran on unlike data. I rebuilt it apples to apples — normalized to the same sources — and their 63 collapsed to about 27 against our 25: a measurement gap, not a product gap.

The arithmetic of the two comparisons — same product, two conclusions:

```
as reported:  63 - 25 = 38 points behind  → "write the product off"
normalized:   27 - 25 =  2 points behind  → "close the last 2 points"
of the apparent 38-point gap, 36 points (95%) were definitional
```

What it means: 95% of a gap that was about to kill a product lived in the measurement, not the product — and reliability later reached 90% against an 80% north star once we added sources, so the 2 real points were the only ones worth working. **What I'd do here:** the same rule on promise accuracy — if Seller Ops' O1 and the DS team's O1 disagree, neither number enters the room until they're reconciled to one event source. The permanent fix was governance, not diplomacy: one definition, one scoreboard, and a standing rule that two

numbers that disagree get reconciled before they get compared. On this team I'd want promise accuracy, OTD, and the scorecard views all deriving from one event source with published definitions.

What does your first 90 days' operating cadence look like?

Days one to thirty: learn the levers — sit in the WBR without touching it, learn the buffer policy, the SSS funnel, which lanes and cohorts are red, and meet Seller Ops, Promise DS, and the carrier PMs. Earn the right before changing anything. Days thirty-one to sixty: own the review — take the output block and the action log, make every red decompose to a named input with an owner and a date. Days sixty-one to ninety: move one number visibly — my candidates in order: buffer recalibration on a pilot lane set where model error is provably low, the SSS funnel's biggest drop-off, or feed accuracy at the top seller nodes.

You won't need to teach me the operating system — I need to learn your numbers, not the method. My 90-day goal is one output metric moved that the WBR can prove.

How do you escalate?

Same day, priced, and framed as a decision — never a complaint. The pattern from my compressed Flexport go-live: when two programs collided over one engineering team, escalations went up as "A costs this, B costs that," and executives ruled fast because the trade-off arrived pre-priced. Three personal rules. Escalate the decision, not the frustration — the ask names the two options and what each costs. Escalate early and small — a two-week-old escalation is a confession. And never surprise your own chain: my manager saw the roadmap math before any room did. The escalation path is part of the operating system, not a failure of it.

How do you keep the numbers honest when they're inconvenient?

By treating a flattering wrong number as worse than an honest bad one. The proof point I use: at Flexport, the final truncated week printed 99.4% automation on three touches — a beautiful slide — and I discarded it as a data miss, because a VP steering on a three-touch week is a decision-quality problem. Same discipline on the red side: the honest red arrives with a cause and a dated ask, which makes it more useful than a green. On this team that instinct maps to things like never letting a thin cohort print a green promise-accuracy week, and labeling every number's grain and date so nobody can quote the flattering version accidentally.

How do you manage cross-team dependencies?

Two rules, one earned expensively. Every cross-functional dependency gets a what-if and a mitigation plan regardless of its effort estimate — I learned that on a "three-day" RF-device task at Sephora that failed on new hardware and slipped training sign-off by a week and a half; the task was scored by effort when it should have been scored by impact, because it sat in front of a go-live exit criterion. And the dependency chain lives on dated, public milestones: UDC's end-to-end test gates — November 22, December 20, January 30 — meant every one of fifteen-plus tracks read the same spine, and a slip surfaced against a date instead of an opinion. Impact decides attention, never effort.

How do you communicate program status to executives?

One page, four moves: the metric, the trend, the driver, the ask — outputs on top, inputs underneath, reds pre-wired. The room gets the answer; the appendix gets the audit. At Sephora I ran UDC on a weekly two-block status — thirty minutes IT, thirty minutes business — one dashboard, one risk register, no side decks, for sixty-plus stakeholders across fifteen tracks. Two habits I'd bring to this team's reviews: the biggest ask is

framed as a decision with a price on each option, and the worst number in the room is mine to say out loud before anyone finds it — volunteering the cost is the highest-trust move in the toolkit.

Section 4 – Behavioral / leadership

Every answer maps to a prepared story arc. Tags reference the story pack: A=Flexport visibility, B=Flexport FF App, C=Sephora UDC, D=Sephora JDA, E=Sephora S/4HANA, F=Elementum, M=Manhattan, G=failures, I=independent build.

The flagships

What's your biggest accomplishment? (story: A1)

Scaling Flexport's 4PL visibility platform from 7 pilot customers to 68 — because it proves the exact motion this role needs: adoption at scale without breaking quality. Leadership set public targets — 68 monthly active customers, 170 monthly active users — off a baseline of seven pilots and a cost-to-serve near a dollar a container. I built the wave machine: an entry bar where every new customer gave us about fifty containers of data for calibration before onboarding, and six published graduation criteria per wave — verification SLA at 90, data quality at 90, ETA accuracy at 80, stability, engagement, client feedback. A wave that failed one didn't move. I held wave three once, in front of everyone, on an ETA-accuracy miss that traced to one bad source. Result: both public targets landed two months early, cost-to-serve closed at 3 cents against the 66-cent baseline — a 95% cut — and support ticket response *improved* from 72 to 90% while volume grew five-fold. The risk we gated against never materialized.

The numbers, worked — because “scaled 10×” means nothing without the quality term:

customers	7 → 68	= 9.7× growth
cost-to-serve	\$0.66 → \$0.03	$(0.66 - 0.03) / 0.66 = 95.5\%$ cut
ticket response	72% → 90%	= +18 points, WHILE volume grew 5×
naive expectation at fixed capacity: 5× volume ⇒ quality falls		
what happened: quality ROSE 18 points ⇒ the gates were doing the work		
calibration toll: ~50 containers × 61 new customers ≈ 3,000 containers of		
pre-onboarding data – the reason ETA accuracy held at the 80% bar		

What it means: the interesting number is not 68, it's the 18-point quality gain *at* 5× volume — that is the signature of a gated rollout rather than a fast one. **What I'd do here:** run SSS the same way, and hold the wave when a graduation criterion misses, which I did once in front of everyone on an ETA-accuracy miss traced to one bad source.

Sales wanted the whole pool boarded at once. I traded a faster headline for a rollout that never had to roll back — and that's the same trade SSS adoption will ask for.

Tell me about a time you failed. (story: G1)

At Elementum, two days from UAT sign-off on the track-and-trace app. Five key users had passed all ten end-to-end scenarios, and I was confident. Then the sponsor's escalation landed: the system was buggy, users couldn't finish a flow. My first move was the wrong one — I re-ran the scenarios, they passed, and I sent screenshots saying things worked fine. Defending instead of digging. The root cause: the sponsor had opened access to all their users, and concurrent logins crashed a system my UAT criteria had never load-tested. I'd tested the explicit requirements and ignored the implicit one — performance at the customer's shape. I took

the miss to the sponsor as mine while the team simulated the concurrency and shipped the fix. The cost was real: the sign-off date passed and confidence had to be rebuilt. The rebuild is permanent: every critical scenario since carries numeric entrance and exit criteria — uptime, page-load, concurrent load — and every launch gate I’ve run since carries a load number. The Flexport launch proof was 500 containers in 20 minutes at 99.6% connectivity, before any customer touched the product.

Tell me about a time you disagreed with your manager. (*story: B2*)

The Flexport launch roadmap. The inherited plan — operator-persona first, eighteen months — was already socialized with leadership, and my manager owned it. My cost-to-serve math said it was in the wrong order: partner-first moved roughly 50% of cost-to-serve versus 25% for operator-first, because the partner backend was the chassis operators would ride later. I didn’t flip it in a meeting — he saw the math first, privately, and pressure-tested it with engineering for two days before any room did. The Bellevue onsite then ratified what the arithmetic survived. We flipped: 66% less effort, about \$5 million avoided, value in six months instead of eighteen. Flipping a privately-checked plan is alignment work; flipping it in public would have been theater.

The math I brought him, in the order I brought it:

cost-to-serve moved:	partner-first ~50% vs operator-first ~25%	→ 2× the return
effort:	66% less	→ the same value for one-third of the build
time-to-value:	18 months	→ 6 months = 12 months of benefit pulled forward
avoided spend:	~\$5M	
per-unit sanity check:	$\$0.66 \text{ CTS} \times 50\% \approx \$0.33/\text{container removed by path one}$ $\$0.66 \text{ CTS} \times 25\% \approx \$0.17/\text{container removed by path two}$	

What it means: the partner backend was the chassis operators would ride later, so sequencing it first bought twice the cost-to-serve movement for a third of the effort — the disagreement was arithmetic, not taste. **What I’d do here:** if the roadmap sequences a promise change I think is out of order, my manager sees the cost-to-serve equivalent privately, and engineering pressure-tests it, before any room does.

Influence and stakeholders

Tell me about influencing without authority. (*story: C4*)

Sephora’s UDC conversion touched about a dozen systems — WMS, ESB and Kafka, OMS, finance, the data warehouse — and the architecture council was a hard gate: no sign-off, no build. I owned none of those systems. The influence was structural. Exhaustive discovery first — sessions with every impacted team March through July, level-of-effort from each. Then the pre-wire: every team saw their row of the systems-impacted table before the meeting, because a sign-off gate is no place for first reads. Then the ask: exactly three decisions — architecture, staged timeline, peak-freeze boundary — with a written out-of-scope fence. Sign-off landed, the approved architecture carried both go-lives, zero customer-facing incidents, and nobody escalated a surprise afterward — roughly eighty named participants had their names in the work.

Tell me about a difficult stakeholder. (*story: A4*)

The defenders of Flexport’s legacy visibility product — still billing 79 customers and about \$186K of ARR — while leadership stared at a comparison showing our new product at 25% schedule reliability against a competitor’s 63%. Unmanaged, that ends in the new product written off and a legacy that never retires. I didn’t argue with the stakeholders; I rebuilt the comparison — normalized to the same data sources, their 63 collapsed to about 27 against our 25. A measurement gap, not a product gap. Then I made retirement

survivable for the defenders: every one of the 79 legacy customers got a per-segment destination and date — grow, white-label, or cancel. The room decided in one forum instead of debating for a quarter, and reliability later climbed to 90% against an 80% north star once we added sources — real work, not just normalized math.

The two numbers that changed the decision, and the one that made it safe:

gap as presented: $63 - 25 = 38$ points → kill the new product
 gap normalized: $27 - 25 = 2$ points → finish the new product
 ARR at risk if retirement went badly: \$186K across 79 customers
 = ~\$2.4K ARR per customer → small per head, which is WHY it was retirable
 but 79 conversations, so it needed a plan per segment, not per account
 end state: reliability 25% → 90% against an 80% north star = 10 points OVER the bar

What it means: the stakeholder wasn't defending a product, they were defending 79 relationships worth \$2.4K each — cheap in dollars, expensive in dignity, so the destination-and-date grid was the real deliverable.

What I'd do here: the same grid for any seller cohort we'd move off a legacy promising path — every seller gets a named destination and a date before the deprecation is announced.

Tell me about delivering under a tight deadline. (story: B3)

Flexport leadership pulled a strategic go-live forward by a month — and pulled engineers off my program to do it. Two programs, one team, an immovable date, about \$200K a month per DC at stake. I ran both as one system: one RAID log, one 45-minute daily forum, and one ruler for both programs — true blocker, non-blocker, or requirements gap in disguise; most "blockers" reclassified within a week. True blockers went into a two-pass war room, onsite and offshore handing off in writing, so issues moved around the clock without anyone working 24 hours. Both landed — the compressed program inside four weeks, mine on its original date — protecting about \$1 million per DC. The honest cost: the team's calendar, absorbed by design instead of a missed date.

The price tag that made the priority call obvious:

delay cost: ~\$200K per month per DC
 a 5-month slip × \$200K = ~\$1M per DC ← the number that was protected
 so the daily forum's real job: keep the slip at 0 months, not "manage stakeholders"
 blocker ruler outcome: most escalated "blockers" reclassified within a week as
 non-blockers or requirements gaps in disguise — the queue shrank by triage,
 not by heroics

What it means: at \$200K per DC-month, a week of ambiguity about whether something is truly blocking costs about \$46K per DC — which is why the ruler had to be published and applied daily, not weekly. **What I'd do here:** price every peak-readiness collision in breach-cost and conversion terms the same way, so one forum can rule on both programs with a number in front of it.

Tell me about operating in ambiguity. (story: D1)

Sephora handed me the dual-DC WMS migration with nothing documented — no strategy, no scope, no milestones, no risk register — a hard eight-month deadline, a \$20 million pipeline of four projects gated behind it, and one more complication: I had never run a WMS migration. My answer was gates instead of experience. I sat with the sponsor first — objectives, assumptions, success criteria — then held discovery's written exit until every flow had a documented current state, even though discovery was the phase everyone

wanted to rush. Every phase after opened on entrance criteria and closed on exit criteria. The go/no-go read three mock cutovers — 7%, 1.8%, 0.3% against a 0.5% tolerance published before mock one. On time, about 3% under the \$1.8M budget, and the pipeline started the next month. Ambiguity yields to written criteria faster than to experience.

What the ambiguity was worth, and what the gates converted it into:

at stake:	\$20M pipeline of 4 projects gated behind an 8-month deadline = ~\$2.5M of downstream value per month of slip		
budget:	\$1.8M, closed ~3% under → ~\$54K under, on a program with no documented scope on day one		
the gate:	mock 1	7.0% = 14.0 × tolerance	NO-GO
	mock 2	1.8% = 3.6 × tolerance	NO-GO
	mock 3	0.3% = 0.6 × tolerance	GO

What it means: with \$2.5M/month of pipeline behind the date, the temptation was to go on mock two's "close enough" 1.8% — the published 0.5% line is the only reason that conversation never happened. **What I'd do here:** publish the promise-policy promotion bar before the first shadow week, for exactly the same reason.

Tell me about a decision that turned out to be wrong. (story: E1)

On Sephora's S/4HANA inventory operating model, my first ruleset for in-transit stock was too conservative — I ruled that channels couldn't promise against units on a truck. It felt safe. It was wrong: it starved dotcom of sellable units and cost real conversion on items that were reliably a day from the building. The data made the error visible, and the fix went through the policy, not a side config: a defined in-transit promise window per lane, built from real lane data, written into the signed model. Two lessons I carry. Conservatism has a price, and it's paid in the same currency as risk — that's exactly the buffer-waste trade this role manages daily. And I'd now test a ruleset against history before publishing it — a week of measurement up front would have skipped the starvation.

The trade I had gotten backwards, in the promise language of this role (illustrative units, the real S/4HANA ruleset was per-lane):

my rule:	in-transit units are not promisable	→ effective σ treated as ∞
reality:	those units were reliably ~1 day out, $\sigma \approx 0.3$ day	
the honest window:	$b = 1.645 \times 0.3 = 0.49$ day of buffer buys ~95% coverage	
so the choice was never "risk vs safety" — it was		
	promise nothing vs promise +1.5 days with 95% confidence	
per 1,000 dotcom sessions at 7.0% CVR, killing a fast promise costs the same		
	15 orders as any other unnecessary pad: 70 → 55	

What it means: "unpromisable" is a $\sigma = \infty$ claim, and almost nothing in a supply chain actually has infinite variance — the fix was a measured in-transit promise window per lane, written into the signed operating model rather than a side config. **What I'd do here:** treat every blanket "we can't promise that" as a hypothesis with a σ attached, and go measure the σ .

Tell me about a conflict within a team. (story: D3)

During Sephora's agile conversion, QA and engineering were headed for a standoff: QA saw the four-week cadence as compressed testing windows — more work in less time — and they weren't wrong to fear it.

Instead of arbitrating, I turned the objection into the design: QA authored the definition-of-done with entry criteria, so the thing they feared losing became the thing they controlled. That's my standing read on team conflict — resistance is usually a design requirement wearing a complaint. The conversion landed in six weeks, the team kept the cadence because it worked, and neighboring teams copied it. The conflict resolved because the people with the concern became the authors of the control.

Feedback, direction, and dates

Tell me about giving hard feedback. *(story: C2)*

On UDC I gave it with evidence, which is the only way I give it. The consolidated issues log showed one owner's queue growing two weeks running while everyone else's burned down. Instead of a hallway "you're behind," I brought the rows: here's your open count trend, here's the class of issues aging, here's what the E2E gate needs from you by its date. The conversation became a resourcing and prioritization discussion instead of a character discussion — some of it was genuinely overload, and the evidence got them help rather than blame. Hard feedback lands when the person can argue with the plan instead of defending themselves. The queue recovered before the gate.

Tell me about receiving hard feedback. *(story: A2)*

Early in Flexport's visibility program, a leader told me flatly the program wasn't demonstrating value — the effort was visible, the proof wasn't. It stung because it was true: I was reporting activity, not evidence. My response became the WBR itself — I stopped narrating progress and built the instrument: baseline automation at about 40%, cost-to-serve at 66 cents, a weekly read with the same definitions every Monday, every claim tied to the workbook. The same critic later steered by those numbers. The feedback rewired a default I still run: when someone questions the program's value, the answer is a measurement system, never a better story.

Tell me about managing up. *(story: D1/A2)*

Two habits. First, executives get instruments, not adjectives: my JDA go-live recommendation to leadership was written, and it cited the mock-cutover scorecard — 0.3% against the pre-published 0.5% line — not my confidence. When mock one came back at 7%, my name was on the note that sent that number up unvarnished. Second, protect their decision quality even when it costs you a slide: I discarded a truncated week that printed 99.4% automation because a VP steering on three data points is a decision-quality problem. Managing up isn't managing impressions — it's making sure the person above you is never the last to know and never deciding on flattering data.

Tell me about changing direction mid-program. *(story: C3)*

Two weeks into Sephora's Locus robotics rollout, the bots measured 94% pick-accurate throughput against the 98% gate I'd set before they arrived. The pressure was to push through — the combined go-live was near, and throttling the flagship automation looked like failure. We throttled: about 30% of volume moved back to manual carts while we traced the miss to zone logic — the bot map and induction rules disagreed about which zone owned which work. Fixed, re-measured, full automation by week three, and retail outbound went live with zero customer-facing incidents. The direction change was cheap because it was designed in advance: the bot-versus-cart split was a dial we'd built, and the fallback was planned before it was needed. Changing direction is only dramatic when you haven't rehearsed it.

Tell me about missing a deadline. *(story: G2)*

On the JDA program, training sign-off — a go-live exit criterion — slipped a week and a half, and the miss was mine. The devices team had estimated the RF-gun conversion at three to four days, straightforward, and I planned it that way: low effort, low attention. My own pre-training device test caught the upgrade script failing on newer hardware, and with key people on COVID night shifts, manual conversion couldn't recover the window. I told leadership straight: training slips 1.5 weeks, my miss — I had scored the task by effort when impact was what mattered; a single-point dependency in front of an exit criterion deserved a what-if plan regardless of size. The rule changed permanently, and the program still landed on time and about 3% under budget. No low-effort dependency has surprised a go-live of mine since.

Tell me about juggling competing priorities. (*story: B3*)

The compressed go-live is my proof case — two programs, one shared engineering team, one immovable date. The mechanics that made it survivable: I refused to run two competing systems, so both programs shared one RAID log, one daily forum, and one blocker ruler — which meant priority calls happened once, in the open, against the same criteria, instead of twice in hallways. When something couldn't be resolved in the room, it escalated same-day as a priced decision. The lesson I carry: don't argue priority — make the collision visible and let one forum rule on both programs. Both landed. On this team, that's how I'd run a quarter where an SSS push, a peak readiness program, and a model migration all want the same engineers.

Tell me about cutting scope. (*story: B1*)

On the FF App export launch, the acceleration envelope was minimum-scope by design: we shipped the partner path first and made Air and LCL wait for the following year, and the operators kept their old screens for two more quarters. That's a real cut with real complaints attached. What made it defensible: the success metric was set before scope — monthly operator interactions eliminated, not features shipped — so every cut was scored against what it cost in that number, and the burn-up tripwire kept the door shut afterward: new work estimated and swapped, never absorbed. The MVP landed in April 2022 — eleven partners, ten countries — and 48 of 53 import partners migrated within a month. Scope cuts fail when they're silent; ours was priced, published, and scored.

Tell me about a risk that materialized. (*story: G3*)

On the Snowflake pipeline that automated Flexport's weekly review, I planned extracts, loads, metric parity, a parallel-run cutover gate — and did not plan the data-access security review. Ticket data carries customer content; the review surfaced mid-build and cost two weeks. I owned it upward immediately as a scoping miss, and re-sequenced instead of stalling: the parallel run kept the Monday review alive on the manual path, so the program absorbed the two weeks without skipping a single delivery. The permanent fix is a standing kickoff question on every data program since: whose approval does this data need, and what's its lead time? Security lead time gets planned, not discovered. Small miss, structural cause — those are the ones worth a post-mortem.

Tell me about data changing your mind. (*story: F2*)

At Elementum I walked into Austria convinced we had a carrier performance problem — the noise said carriers were late everywhere. I pulled three years of shipment history myself, and the data said something more embarrassing: we couldn't even measure lateness. True data quality was about 55% at the measured grain against the 40% the reports showed — missing rates, lanes keyed at the wrong grain, carrier names in five spellings, transit times copied from brochures. So the program flipped: the routing engine waited for the data, quality climbed to 99% by defect class before scale, and the savings method was signed with finance

before a dollar was claimed. Result: about \$8 million in freight savings across 25 locations, every dollar auditable to a shipment. The noise said carriers; the data said measurement. I believed the data.

The measurement gap, and why it had to be closed before the engine shipped:

reported data quality	40%	what the dashboards claimed
true quality at grain	55%	what three years of history actually showed
→ the reports were wrong in BOTH directions; they were not measuring the same thing		
post-remediation	99%	by defect class, before any scale-up
savings	~\$8M across 25 locations = ~\$320K per location	

What it means: at 55% true quality, an optimizer's recommendation is a coin flip with a confident interface — routing decisions made on brochure transit times and five spellings of a carrier name would have been indistinguishable from noise, and the savings would not have been auditable to a shipment. **What I'd do here:** the same gate on any promise-model change — feed accuracy (O5) and definition integrity clear first, model second.

People, positions, and trust

Tell me about developing someone. (*story: D3*)

The agile conversion was development at team scale, but the individual version lived inside it. The QA lead was the most skeptical person in the room — and I gave her the most load-bearing artifact: the definition-of-done with entry criteria. I coached the frame — what a gate needs to survive pressure, how to write criteria that can say no — and she authored the content. Within two cycles she was defending the gate to engineering herself, and the ceremony survived my leaving the program because it was hers, not mine. That's my development model: give people an instrument to own, coach the craft of it, publish their wins in the evidence. The team converted in six weeks and other teams copied it — which made her work the template.

Tell me about handling underperformance. (*story: C2 / MetricStream*)

Honestly, with evidence and a split diagnosis. Running MetricStream's onshore-offshore delivery teams, when a workstream underperformed, the first question was never "who's weak" — it was "is this a capability gap, a load problem, or an unclear definition of done?" Each has a different fix: pairing and review for capability, rebalancing for load, written acceptance criteria for clarity. On UDC, the issues log made it unemotional — an owner's queue growing two weeks running triggered a conversation with the trend attached, and more often than not the fix was resourcing, not reprimand. Where it *is* individual, I'm direct early — the kindest thing you can do with a performance gap is name it while it's small. What I won't do is manage performance by mood; the log decides when we talk.

Tell me about taking an unpopular position. (*story: Manhattan OMS-WMS discrepancy*)

At Manhattan, as the Distribution OMS consultant on a retail account, we found an OMS-to-WMS handshake discrepancy — a couple hundred orders, roughly \$2.5 million, where the two systems disagreed. My manager's instinct was to correct it quietly and move on. I argued the unpopular route: full transparency with the customer — show them the discrepancy, let them prioritize the affected orders, and fix the class with a bulk patch rather than silent record surgery. Uncomfortable meeting, short-term embarrassment. But the customer got to protect the orders that mattered to *their* business, the fix was auditable, and the trust it built contributed to winning a much larger initiative with that client. Silent corrections are loans against trust at a terrible interest rate.

Tell me about earning trust from skeptics. (story: F1)

Elementum's routing guide asked planners to trust a system over their own memory — and the skeptics were often *right* about specific lanes while wrong about the system. Two moves earned them. First, value-phasing: their highest-spend lanes onboarded first, so the guide proved itself on the lanes they cared about, gated by a data-quality toll — a lane missing its rate didn't onboard. Second, the override path: planners could reject a tender with a reason code, and those reasons flowed back into the guide. Overrides with reasons are data; silent workarounds are sabotage. The leaners-in became cohort champions, completeness climbed from about 60% to 95%, and adoption followed the evidence. Sellers skeptical of SSS will be won exactly this way — proof on *their* orders, and a dignified path to disagree.

Tell me about deciding with incomplete data. (story: M1)

Cyber Monday, 4 AM, at Manhattan. The VP called: backorder lines piling up despite sufficient inventory — millions in sales at risk, and the full diagnostic picture an hour away. I made the triage call on what we had: rank by promise-at-risk, rebalance allocation across the three DCs so no node drowned while others idled, one bridge, one log, every decision signed with a timestamp. The trace later showed the allocation scheduler had died silently at the daylight-saving change — inventory fine, nothing draining the queue. Fill rate held at 99% and roughly \$8 million was protected. Incomplete data with a decision and a timestamp beats complete data an hour late — and the post-mortem made silence itself an alarm, so the next peak ran boring.

The arithmetic that made “decide now” the right call at 4 AM:

```
waiting for the full diagnostic: ~1 hour
Cyber Monday order rate is the peak of the year – an hour of the day's volume
  sits in that queue, and backorder lines were growing, not flat
so the choice was: act on rank-by-promise-at-risk now
                   vs act on complete information after an hour of accrual
the triage rule: rank by promise-at-risk, NOT by queue order (FIFO would have
  cleared cheap orders while the at-risk promises aged)
rebalance across 3 DCs so no node drowned while others idled
outcome: fill rate 99%, ~$8M protected
root cause: allocation scheduler died silently at the daylight-saving change
  – inventory was fine; nothing was draining the queue
```

What it means: the expensive thing was never the diagnosis, it was an hour of peak-rate accrual against a FIFO queue — so the decision that mattered was the *ordering rule*, which needed no diagnosis at all. **What I'd do here:** the same triage grammar on a promise-engine incident — rank by promises at risk, and make silence an alarm, because a heartbeat that stops is the only failure mode nobody pages about.

Tell me about going above and beyond for a customer. (story: F3)

Elementum's planners were learning about shortages when the production line stopped — the ask was an alert, and the easy version was a threshold email nobody would trust. I ran it as a trust program instead. Fixed the inputs first: forecast accuracy climbed from 70 to 85% at the grain planners actually plan at — SKU-location-week — because aggregate accuracy hides the misses that stop lines. Built the alert on coverage math with a reason line on every alert, checkable in thirty seconds. Then treated trust as a metric: false alarms counted and burned down publicly, like defects. Live in three months. The alert earned its place in the weekly S&OP rhythm — acted on, not routed around — and about \$2 million of safety stock came out as the uncertainty shrank.

Why fixing the input came before building the alert:

forecast accuracy at SKU–location–week: 70% → 85% = +15 points
 error term: 30% → 15% = the noise the alert had to see through, HALVED
 alerting on a 30%–error forecast: every other alert is the forecast's fault,
 and planners learn to ignore the channel – trust dies at the second false alarm
 false alarms were counted and burned down publicly, like defects
 safety stock released: ~\$2M, because safety stock scales with uncertainty

What it means: an alert's precision is capped by its input's error, so building the alert first would have shipped a channel that was ignored by design — the 15-point forecast gain *is* the alert. **What I'd do here:** the promise-engine analog is O5 feed accuracy and the transit model's MAE — I don't ask sellers to trust a promise built on a feed I haven't cleaned.

Tell me about simplifying something complex. (story: A3)

Flexport's Monday review ran on four systems that didn't talk, stitched together in Excel by one person — me — every week. Single point of failure, about two hours a week, and system accuracy at a 60% baseline. The simplification wasn't heroic engineering; it was a document. I wrote the BRD myself, froze the seven metric definitions exactly as the team used them, priced doing nothing at over \$100K a year so the infra team could prioritize it themselves, and gated the cutover on a parallel run — manual versus pipeline within one percent for consecutive weeks. Four systems became one Snowflake pipeline, accuracy went 60 to 90%, and its first automated catch — a 4% ticket-sync drop — was invisible in the manual era. Programs like this die as complaints and ship as documents.

The price of doing nothing, which is the only number that got it funded:

my time: ~2 hours/week × 52 = ~104 hours/year of one senior program owner
 plus: a VP's weekly decisions running on stale joins
 plus: reconciliation risk nobody else could re-derive (bus factor = 1)
 priced at the infra team's OWN loaded rates: > \$100K/year, recurring
 cutover gate: manual vs pipeline within 1% for consecutive weeks
 accuracy: 60% → 90% = +30 points, i.e. errors 40% → 10% = a 4× reduction
 first catch: a 4% ticket-sync drop – undetectable by hand

What it means: “reporting is painful” gets funded never; “the status quo costs you \$100K a year at your own rates” got prioritized by the infra team themselves. **What I'd do here:** price the manual work behind any promise-ops process the same way, at the platform team's rates, so the business case survives their review rather than mine.

Tell me about handling a crisis. (story: M1)

The 4 AM Cyber Monday call is my template, and the mechanics matter more than the drama. One bridge, one log, one decision list — every call signed with a timestamp. First-principles triage before spending money on symptoms: the noise said capacity, throw staff at the backlog; the trace said the allocation scheduler had stopped at the daylight-saving change. Triage ranked by promise-at-risk, not queue order. Leadership got order-status reports every two hours — orders at risk, orders saved, time to clear — until 9 PM. Fill rate held at 99%, about \$8 million protected. And the crisis isn't over at recovery: the post-mortem shipped a scheduler heartbeat, a daylight-saving checklist, and a rehearsed rebalance playbook. The next peak ran boring. Boring is the deliverable.

Tell me about persuading senior leadership. (*story: A3*)

The Snowflake pipeline got funded by a price tag, not a pitch. Infrastructure teams hear “reporting is painful” weekly and fund none of it — so the BRD priced doing nothing: my loaded hours, the VP’s weekly decisions running on stale joins, and the error risk of a reconciliation nobody else could re-derive — over \$100K a year, built with the infra team’s own rates so it survived their review. I also shrank the ask: definitions frozen, new metrics explicitly out of scope — a narrower fundable pipeline this quarter over a perfect one someday. They prioritized it themselves. Persuasion upward is mostly pricing and shrinking: price the status quo, shrink the decision.

Tell me about taking ownership outside your lane. (*story: F4*)

A logistics manager asked me for a report — lead-time variability at lane level. The report was the ask; the problem was bigger: the entire inventory plan ran on carrier-brochure lead times, single optimistic numbers nobody had measured, so every safety-stock buffer built on them was fiction. Nobody owned that seam, so I took it. I measured actual transit distributions across 200 lanes from shipment history — mean, spread, tail — set the lead-time promise at the 99th percentile per lane, and fed measured spread into the safety-stock math. Then I pitched leadership with mockups to make it product, not a one-off. Result: 99% lead-time accuracy, 580K units and about \$1.5 million of buffer released — from a feature request. The promise at the tail, not the brochure — which happens to be this team’s entire business.

The tail, defined precisely — because “p99” is the part people say loosely:

p99 = the smallest d such that $P(L \leq d) \geq 0.99$ (CUMULATIVE, not per-day)
 it is NOT “the day where 1% of shipments land” — it is the day by which
 99% have ALREADY landed

one measured lane, right-skewed, $\sigma \approx 2.6$ days (illustrative shape, real method):

days	n	cum	$P(L \leq d)$	
9	5	5	0.05	
10	14	19	0.19	
11	21	40	0.40	
12	21	61	0.61	← p50 = 12
13	15	76	0.76	
14	9	85	0.85	
15	6	91	0.91	← p90 = 15
16	3	94	0.94	
19	3	97	0.97	
22	3	100	1.00	← p99 = 22

mean 12.50 · σ 2.64

normal approximation: $12.50 + 2.326 \times 2.64 = 18.64$

empirical p99: 22 → the normal form under-promises the tail by
 $22 - 18.64 = 3.4$ days

And the safety-stock arithmetic that turned it into inventory: safety stock = $z \times \text{demand/day} \times \sigma_L$, so

brochure-era $\sigma_L \approx 4.0$ days: $2.33 \times 100/\text{day} \times 4.0 = 932$ units held

measured $\sigma_L \approx 2.6$ days: $2.33 \times 100/\text{day} \times 2.6 = 606$ units held

freed per node-SKU cell: $932 - 606 = 326$ units

across the program: ~580K units and ~\$1.5M released

What it means: the brochure number was a *point*, and a point cannot be promised against — the distribution's spread is the entire asset, because it sets both the honest promise day and the inventory you must hold to defend it. **What I'd do here:** exactly this, per lane, and flag drift when a lane's σ moves — which is the same drift flag Part 12's dashboard runs on.

Tell me about sacrificing short-term results for long-term value. (*story: A1*)

Every wave of the Flexport rollout, sales pushed to board the whole customer pool at once — the headline would have been spectacular for a quarter. I held the graduation gates instead: every cohort waited for its criteria, every new customer paid the fifty-container calibration toll, and I once held wave three publicly on an ETA-accuracy miss. The short-term cost was real and visible — slower logo count, awkward pipeline reviews. The long-term return: the rollout never rolled back, support ticket response improved from 72 to 90% while volume grew five-fold, and the product's reputation survived scaling — which is the asset that made 68 customers possible at all. I'll make that trade every time, and SSS will demand exactly it: enrollments are the headline; 30-day retention is the business.

Tell me about learning something fast. (*story: I1*)

Last year I decided AI-in-operations was something I'd own hands-on, not read about — so I built an LLM triage agent for incident queues myself: Python, FastAPI, tool-use, the whole stack. The learning method was the same one I use for programs: instrument first. I built the eval set before the agent — graded historical incidents with known-correct outcomes — and gated every prompt, model, or routing change on the past continuing to pass. Then I engineered cost like a budget line: \$25 an incident was a demo, so I routed easy classifications to small models and reserved expensive calls for the hard tail — down to 40 cents, a 98% cut, with accuracy held and citation coverage at 100%. That's how I learn: build the scorecard, then let it teach me.

The cost engineering, worked — it's the same tiering logic as a promise fallback:

```
before: $25.00 per incident      after: $0.40 per incident
cut:    (25.00 - 0.40) / 25.00 = 98.4%      ratio: 62x cheaper
at 1,000 incidents/month: $25,000/mo → $400/mo = $295K/year avoided
the mechanism, not magic: route the easy majority to a small model, reserve
    the expensive call for the hard tail – cost follows the tail, like buffer does
the guardrail: accuracy held on a frozen eval set, citation coverage 100%,
    so "cheaper" never quietly meant "worse"
```

What it means: a 62× cost reduction with accuracy held is a routing decision, not a modeling breakthrough — and the eval set built *before* the agent is what let me prove the second half of that sentence. **What I'd do here:** the promise engine's p99-latency fallback is the identical shape, which is why I'd price the fallback's padding in lost orders rather than treating a timeout as free.

Section 5 – Supervisory / people leadership

The JD asks for 2+ years of supervisory experience. The honest shape of my leadership is a cross-functional pod — program owners, engineers, QA, ops — plus onshore-offshore delivery teams at MetricStream. I will not invent a direct-report headcount I didn't have.

How many people have you managed?

Direct reports in the classic HR sense: the MetricStream years — I ran onshore-offshore delivery teams across concurrent client programs, owned the SOWs, and was the person who assigned work, managed risk, and sat in the go-live chair. Everywhere else the leadership is matrix: at Sephora I converted a 20-person Product/Eng/QA team to a four-week agile cadence and ran UDC across 15+ named tracks with 60-plus stakeholders. At Flexport I sequenced nine teams / ~100 people on the visibility program through daily standups and a scrum-of-scrums. The Senior Manager seat here is a pod of program/ops owners — that is the shape I already run.

I will not dress matrix leadership up as a headcount I didn't have. I will show you the pod I already ran, and the gates that made it deliver.

How would you structure this Tech Ops pod?

By output, not by activity. One owner of Promise Yield (the joint of accuracy × fast-promise share) who is me. Under that, three input-owning seats matching the WBR tabs: Speed Programs & Adoption (SSS, eligibility, onboarding throughput), Promise Engine (model MAE, buffer policy, latency), Seller & Carrier Execution (handling adherence, carrier SLA, inventory-feed accuracy). Each seat owns named input lines, pre-wires their reds before Tuesday, and carries one dated action item at a time that the WBR can verify. I would not create a fourth seat for “special projects” — special projects become action items on the existing three.

How do you set goals for people who don't report to you?

I set the instrument, not the person. A named input line, a numeric target, a published grain, and a date the WBR will read it. At Flexport the wave-graduation criteria were the goals — verification SLA 90, data quality 90, ETA accuracy 80 — and a wave that missed one didn't board, regardless of whose org the miss sat in. At Sephora the Locus 98% pick-accuracy bar was the goal for a robotics vendor I did not employ. People respond to a number that can stop the program. They do not respond to a slide about “alignment.”

How do you run 1:1s?

Short, instrument-first, two blocks. Block one: their input lines — what's red, what's the cause, what's the dated ask. Block two: them — load, a skill they want, a decision they're stuck on. I learned this the expensive way: when I scored an RF-device task by effort instead of impact, the person doing it had no channel to tell me the script was going to fail on new hardware. A 1:1 that only reviews status is a status meeting wearing a nicer name. The 1:1 exists so the thing that will break next week has somewhere to land this week.

How would you hire for this pod?

I hire for the decomposition habit. Can they look at a red output and name the input, the owner, and the test that would falsify their cause — in five minutes, on a whiteboard, without a deck? I would rather a strong

operator who has run a WBR than a brilliant generalist who has not. The screen is a 30-minute case: “promise accuracy dropped three points last week — first hour.” Candidates who jump to a solution fail. Candidates who verify the number, factor the miss (seller × carrier × model), and refuse to act until they know which factor moved, pass.

How do you handle a direct report — or a pod owner — who disagrees with you?

The same way I flipped a roadmap: privately, with the math, before any room. If they have a better decomposition I take it and I say so in the WBR so the credit is public. If we still disagree after the evidence is on the table, I decide, they commit, and the next Monday’s number is the appeal. I have never won a disagreement by seniority and I have never lost one by updating my mind in public — the Flexport operator-first plan was already socialized when the cost-to-serve table killed it.

How do you develop a program manager?

Give them an instrument to own, coach the craft of the gate, publish their wins in the evidence. The QA lead on the agile conversion was the most skeptical person in the room; I gave her the definition-of-done — the load-bearing artifact — and coached how to write a criterion that can say no. Within two cycles she was defending the gate to engineering herself, and the ceremony survived my leaving because it was hers. Development that depends on me staying is not development.

How do you delegate?

I delegate the lever, not the task. The person who owns I3 (handling adherence) owns the decomposition of I3, the action item on I3, and the Tuesday sentence about I3. I do not delegate “please look into late shipments.” I keep: the output block, the action-item log, the speed-versus-accuracy dial, and any call that spends customer trust. Everything else is a named input with a named owner. Undelegate-able is a short list on purpose.

How do you handle underperformance in a matrix?

Evidence, then a split diagnosis: capability, load, or unclear definition of done. Each has a different fix — pairing, rebalancing, written acceptance criteria. The UDC issues log made this unemotional: an owner’s queue growing two weeks running triggered a conversation with the trend attached, and more often than not the fix was resourcing. Where it is individual I am direct early. What I will not do is manage performance by mood, or escalate a person before I have escalated the instrument.

What is your coaching philosophy?

Standards are the coaching. A published gate, a published tolerance, a published WBR definition — then I coach people on how to hit them and how to write the next one. I do not coach attitude. I coached the QA lead on how a gate survives pressure; I coached Flexport’s wave owners on how to hold a cohort that missed ETA accuracy in front of sales. The philosophy in one line: make the standard visible, then make the person the author of the control that enforces it.

How would you take over a pod that is already in flight?

Thirty days of not touching the machine. Sit the WBR, map each metric to its owner and its decomposition, meet the five sellers who refused SSS — the refusals teach more than the enrollments. Identify the one metric the room trusts least; that is the starting point, the same way it was at Flexport, Elementum, and Sephora. Then take the output block and the action log. A new leader who rearranges the org chart in week two is managing their anxiety, not the promise.

The honest gap — you have not been a people manager with a large direct-report org. Why should we trust you with a Senior Manager seat?

Because this seat is a pod of program owners measured on outputs, which is the job I have already done. I converted a 20-person team, ran 15+ tracks through a 15-month go-live, sequenced ~100 people on a visibility program, and ran onshore-offshore delivery teams with SOW ownership. What I have not done is a 12-person HR org with compensation cycles, and I will not pretend that is the same muscle. What I will do is hold named owners to published numbers, develop them by giving them the gate, and take the red as mine before anyone else finds it.

The gap is real. The adjacent strength is the actual job description.

Section 6 – Business acumen & strategy

How does Walmart make money on Marketplace?

Take rate on GMV, plus ads, plus fulfillment services (WFS), plus the halo that a broader assortment puts on the 4,600-store flywheel. SFS is the hard dollar: Walmart collects the take rate on inventory it never touches, but it also writes the delivery promise the customer will remember. Every basis point of conversion on a fast, kept promise is take-rate GMV; every broken promise is a contact, a refund risk, and a customer who discounts the next Walmart EDD they see. This role's P&L is conversion lift minus breach cost, scaled across seller-fulfilled GMV.

Describe the Marketplace flywheel.

Faster + more accurate promises → higher conversion → more seller GMV → sellers invest in SSS, better nodes, tighter handling → better inputs to the promise engine → faster promises. The counter-rotation is as real: one bad promise week → scorecard reds → suppressed listings → seller churn → thinner assortment → slower promises. My job is to keep the wheel spinning forward while the inventory stays in someone else's building.

Where would you invest a fixed quarterly budget?

Promise Yield per dollar, diminishing returns explicit. In the Solver-2 model the first dollars go to SSS default-on (adoption compounds into every other lever), then buffer recalibration (speed for free where σ is already small), then inventory-feed accuracy (kills cancellations, which are worse than lateness). Promise-engine cache work is last — it is real, but it is infra wearing a revenue costume and it saturates fast. I would not split the budget evenly across seven workstreams; equal split is how committees spend.

What is the biggest risk to SFS?

Trust, spent once. GMS lost to a slow promise comes back next quarter; trust lost to a broken one compounds. The operational version of that risk is SSS adoption stalling at ~40% of GMV so the majority of promise risk still rides on optimistic seller-declared handling. The competitive version is Amazon's SFP speed floors resetting customer expectations — 40% of standard-size page views at 1-day, 75% at 2-day as of July 2026 — whether Walmart likes the curve or not.

How do you think about marketplace growth versus customer trust?

They are not a trade if you grade the pod on Promise Yield. Growth-only scores penetration and will expand eligibility into cohorts that cannot execute; trust-only scores accuracy and will pad every buffer until conversion dies silently. Promise Yield = fast share × accuracy makes the fake improvements — padding, cliff-chasing, easy-lane mix shift — fail the KPI. I would rather grow slower on kept fast promises than grow faster on promises we underwrite as losses.

What would you kill?

Two things. First, any eligibility expansion that is not gated on a seller-lane cell's 7-day breach rate — that is how you manufacture O1 reds two quarters after you celebrate O3. Second, the 30-day-only scorecard as the sole enforcement clock. Keep it as the policy floor; add a 7-day Andon so a degrading cell does not burn customers for a month. I would not kill the accuracy floor to buy speed. That is borrowing at loan-shark rates.

What does good look like in 12 months?

Promise Yield up — illustratively from ~41.5% toward 50% — with mix-adjusted twins so we did not get there by shifting volume to easy metros. SSS GMV share compounding, 30-day retention holding. Buffer policy reduced to one governed k-dial, re-solved for Q4 in October, not November. One COE written for every accuracy incident above a published bps threshold, each ending in a mechanism change. And a standing SFS→WFS funnel for cells that structurally cannot hit the fast band.

How would you know your program worked?

The WBR can audit the cause. An output moved, the named input moved first, the residual in the $O1 \approx I3 \times I4 \times \Phi(b/\sigma)$ decomposition shrank, and a mix-adjusted twin confirms it was not Simpson's paradox. If the input moved and the output did not, the lever map is wrong — that is a finding, and it is how the map stays honest. I will not accept a green quarter whose story is "a lot of things got better."

How does this role create GMS?

Three paths, in order of how fast they pay. (1) Fast-promise penetration on already-eligible items — buffer waste harvested, conversion elasticity $\times$ volume, weeks not quarters. (2) SSS adoption — more GMV under Walmart-managed transit, which both speeds the promise and protects the scorecard, compounding. (3) Eligibility expansion that has earned the badge — slower, only after (1) and (2) have made the new cells executable. Path 3 without 1 and 2 is how you buy GMS with trust.

What is the cost of a broken promise?

WISMO contact + refund / return risk + the discount customers apply to the next EDD they see. In the Solver model I priced the bundle at \$12 against a \$6.50 contribution margin; at those numbers the profit-optimal policy holds 97.8% accuracy — *above* the 95% floor. That is the point: either the floor is set below what the economics want, or the \$12 is too high, and finding out which is a governance decision this role should force with Finance. An unpriced breach makes the speed-versus-accuracy argument theological.

How does Amazon think about this problem, and what would you copy?

Amazon converted seller-fulfilled adoption into an insurance contract. Late delivery does not count against OTDR only if Automated Handling Time, Shipping Settings Automation, and an OTDR-protected Buy Shipping label are all on — all-or-nothing accountability transfer, and they still will not excuse a late warehouse handoff. Listing-level deactivation at OTDR < 90% (Feb 2026) is surgical enforcement at the grain where the variance lives. I would copy the contract structure and the grain. I would not copy Amazon's churn tolerance — Walmart Marketplace is still in assortment-growth mode, and WFS is not yet FBA.

Where is Walmart's unfair advantage versus Amazon on seller-fulfilled delivery?

Four thousand six hundred stores. Store-adjacent same-day, returns density, local courier integration — Amazon cannot match that network. A roadmap that only clones SFP leaves the stores on the table. Cutoff arbitrage at store-as-node, and a returns experience that is easier than Amazon's, are Walmart-native levers. I would spend my first 90 days finding which of those the promise engine actually consumes today, because I would bet the answer is "less than it should."

What would you do with a seller who is strategically important and operationally weak?

Do not buffer the whole lane to hide them — that taxes every customer on that lane. Surgical: pull the fast badge at the seller-lane cell, keep them in the standard band, put a named Seller Ops owner on a 30-day

handling plan with a published return-to-badge bar, and offer the SSS bundle as the off-ramp (Walmart takes carrier risk, they give up declared handling). If they still cannot execute, the honest conversation is WFS. Strategic does not mean exempt from the customer promise.

How do you think about take rate versus experience investment?

Take rate is the harvest; experience is the orchard. Raising take rate on a marketplace whose promises are slipping is extracting from a shrinking base. I would fund SSS onboarding, buffer recalibration, and feed accuracy out of the conversion lift they produce — the Solver-1 model shows ~\$20M/year of profit from buffer policy alone on an illustrative million-order book — and I would fight any take-rate increase that lands on sellers we are simultaneously asking to adopt SSS. You cannot tax adoption and buy it at the same time.

If you had to pick one number to put on your own scorecard, what is it?

Promise Yield. Fast-promise penetration $\times$ promise accuracy. It is the only single number that refuses to let me game speed against trust. I would still report the factors underneath it every week — a composite that cannot be decomposed is a hiding place. But if you ask what I am willing to be fired for, it is that joint.

Section 7 – Case / whiteboard prompts

Promise accuracy dropped 3 points last week. Walk me through your first hour.

Do not touch the buffer. (1) Verify the number — grain, denominator, cancelled orders leaving the base, mix shift into hard lanes, pipeline freshness. I have seen more red weeks caused by measurement than operations. (2) Factor: $O1 \approx I3 \times I4 \times \Phi(b/\sigma)$. Compute each factor's WoW delta; the biggest mover names the family. (3) Segment: one carrier, one region, one seller cohort, or everywhere? A Southeast-only miss after a hub consolidation is C-family; a miss everywhere with I3/I4 green is model or buffer. (4) The room leaves with a cause, an owner, and a date — not a theory. At Flexport a 78%-manual week looked like lazy operators and was a carrier format change.

Fast-promise penetration has been flat for two quarters. What do you do?

Penetration is a chain: eligibility $\times$ (handling + predicted transit + buffer ≤ 2 days). Work it cheapest-first. (1) Buffer Waste Index — mean days delivered early on kept promises, ranked by BWI $\times$ volume. That list is the trim roadmap; in the illustrative model, moving from a uniform 1.10-day buffer to σ -proportional bought +32 points of fast share at the same 95% accuracy. (2) Handling-time truth — declared vs observed; ML-recommended lag. (3) Transit-model MAE. (4) Eligibility last, and gated — expanding into cells that cannot execute is how you buy O3 with O1. Also mix-adjust: flat penetration with improving metros and worsening rural is not flat, it is a mix.

Design the WBR for this team.

One page. Outputs on top: Promise Yield, then O1 accuracy, O3 fast share, O2 OTD, O4 late shipment, O5 cancel, O6 WISMO — 6 weeks + WoW + RAG, hard reds on policy floors. Inputs underneath in three tabs matching the pod: Speed Programs (I1, I8, I10), Promise Engine (I2, I5, I6, I9), Seller & Carrier (I3, I4, I7). Every input has a RollUp column to the output it drives. Action log with owner + date + age. Call Outs are pre-wired before Tuesday — a red that is discovered in the meeting is a process miss. Mix-adjusted twins sit next to raw outputs so Simpson cannot hide. I have run this cadence; I do not need to invent it.

A seller cohort is at 20% late shipment. Go.

20% is not a coaching problem, it is a control-loop problem. (1) Confirm measurement — first-scan vs tender vs movement; phantom labels. (2) Split the 20%: handling miss vs carrier pickup miss vs capacity spike vs weekend-blind handling. (3) Immediate customer protection: pull the fast badge on that cohort's cells within 24 hours (Andon), widen buffer on remaining standard promises so O1 does not eat the miss. (4) Seller conversation with the specific defect evidence, not a generic scorecard letter. (5) Return-to-badge bar published: 7-day late-shipment back under 5% for two consecutive weeks. Do not buffer the rest of the lane to hide 20% of sellers.

We want same-day on seller-fulfilled in three metros. How do you scope it?

As a go-live, not a feature. Entrance: node coverage, cutoff times that actually recover same-day given the intraday order curve, sellers who have demonstrated same-day handling on observed (not declared) data, carrier pickup windows that exist on the calendar including Saturday. Trial gate analog to Amazon SFP: 100 packages at full standards, badge withheld during trial, no peak enrollment. Shadow the promise for four weeks — log it, do not display it — and promote only if accuracy clears 95% and WISMO does not rise.

Hyperscore with numeric exit. The metros I would pick are the ones where store-as-node coverage already exists; I would not pick the three largest GMV metros if their σ cannot support same-day.

Leadership wants +5 points of fast-promise penetration this quarter. How do you answer?

I price the ways. Path A: harvest Buffer Waste — σ -proportional buffers, likely most of the five points on already-eligible volume, accuracy held. Path B: eligibility expansion — faster headline, accuracy risk two quarters later unless gated. Path C: handling-time truth + cutoff arbitrage — real, slower. I walk in with the three paths costed on Promise Yield, not on penetration alone, and I ask which loss ratio they are willing to write. A +5 that is all Path B is not a target I will commit to without an Andon in front of it.

The transit model MAE improved but promise accuracy got worse. What's going on?

MAE is the middle of the error distribution; breaches live in the tail. A model can fit the median better and get the p90 wrong — especially if you trained on MAE and the loss function is not a quantile at the accuracy target's implied percentile. Second possibility: mix. MAE fell because easy lanes gained volume while a fat-tailed rural set quietly degraded. Third: buffer was cut on the MAE news before the tail was checked. I would put p90 error and buffer-exceedance rate on the WBR next to MAE this week, and I would not let I5 (MAE) be the only model health line.

A VP says SSS adoption is a Product problem, not Tech Ops. Respond.

SSS is an incentive-design problem whose output is a Tech Ops number. Product can ship the toggle; Tech Ops owns whether enrolled GMV actually produces faster kept promises — which is I1 rolling up to O1 and O3. If Product owns the feature and nobody owns the insurance contract (observed handling + accountability transfer + priced GMV upside in Seller Center), adoption stalls at 40% and my output block stays red. I will partner with Product on the surface and I will own the contract, the scorecard protection rules, and the 30-day retention line. Features without the contract are how we got to 40%.

Show me how you would set k this week.

k is the single dial in $b = k \cdot \sigma$. I set it so the order-weighted $\Phi(k)$ equals the accuracy floor — for 95% under normal errors, $k = 1.645$. I do not set it per lane. I do re-solve it when the σ regime changes (Q4 multiplier ~ 1.45 in the model; frozen summer k broke the floor, accuracy fell to 91.9%). Production uses the empirical quantile, not the normal, and k is published in the WBR header every Monday so the business sees the trade. Shadow for four weeks before the displayed promise uses it.

Finance asks whether we should raise the accuracy floor from 95% to 97%. Your recommendation?

I would not raise a floor by fiat. I would run the profit model: at \$12 breach cost and \$6.50 margin, the unconstrained optimum already sits at 97.8% — so the economics are already asking for ~ 97 , and a 95% floor is slack. If Finance believes breach cost is lower (say \$6), the optimum falls and a 97% floor would spend conversion for accuracy the customer is not paying us to hold. Recommendation: price the breach with Finance this quarter (contact cost + refund rate $\times$ refund \$ + a measured LTV discount from A/B on broken-promise customers), then set the floor at or slightly below the economic optimum so the constraint is an emergency brake, not a substitute for the pricing.

Section 8 – Questions he should ask

Ask two. Three if they offer. These are ranked.

- 1. Which output metric is red right now — and when it went red, how long did it take to know which input caused it?** Why it reads senior: you think in decomposition, and you are diagnosing their operating system, not fishing for gossip. Whatever they name, you have the matching family in the issue tree.
 - 2. Where does the buffer decision actually live today — DS model, policy table, or negotiation — and who owns the speed-versus-accuracy dial?** Why it reads senior: you already know the failure mode (it lives everywhere, which means it lives nowhere). Their answer is your opening for the one-dial governance story.
 - 3. What does the SSS enrollment funnel lose the most sellers to — fear of cost, loss of carrier control, or distrust of the scorecard?** Why it reads senior: you are treating adoption as an incentive problem with a named drop-off, not a feature-marketing problem.
 - 4. How is this pod measured at six months — Promise Yield, GMS, SSS share, or something else?** Why it reads senior: you refuse to start a job whose scoreboard you have not seen.
 - 5. How many program/ops owners sit in the pod, and who owns I3 versus I4 versus I6 today?** Why it reads senior: you are already placing named levers on named people.
 - 6. How does Tech Ops sit between Promise Optimization (DS), Product, and Seller Ops — who can stop a launch?** Why it reads senior: you are mapping the real org chart, the one with vetoes.
 - 7. What happened the last time a speed-program launch missed its accuracy floor?** Why it reads senior: you are asking whether gates have teeth.
 - 8. How is Q4 promise policy set — and when is it published to the business?** Why it reads senior: you already believe peak is a different statistical regime.
 - 9. What share of seller-fulfilled GMV is on SSS today, and what did 30-day retention look like on the last two enrollment waves?** Why it reads senior: you pair throughput with retention; you will not celebrate a vanity enrollment number.
 - 10. Where does WFS conversion sit relative to this team's goals — is a cell that cannot hit the fast band our problem to migrate, or someone else's?** Why it reads senior: you are willing to hand an adjacent org a ranked funnel instead of defending SFS share.
 - 11. What does success look like at 12 months for the person who gets this seat?** Why it reads senior: you are interviewing them back.
 - 12. If I took this job, what would you need me to be right about in week six that a new hire is usually wrong about?** Why it reads senior: you are asking for the hidden curriculum — the thing that is not in the JD.
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Section 9 – Risk flags and how to defuse

No large direct-report headcount. Reframe: this seat is a pod of program owners measured on outputs. I converted a 20-person team, ran 15+ tracks, sequenced ~100 people, and ran onshore-offshore delivery with SOW ownership. I will not invent a 12-person HR org. I will hold named owners to published numbers.

Never worked at a marketplace. Reframe: I have sat on every side of the promise — retailer (Sephora ATS/promising), logistics platform measuring carriers (Flexport, Elementum), OMS consultant on the handshake (Manhattan). Marketplace is those three in one P&L. The motion I have not done is take-rate and seller acquisition; I will learn those from the business in days 1–30. The motion I have done is the one this JD names.

No Walmart systems experience. Reframe: I would not arrive with a target. I would arrive with a method. First thirty days: the five numbers this team runs on, who owns each definition, which number the room trusts least. I learned WMS migrations by running one and robotics by landing one. Walmart's stack is a 30-day problem; the operating system is the career.

Five companies in fifteen years. Reframe: look at the shape, not the count. Every move went toward a bigger, more physical operations problem. Sephora is three-plus years and the biggest program on their slate. I stay where the problem is big and physical. Marketplace SFS is that problem at the scale that is not a stepping stone.

Not an ML practitioner. Reframe: I am the person who governs the ML, not the person who trains it. I have owned the loss-function conversation (MAE vs tail), the buffer as a policy instrument, and the shadow-mode promotion bar. Promise Optimization DS builds the model; I own whether its output is allowed to touch a customer. That is the correct split, and it is the one I ran with data-science partners at Flexport.

No formal same-day / last-mile ownership. Reframe: same-day is a go-live with a cutoff calendar, not a last-mile network I have to have built. I have owned cutoffs, handling windows, and peak regimes. Store-as-node same-day is a Walmart-native asset I would learn from the network team; the promise math and the launch gates are mine on day one.

Consulting years at Manhattan and MetricStream. Reframe: Manhattan was distribution OMS — allocation, promising, the OMS–WMS handshake — on the client's peak, including a 4 AM Cyber Monday. MetricStream was SOW ownership and onshore-offshore delivery. Neither was slideware. Both are why I can sit with a warehouse associate in the morning and a VP in the afternoon.

Enterprise-retail scale versus marketplace scale. Reframe: Sephora is a closed inventory system; Marketplace is an open one. The closed system taught me promising governance (what “sellable” means). The open-system analog is SSS and seller scorecards, which I have the adoption-wave muscle for from Flexport (7→68, gated) and the lane-analytics muscle for from Elementum (1,000 routes, 60→95% completeness). Scale of GMV is larger here; the mechanism is not foreign.

You have a model with illustrative data. What if our real σ looks nothing like yours? Reframe: then k changes and the ranking of lanes changes. The argument that does not change: buffers should be risk-proportional, the last buffer-day should buy the same accuracy everywhere, and a floor that disagrees with the economics is a governance finding. The parameters are yours to correct. I built the model so we could have that argument with numbers instead of adjectives.

Why would you leave a live omni-DC program at Sephora? Reframe: I delivered what I went there to do — the WMS migration, the omni DC, the promising governance are live. I am not running from anything. I am running toward a basis point that moves millions of orders instead of thousands.

Section 10 – The 30-minute clock

Minute by minute

Minutes	What happens	Your job
0–2	Rapport, their framing	Listen for the pain word — accuracy, adoption, peak, SSS. That word picks the Part 4 proposal you will spend.
2–5	“Tell me about yourself”	90-second TMAY. Land: three sides of the promise — retailer, logistics platform, data layer.
5–20	Their questions	Defaults: F4 (tail not brochure), A1 (wave gates), A2/ A3 (the WBR). One answer must include the free-lunch line: σ -proportional buffers, +32 pts of fast share at the same 95%.
20–25	Your questions	Q1 (what’s red, how long to attribute) and Q2 (where the buffer decision lives).
25–29	Close	90-day plan in 45 seconds, then the closing line.
29–30	Logistics	Next steps, thank you, out clean.

Three stories to lead with: F4 (promise at the tail), A1 (adoption gated), E1 (policy before config). **Reserves:** B2 (roadmap flip — if they probe influence), C3 (Locus throttle — if they probe gates/supervisory), G1 (UAT crash — if they ask for a failure; do not volunteer it). **Two must-ask questions:** (1) which output is red, and how long to know the input; (2) where the buffer decision lives and who owns the dial. **30-second close:** “The delivery promise is an insurance policy Walmart writes on sellers it doesn’t control. I have priced that risk at the lane, gated the adoption, and run the weekly review that keeps the loss ratio honest. Ninety days: one output metric moved by a mechanism your WBR can audit. I’d like to do that here.”

If it goes sideways

- **Interviewer arrives late / short on time:** thesis → F4 in 90 seconds → 90-day plan → question 1. Ten minutes, complete.
- **Interviewer is quiet / lets you drive:** run the clock yourself in the order above. Quiet interviewers reward structure; do not fill silence with a second story they did not ask for.
- **Interviewer only wants supervisory:** honest gap script (Section 5 last question), then bridge to C3 — throttling your own robots at 94% against a 98% bar is what pod leadership looks like when the gate has teeth.
- **They challenge the math:** Part 5.3 humility, then hold the structure: “the parameters are yours to correct — risk-proportional buffers survive any parameter set. What would you like to change first, σ or the breach cost?”
- **They only want to talk Amazon:** copy the insurance contract (AHT + SSA + Buy Shipping bundle), do not copy the churn tolerance, and name the store network as the thing Amazon cannot match. Then ask question 3 — SSS funnel drop-off — so the rest of the time is about their sellers.

The three sentences that must get said

1. The promise is underwriting; the buffer is priced risk.
2. Risk-proportional buffers buy ~30 points of fast-promise share at the same accuracy — I built the model.

3. Ninety days: one output metric moved by a mechanism the WBR can audit.

Section 11 – Amazon Leadership Principles (full set, Walmart-mapped)

Ex-Amazon interviewers will not say “tell me about ownership.” They will say “**Tell me about a time...**” and then drill **What did YOU do? Why not X? How did you measure? What would you do differently? Walk me through the data pull.** Every LP below has a story tag from the pack (A=Flexport visibility, B=Flexport FF App, C=Sephora UDC, D=Sephora JDA, E=Sephora S/4HANA, F=Elementum, M=Manhattan, G=failures, I=independent build). Spoken answers are ~90 seconds. Do not stack two stories unless they ask.

Customer Obsession

Tell me about a time you obsessed over a customer problem the rest of the room had normalized. (F4)

The customer was a planner who asked me for a lead-time-variability report. The room had normalized carrier-brochure lead times — one optimistic number per lane, never measured. I treated the brochure as a lie the customer was paying for in inventory. I pulled shipment history across 200 lanes, built mean / spread / tail per lane, and moved every promise to the tail, not the average. Result: 99% lead-time accuracy held, and because measured spread was *smaller* than assumed on most lanes, 580,000 units and \$1.5M of safety stock came free. Then a drift flag per lane so a widening spread triggered review instead of silent re-planning.

The customer of this role is Maya on the PDP and the seller who has to execute. Obsessing over one and ignoring the other is how you pad buffers or burn sellers.

Give me an example of sacrificing an internal metric to protect the customer. (C3)

Two weeks into Locus robotics, bots were 94% pick-accurate against a 98% gate. The internal metric that wanted to be green was “automation live at go-live.” I throttled robots to ~30% of volume, kept the customer promise on manual process, fixed zone logic, restored in week three. Processing efficiency ended +15%, zero customer incidents.

Follow-ups. *What did YOU do?* I published the 98% gate before the bots arrived, made the throttle call, sat the zone-logic debug. *Why not push through?* A gate that cannot stop the program is decoration; I will not ship a customer-facing miss to protect a robotics narrative. *How did you measure?* Pick-accurate throughput vs the pre-published 98%; customer incidents = zero. *Differently?* I would have shadowed bots on a single zone for a week before they touched 100% of volume. *Data pull?* WMS pick-confirm vs gold-standard location, by zone, by hour — not a building average.

Ownership

Tell me about a time you owned a problem that was not in your job description. (F4)

Same 200-lane story — the ask was a report. Nobody owned the seam between brochure lead times and the inventory plan. I took it, measured it, productized it, pitched leadership with mockups. 99% LT accuracy, \$1.5M released. Ownership is taking the seam, not writing the ticket.

Tell me about a time you delivered bad news that was yours. (G2)

JDA training sign-off — a go-live exit criterion — slipped 1.5 weeks. I had scored the RF-gun conversion as low-effort. My own pre-training test caught the upgrade script failing on newer hardware; COVID night shifts killed the recovery window. I told leadership: training slips 1.5 weeks, my miss. Program still landed on time, ~3% under the \$1.8M budget. The rule: a single-point dependency in front of an exit criterion gets a what-if plan regardless of size.

Follow-ups. *What did YOU do?* I ran the device test, made the call to slip, wrote the new rule. *Why not blame devices?* I had accepted their estimate without a what-if; the miss is the scoring, not the script. *Measure?* Training sign-off date vs plan; budget 3% under; go-live still on the 8-month clock. *Differently?* Dual-path conversion on day one of the devices workstream. *Data pull?* Device model × OS × script pass/fail table before training week.

Invent and Simplify

Tell me about a time you simplified a complex process. (A3)

Flexport's Monday review ran on four systems stitched in Excel by me — two hours a week, 60% system accuracy, single point of failure. I wrote the BRD, froze seven metric definitions as the team already used them, priced doing nothing at >\$100K/year in the infra team's own rates, gated cutover on a parallel run (manual vs pipeline within 1% for consecutive weeks). Four systems → one Snowflake pipeline, accuracy 60% → 90%. First automated catch: a 4% ticket-sync drop the manual era never saw.

Invent and Simplify here is Promise Yield and one k-dial — not a new dashboard. New metrics without killing old ones is complexity wearing a badge.

Tell me about inventing a mechanism, not a slide. (F3)

Planners learned about shortages when the line stopped. I did not ship a threshold email. Forecast accuracy 70% → 85% at SKU-location-week, coverage-math alerts with a 30-second reason line, false alarms burned down in public. Live in three months; ~\$2M safety stock out; the alert entered the weekly S&OP because it was trusted.

Follow-ups. *What did YOU do?* BRD, frozen definitions, priced status quo, parallel-run gate. *Why not a nicer Excel?* Excel was the single point of failure; the invention was the frozen contract plus the pipeline. *Measure?* Accuracy 60→90; hours/week; first catch 4%. *Differently?* Security-review lead time on the kickoff checklist from day one (G3 taught me that). *Data pull?* Seven frozen definitions; week-by-week parallel delta.

Are Right, A Lot

Tell me about a time you were 70% sure and still decided. (M1)

Cyber Monday, 4 AM. VP call: backorder lines piling up, millions at risk, full diagnostic an hour away. I triaged on what we had: rank by promise-at-risk, rebalance allocation across three mega-DCs, one bridge, one log, every decision timestamped. Trace later: allocation scheduler died at the daylight-saving change — inventory was fine. Fill rate held 99%, ~\$8M protected. 70% is enough when waiting is a customer miss.

Give me an example of being wrong, and how you knew. (E1)

S/4HANA in-transit rules: I forbade channels from promising against units on a truck. Felt safe. Starved dotcom of sellable units. Data showed the miss; fix went through the signed policy — a defined in-transit promise window per lane from real lane data. Conservatism has a price, paid in the same currency as risk — that is buffer waste in inventory clothes.

Tell me about changing your mind because of data. (F2)

Austria: noise said carriers were late. I pulled three years of shipments. True data quality ~55% at the measured grain vs 40% the reports showed. Program flipped from “fix carriers” to “fix measurement.” Quality to 99% by defect class; ~\$8M freight savings, finance-signed, auditable to a shipment.

Follow-ups. *What did YOU do?* I pulled the history myself; I did not send an analyst. *Why not trust the reports?* Reports were at the wrong grain (F2). *Measure?* Completeness by defect class; savings method signed before a dollar claimed. *Differently?* Start every new domain with a measurement audit in week one. *Data pull?* Shipment-level: rate present, lane key, carrier spelling, transit source (brochure vs scan).

Learn and Be Curious

Tell me about learning something you had never done. (D1)

Sephora handed me a dual-DC WMS migration — I had never run one — hard eight months, \$20M pipeline gated behind it, nothing documented. I did not fake experience. I sat the sponsor for objectives/assumptions/success criteria, held discovery’s written exit until every flow had a current state, then entrance/exit criteria every phase. Three mocks against 0.5% reconciliation tolerance published *before* mock one: 7% → 1.8% → 0.3%. On time, ~3% under \$1.8M budget.

Tell me about teaching yourself a new stack because the job needed it. (I1)

I built an LLM triage agent (Python, FastAPI, tool-use) with the eval set *before* the agent. Every prompt/model change gated on the known past still passing. Cost from \$25/incident → \$0.40 (98% cut) with accuracy held and 100% citation coverage. Same operating system: instrument first, then let the scorecard teach.

Follow-ups. *What did YOU do?* I wrote the tolerance with finance before mock one; I read the 7% as a scorecard, not a mood. *Why not hire a WMS specialist and hide?* I did use specialists; I owned the gate. *Measure?* Mock miss 7→1.8→0.3 vs 0.5% line; date; budget. *Differently?* I would have dual-pathed RF devices from day one (G2). *Data pull?* Unit-level reconciliation: expected vs actual inventory by location after mock.

Hire and Develop the Best

Tell me about developing someone. (D3)

QA lead was the most skeptical person in the agile conversion. I gave her the load-bearing artifact — definition-of-done with entry criteria — and coached how to write a gate that can say no. Within two cycles she was defending it to engineering. Ceremony survived my leaving. Development that depends on me staying is not development.

How do you hire? (pod screen)

30-minute case: “promise accuracy dropped three points — first hour.” Jump-to-solution fails. Verify the number, factor seller × carrier × model, refuse to act until they know which factor moved — that is a hire.

Follow-ups. *What did YOU personally coach?* How a criterion survives pressure; she authored the content. *Why not just tell QA the cadence?* They would have complied and then routed around it. *Measure?* Conversion in six weeks; neighboring teams copied the DoD. *Differently?* I would have put her name on the WBR line from week one. *This role's gap:* I have not run a 12-person HR org with comp cycles. I will not invent that. I have developed owners by giving them the gate.

Insist on the Highest Standards

Tell me about a time you held a bar when it was expensive. (A1)

Flexport visibility: six published graduation criteria per wave (data quality $\geq 90\%$, milestone/ETA accuracy $\geq 80\%$, verification SLA, engagement, stability, CSAT). I held wave three in front of everyone on an ETA-accuracy miss from one bad source. Sales wanted the pool boarded. We landed 68 MACs two months early, CTS \$0.66 → \$0.03, ticket response 72% → 90% while volume grew 5×.

Give me an example of a numeric gate with teeth. (C3 / D2)

Locus 98% pick-accuracy; JDA 0.5% mock tolerance. Both could stop the program. 94% got throttled; 7% mock did not go live.

Highest standards here is k published before the quarter, Andon in 24 hours, shadow four weeks. Standards without a stop-the-line are posters.

Follow-ups. *What did YOU do?* I published the six criteria; I held the wave; I named the source. *Why not a private hold?* Public hold teaches the mechanism; private hold teaches politics. *Measure?* Criteria table per wave; CTS; tickets; 68 vs target. *Differently?* I would have put the bad source's kill switch in the platform earlier. *Data pull?* Per-customer ETA accuracy vs the 80% line, by source.

Think Big

Tell me about a time you thought bigger than the ask. (F4 + Promise Yield)

The ask was a report. The product was lane-level promising as a system. In this seat the equivalent is: do not “improve O3.” Install Promise Yield so speed and accuracy cannot be gamed apart, one k-dial so buffer is a policy instrument, and an SFS→WFS funnel for cells that structurally cannot hit the fast band. Think Big is the mechanism that still works at 10× volume, not a bigger slide.

Tell me about a multi-year bet you sequenced. (B1/B2)

Inherited 18-month operator-first launch. Cost-to-serve math said partner-first: ~50% of CTS vs ~25%, value in 6 months not 18. Flipped: 66% less engineering effort, \$5M avoided. Think Big without sequencing is a slogan.

Follow-ups. *What did YOU do?* Two-column CTS table; private pressure-test with my manager two days; Bellevue onsite ratified. *Why not both in parallel?* We did not have the engineers; sequencing is the bet.

Measure? Effort –66%; \$5M; time-to-value 6 vs 18 months. *Differently?* I would have priced breach/conversion the same way I now price buffers — explicit economics, not only CTS.

Bias for Action

Tell me about a time you moved before you had perfect information. (M1)

4 AM Cyber Monday — covered above. Bias for Action is a timestamped decision plus a post-mortem that makes silence an alarm (scheduler heartbeat, DST checklist, rehearsed rebalance). Action without a mechanism is heroics; I do not want a second 4 AM.

Tell me about a two-way-door decision you made fast. (C3)

Throttling robots to 30% was a two-way door: the split was a dial we had built. I turned it the same day 94% printed. One-way doors (cut over JDA, display a new buffer policy to customers) get mocks and shadow.

Follow-ups. *What did YOU do?* Throttle call, war-room, restore on re-measure. *Why not wait for a committee?* The gate was pre-authorized; waiting was a customer-incident bet. *Measure?* 94% → 98%+; week-three full volume; zero customer incidents; +15% processing. *Differently?* Shadow one zone first. *Data pull?* Hourly pick accuracy by zone vs 98%.

Frugality

Tell me about accomplishing more with less. (A1 / B2 / A3)

Three currencies. Flexport CTS \$0.66 → \$0.03 while scaling 7 → 68. Roadmap flip: \$5M avoided, 66% less effort. Snowflake: killed two hours/week of hero Excel and a \$100K/year status-quo. Frugality in this role is **buffer waste** — days of pad the customer paid for in slower promises. Elementum: measured spread < assumed spread → 580K units / \$1.5M free. Same muscle.

I do not confuse cheap with under-instrumented. The WBR is cheap because definitions are frozen, not because we skip measurement.

Follow-ups. *What did YOU do?* Wave gates that cut support load; CTS table that killed 18 months of the wrong build; BRD that funded the pipeline. *Why not more headcount?* Headcount would have scaled the waste. *Measure?* CTS, effort, pipeline accuracy. *Differently?* Put Buffer Waste Index on the WBR in month one here — it is the frugality metric of promising.

Earn Trust

Tell me about earning trust when you had none. (Manhattan OMS–WMS; F1)

Manhattan: OMS–WMS handshake discrepancy, a couple hundred orders, ~\$2.5M. Manager instinct: correct quietly. I argued full transparency — show the customer, let them prioritize, bulk-patch the class. Uncomfortable. They protected the orders that mattered to *their* business; we later won a larger initiative. Silent corrections are loans against trust.

Elementum routing guide: skeptics were often right about specific lanes. Value-phase highest-spend lanes first behind a data-quality toll; overrides with reason codes (data, not sabotage). Completeness 60% → 95%.

Tell me about being transparent with a bad number. (D1 / A2)

Mock one at 7% went up unvarnished with my name on it. A truncated week that printed 99.4% automation got discarded because a VP steering on three data points is a decision-quality problem.

Follow-ups. *What did YOU do?* I put the discrepancy on the table; I discarded the flattering week. *Why not quiet-fix?* Trust compounds; quiet-fix spends it. *Measure?* Customer still there; larger initiative won; 7% → 0.3% on a published line. *Differently?* I would have sampled OMS–WMS handshake daily in hypercare from day one. *Data pull?* Order_id in OMS vs WMS, \$ at risk, aged hours.

Dive Deep

Tell me about a metric that looked fine until you changed the grain. (F2 / A3 / Flexport 78%)

Austria: reports said 40% quality; at the planning grain it was ~55% and still not usable. Flexport: a week printed 78% manual — looked like operators; root cause was a carrier format change. Snowflake: 4% ticket-sync drop invisible in Excel. Dive Deep means I pull until the unit is a shipment, an order, a lane, or a seller — not a national average.

Walk me through the data pull on a red O1.

(1) Recompute O1 from promise_log $\bowtie$ delivery scan, delivered-only, order-week Mon–Sun. (2) Mix: same cohort weights as last week? (3) Factor I3 (order, first scan vs handling window), I4 (lane, not order), implied Φ . (4) Segment carrier $\times$ region $\times$ seller decile. (5) Stop when one cell explains the delta. I have seen more red weeks from measurement than from operations.

Follow-ups. *What did YOU do?* I wrote the SQL / sat the extract; I did not accept the dashboard. *Why not trust I4 as P(late)?* I4 is lane-grain; treating it as package survival double-counts geography (Part 2). *Measure?* Residual Actual O1 – I3 $\times$ I4 $\times\Phi$; contribution-to-change shares. *Differently?* Mix-adjusted twins on page 1 from week one (proposal 8).

Have Backbone; Disagree and Commit

Tell me about disagreeing with your manager. (B2)

Inherited operator-first 18-month plan, already socialized. CTS math said partner-first. I did not flip it in a meeting. He saw the math privately, pressure-tested with engineering two days, Bellevue ratified. 66% less effort, \$5M, 6 months vs 18. Backbone is the private table. Theater is the public ambush.

Disagree with me right now on SSS. (use in the room)

If SSS is a settings toggle, it will stall. Amazon's version is an all-or-nothing insurance contract: observed handling + transit authority + protected label, or you keep the risk. Walmart sample I1 is still ~45% of GMV after six weeks of compounding. I will not celebrate activations (I10) without 30-day retention, and I will not pad every buffer to hide the unenrolled 55%. Copy the contract; do not copy Amazon's churn tolerance —

WFS is not FBA yet. If you disagree after that, I will commit to *your* enrollment target and I will still put retention and Promise Yield on the same page, so the argument has a number next Monday.

Follow-ups. *What did YOU do?* Private math, not a slide war. *Why not go around him?* Because commit after the room is the LP; ambush is not. *Measure?* Effort, \$5M, time-to-value. *Differently?* I would have forced the economic disagreement (floor vs profit-opt 97.8%) into the same two-column format.

Deliver Results

What is the result you are proudest of? (A1)

7 → 68 customers, two months early, CTS \$0.66 → \$0.03, tickets 72% → 90% at 5× volume. Results with a rollback are not results. The gates are why it counted.

Tell me about delivering on a hard date. (D2)

JDA dual-DC, eight months, on time, on budget, mocks 7% → 1.8% → 0.3% vs 0.5%, unblocked \$20M ROI pipeline, 4-week hypercare with numeric exit.

Follow-ups. *What did YOU do?* Wave machine; mock scorecard; go/no-go. *Why not a heroic crunch?* Crunch is not a mechanism; mocks are. *Measure?* Date, budget ~3% under \$1.8M, pipeline start next month, hypercare exits. *Differently?* RF-device what-if (G2). *Data pull?* Reconciliation miss by mock; RAID ageing.

Strive to be Earth's Best Employer (if asked)

Tell me about making work sustainable, not heroic. (B3 / D3)

Compressed go-live: two programs, one engineering team, ~\$200K/month per DC. One RAID log, one 45-minute daily forum, two-pass war room so issues moved around the clock *without* anyone working 24 hours. Both landed; ~\$1M per DC protected. Heroics are a process miss. The QA-lead story is the people version: she owned the gate, so the cadence survived.

I will not claim a 12-person people-manager brand I did not hold. I will claim: named owners, published bars, 1:1s that surface the thing that breaks next week.

Success and Scale Bring Broad Responsibility (if asked)

Tell me about scaling without dumping the cost on someone else. (A1)

We scaled 10× while ticket response *improved*. The cost of scale was paid in gates, not in support's backlog and not in the customer's ETA. Here: SSS enrollments that dump breach cost onto Maya are not scale. Promise Yield and 30-day retention are how I refuse that.

Follow-ups. *What did YOU do?* Held wave three; paid the 50-container calibration toll. *Why not grow faster?* Headline without retention is someone else's page next quarter. *Measure?* Tickets 72→90 at 5×; CTS -95%.

Section 12 – Dashboard / WBR design (blank-page to VP)

Whiteboard-ready. Sample figures = Wk 28–33 illustrative WBR. Policy A/B/D = `model_reference.py`. Draw Part 8.5 while you talk.

12.1 Design the WBR / dashboard for this engagement from a blank page

Architecture (say in 2 minutes, then draw). Audience = Tuesday pod + any VP who walks in — same page. Cadence = Mon–Sun week, Monday freeze, Tuesday review. Page 1 grain = portfolio order-weighted with mix-adjusted twins under each output; dive-deep grain = seller × lane × program cell in owner tabs, never on page 1. Outputs on top (O1–O7 + **Promise Yield**), inputs underneath in three owner blocks (I1 / I8 / I10, I2 / I5 / I6 / I9, I3 / I4 / I7). RAG: $\pm 5\%$ relative WoW, plus hard reds O1<95, O2<90, O4>5, O5>2. Mix-adjusted twins kill Simpson. Promise Yield = $O3 \times O1$, Wk 33 sample $45.1\% \times 95.4\% = 43.0\%$, 4-quarter target 50%. Action log: owner, metric that will go green, date, age. I refuse on page 1: MAE histograms, seller leaderboards, p99 charts, GMV, NPS, experiment dumps — those are appendix. Challenger-vs-champion shadow line in the header once D1 (promise-at-display log) is airtight.

A WBR is a decision system. If a red can arrive without a cause, an owner, and a date, you built a viewing.

Follow-ups. What did YOU do at Flexport? I ran this grammar by hand on four systems, then Snowflake 60→90% so the cadence had boring data. Why not one metric? Because O1 and O3 trade; PY is the joint, factors still visible. How measured? Same definitions every Monday; grain and date on every number. Differently? Twins and Buffer Waste Index in the first 60 days (proposals 8 and 3).

12.2 Walk me through how you would instrument promise accuracy this week

Day 1: confirm `promise_log` has promised EDD, model version, buffer, node, lane, at **display** (not at ship). Day 1–2: join delivery scan; denominator = delivered orders, cohort = order date Mon–Sun — not delivery date (that mix-shifts). Day 2: I3 = first carrier scan vs configured handling window (scan, not manifest). Day 2: I4 at **lane-week**, not order. Day 3: implied $\Phi = O1 / (I3 \times I4)$; expect it can exceed 1 — I4 is the wrong grain to multiply as a survival probability (Part 2). Day 3: mix-adjusted O1 with last-quarter cohort weights. Day 4: 7-day CUSUM per seller-lane cell for Andon. Day 5: one Call Out in the WBR with the factor that moved. I have done the analog: Elementum 200 lanes from scans; Flexport freeze-seven-definitions then automate.

Failure mode this week: cancellations silently dropped from the base making O1 look better; or someone uses rolling 30-day O2 as if it were O1.

12.3 An output is green and an input is red – what do you do?

Do not celebrate. The output is delayed or buffered. Wk 32 sample: I4 89.6% **RED** (SE hub), O1 95.0% still above floor; **O4 5.2% RED** — seller clock, no customer pad. Action: contain the input (widen buffer or pull fast badge on the affected cells within 24h — proposal 5), dated owner on I4, verify next Monday that the *input* recovered, not only that O1 stayed lucky. If inputs are all green and output is red, the input set is incomplete — that is a finding, add the missing lever.

12.4 How do you stop teams gaming a metric?

(1) Joint KPIs: Promise Yield so padding or cliff-chasing fails. (2) Dual grain: order-weighted *and* page-view-weighted O3. (3) Mix-adjusted twins. (4) Scan not manifest for I3. (5) Pair I5 MAE with p90 / exceedance. (6) I10 = activations **and** 30-day retention. (7) Hard floors that cannot be WoW-washed. (8) COE on accuracy incidents $\geq X$ bps that ends in a mechanism change. Gaming is what happens when one line has an owner and its twin does not.

12.5 What is the north star vs the guardrails?

North star: **Promise Yield** $\rightarrow 50\%$. Guardrails: $O1 \geq 95\%$, $O2 \geq 90\%$, $O4 \leq 5\%$, $O5 \leq 2\%$, and I will not expand I8 eligibility into cells whose 7-day breach rate is already on fire. North star without guardrails is growth-on-trust-credit. Guardrails without a north star is padding forever.

12.6 Page-view-weighted vs order-weighted penetration – which and why?

Both, on the same page. **Order-weighted O3** is what operations executed and what conversion already selected. **Page-view-weighted** is what Maya *saw* — Amazon's SFP floors are % of page views (40% at 1-day / 75% at 2-day standard, Jul 2026 published). If metros convert and order more, order-weighted O3 can print green while mountain Mayas saw 7 days. Demand-weighted is the honest customer grain; order-weighted is the honest ops grain. I will not pick one so the other can hide.

ILLUSTRATIVE: 100 PDP views, 70 California-like, 30 mountain-like. If only Californians buy, order-weighted fast share looks like California. Page-view O3 still sees the 30.

12.7 How would you explain this dashboard to a VP in 90 seconds?

Use Part 8.5.6 verbatim. Land: Yield 43%, floors, ten named inputs, Wk 32 hub as proof the room contains *inputs* not stories.

12.8 What leading indicators give you 7 days of warning before O1 breaks?

I3 first-scan (same-day grain), I4 7-day CUSUM not 30-day scorecard, I5 p90 error (MAE can improve while the tail dies), I7 feed freshness, I9 cache-hit collapse (stale fast promises), I10 30-day retention on the last SSS wave, Buffer Waste Index falling while σ is rising (you already spent the pad). O6 WISMO is lagging but severity (days late) leads the next O1 print. O2 30-day is the *slowest* alarm — never operate on it.

12.9 Build the causal tree from I-levers to O-metrics on the whiteboard

Draw Part 8.5.4. Speak the identity: $O1 \approx I3 \times I4 \times \Phi(b/\sigma)$ as a flashlight for *family*, not a conservation law (implied Φ can exceed 1). $PY = O3 \times O1$. $O6 \leftarrow$ broken O1 + tracking holes. $O7 \leftarrow$ O3 via A/B elasticity. $I8 \rightarrow$ O3 only if I3/I4 can execute.

12.10 Your Promise Yield went up but customers are angrier – diagnose

Decompose PY. If O3 up and O1 flat, you added fast badges on pages that still *feel* late: (a) severity — kept-but-close misses and Sunday-vs-Friday are invisible to O1 binary; (b) mix — fast share gained on cheap SKUs while a high-intent cohort broke; (c) page-view vs order — orders got faster, views did not; (d) WISMO taxonomy / tracking gaps (O6 up while O1 green); (e) cancellations (O5) never enter O1. Pull O6 severity, O5, page-view O3, and mix-adjusted O1 before you touch k. Yield is necessary; it is not sufficient if the binary keep hides a nastier tail.

12.11 Excel Solver: show me how you'd set buffers

Open **Solver 1**. Speak the four-policy table (illustrative 12-lane, ~1M orders / week, $\varepsilon \sim N(0, \sigma^2)$):

Policy	Accuracy	Fast $\leq 2d$	Meaning
A uniform $b=1.10d$	95.8%	7.6%	Same pad on a commute and a mountain pass
B $b=1.645\sigma$	95.0%	39.6%	One dial; 95% floor exact under normals; +32 pts vs A
C min-days KKT	95.8%	22.4%	Last buffer-day buys equal accuracy on every lane
D profit opt (Solver)	97.8%	22.9%	Breach \$12 vs margin \$6.50 — floor is <i>slack</i>

Script: “I do not set twelve buffers in a meeting. I set k . This week $k=1.645$ if the floor is 95% and errors are roughly normal. Production uses the empirical quantile per lane; k still one number. Shadow four weeks (proposal 7). Peak: re-solve; frozen summer k broke the floor in the Q4 tab (accuracy 91.9% in the model). D vs B is governance: B maximizes the 2-day badge, D buys accuracy because breaches are expensive. I will not pick B in secret.”

Follow-ups. *What did YOU do?* I wrote `model_reference.py`, verified KKT spread $< 1e-15$, reproduced in Excel. *Why not per-lane ML buffers?* A hundred micro-decisions is a lobby; one dial is a policy. *Differently?* Empirical quantiles in production, normals on the whiteboard. *Data pull?* Per-lane σ from scan residual; order weights; $\Phi(b/\sigma)$.

12.12 What does a good action item look like vs a rotten one?

Good (from the workbook): “Recalibrate promise buffers on 42 Southeast lanes post hub consolidation. Owner: Promise DS. Created 2026-08-05. Due 2026-08-12. Success = O4 back under 5% *and* I4 $\geq 90\%$. Update: shipped Wk 33; O4 4.6%.” One owner, one date, two metrics that can go green, age visible.

Rotten: “Align with DS on accuracy.” No owner of a lever, no date, no verifying number, not deferral-in-writing. I ran UDC on 836 log rows over 14 months, 85.4% closed — the grammar is why.

Section 13 – Bar-raiser / stress / “are you L7”

These are designed to see if you are mechanism-level under heat. Answer in this order: **protect the customer this week** → **name the input** → **owner + date** → **what you will not do**. Resume numbers stay canonical. Marketplace arithmetic stays **illustrative**.

Peak week, accuracy down 4 points

Contain first: freeze non-essential promise changes; Andon on cells with 7-day breach $\geq 2 \times$ baseline ($n \geq 30$) — widen buffer or pull fast badge within 24h; do not wait for the 30-day scorecard. Decompose O1: I3 (seller handling under volume), I4 (carrier saturation), Φ (σ inflated — model $\sim 1.45 \times$ peak in the Q4 tab). If I3/I4 are the miss, do not “add 0.5 days nationally.” If Φ is the miss, you are in a regime change: that is proposal 10, which should have been published in October. I have run peak: Manhattan Cyber Monday, 99% fill, $\sim \$8M$ protected; Sephora UDC October–December freeze. After contain, one COE: why k was still summer. Next peak: seasonal k declared in advance; zero emergency buffer interventions Thanksgiving–Christmas is the success metric.

Follow-ups. *What did YOU do at peak?* 4 AM bridge, promise-at-risk rank, timestamped log, heartbeat after. *Why not pad everything?* National pad spends O3 to hide a cell problem; Andon is surgical. *Measure?* O1 vs 95% floor by day; cells Andon’d; WISMO. *Differently?* Publish peak k in October — don’t discover regime in November.

Legal wants to pad all buffers

Two-way vs one-way: displaying slower promises site-wide is a one-way-ish customer-trust door. I will not refuse Legal’s *risk*; I will refuse an unpriced blanket. Counter: σ -proportional k raised to a published quantile they pick (e.g. 99% → $k \approx 2.33$ under normals), not +1 day everywhere (Policy A is the cautionary 7.6% fast share). Shadow for two weeks; mix-adjusted O3 and PY on the page so we see the conversion we are buying Legal. Disagree and commit: if General Counsel still wants uniform pad, I commit, I log the PY we spent, I review in 30 days with data. Frugality + Customer Obsession: over-insurance is a tax on Maya’s Thursday.

A director wants eligibility +5 pts this quarter

Price three paths (Section 7). Path A: Buffer Waste / Policy B — **illustrative +32 pts** fast share at the *same* 95% on already-eligible volume; likely most of a +5 without I8. Path B: I8 expansion — headline, O1 risk two quarters later unless gated on cell 7-day breach. Path C: I2 handling truth + cutoff arbitrage. I will not commit to +5 that is all Path B without Andon. Size of prize: O7 $\times$ expanded sessions, minus expected breach cost. Two-way door: eligibility flag can turn off; I still want a trial gate (Amazon SFP analog: standards during trial, badge withheld).

Seller with 8% of GMV is the worst handler

Do not buffer the lane to hide them — that taxes the other 92% of customers on that lane. Surgical: pull fast badge at **this seller’s cells** within 24h; keep them in standard band; named Seller Ops owner; 30-day handling plan; return-to-badge bar = 7-day late-ship under 5% for two consecutive weeks. Offer full SSS bundle as the off-ramp (Walmart takes carrier risk; they give up declared handling). If they still cannot execute: WFS

conversation. Strategic ≠ exempt from Maya. Flexport analog: I held wave three on one bad source rather than boarding the pool.

DS model MAE improved but O1 fell

MAE is the middle; breaches live in the tail. Causes: (1) loss function is MAE, not a quantile at the floor's percentile; (2) mix — easy lanes gained volume; (3) someone cut buffer on the MAE news. This week: p90 error and buffer-exceedance on the WBR next to I5; mix-adjusted O1; freeze buffer cuts. I will not let I5 be the only model-health line. Eval-set discipline from my LLM agent: known past must still pass before the live present.

International expansion

σ changes shape: longer transit, fatter tails, customs as a variance spike. Measurement first — Elementum 1,000 routes, completeness 60% → 95%; nothing works until the scan pipe is honest. Equimarginal: many cross-border lanes cannot live in the fast band — promise slower, keep O1. New cells start padded (Amazon new-seller probation analog). Do not copy US k. WFS/SFS boundary is sharper: cells that cannot hit the band go to the conversion funnel (proposal 9), not heroic buffers.

Why should I hire you over an Amazon SFP internal transfer?

They have SFP scar tissue. I have the **operating system this team already uses** (Flexport input/output WBR), the **retailer promising governance** (Sephora S/4HANA ATS), and the **lane-tail math** (Elementum 200 lanes, 99% LT, \$1.5M). I will copy Amazon's insurance contract and listing-level grain. I will not copy FBA-backed churn tolerance, and I will use 4,600 stores — an asset an SFP transfer can undervalue. Honest gap: I have not sat SFP policy. Honest edge: I will not arrive trying to rebuild Walmart as a clone; I will arrive with k, PY, Andon, and a 90-day "one output moved by an auditable mechanism."

You have no marketplace on your resume

Correct. I have not owned take-rate or seller acquisition. I have sat every side of the *promise*: retailer (Sephora ATS), logistics platform (Flexport, Elementum 1,000 EDI routes), OMS handshake (Manhattan, 99% fill, ~\$8M). Marketplace SFS is those three in one P&L. Days 1–30 I learn *their* numbers, including the five sellers who refused SSS. I will not invent a marketplace logo.

This needs 2 years supervisory

Honest shape: MetricStream onshore-offshore delivery + SOW ownership; everywhere else **matrix/pod** — 20-person agile conversion, UDC 15+ tracks / 60+ stakeholders, Flexport ~100 people sequenced. I will not invent a 12-person HR org. This seat is named input owners measured on outputs — that is the job I already run. Proof of teeth: I throttled my own robots at 94% vs 98%.

What's the stupidest metric on this WBR?

I6 as currently defined — % of lanes with buffer in [p50, p90] of error. It is a sanity band, not a policy. It can be 70% "in band" while every lane is uniformly over-padded (Policy A). I keep it as a hygiene line for a quarter; I replace the *decision* with k and Buffer Waste Index. Second nominee: operating off **O2 30-day** as if it were O1. I would not delete O2 — sellers need a stable clock — I refuse to steer the week on it.

Kill one program today

Kill **ungated eligibility expansion** (I8 growth without cell-level 7-day breach gates). It manufactures O1 reds two quarters after the O3 party. Alternate: kill 30-day-only enforcement as the *sole* clock — keep it as policy floor, add 7-day Andon. I will not kill the 95% accuracy floor to buy speed.

Write the PR/FAQ for σ -proportional buffers (*working backwards*)

PR headline: Fast promises, same accuracy: one buffer dial. **Problem:** Uniform pads waste speed on tight lanes and under-protect noisy ones (illustrative A: 7.6% fast at 95.8%). **Solution:** $b = k \cdot \sigma$, k set to the accuracy floor ($1.645 \approx 95\%$ under normals). **Customer:** faster honest dates on metros; rural stays honest-slow. **Seller:** no new config if on SSS; observed handling still theirs. **FAQ:** What if σ is wrong? Shadow 4 weeks, backtest per lane. Peak? Re-solve k in October. Why not profit-opt D? D holds 97.8% — we will pick B vs D with Finance once breach cost is priced. **Tenets:** one owner of k ; never uniform days; never cut k on MAE alone; Andon still wins in-week.

Correction of error – last time you were the mechanism (G1)

Elementum UAT: five users, ten scenarios, I sent screenshots instead of digging. Concurrent load crashed; sign-off missed. COE: implicit requirement (performance at the customer's shape) was not in the entrance gate. Mechanism since: numeric load criteria on every critical UAT; Flexport launch proof 500 containers / 20 min / 99.6% connectivity *before* customers. I was the mechanism that failed; I wrote the mechanism that replaced me.

Estimation – how many seller-fulfilled orders? (see §15)

Do not bluff a Walmart actual. Walk a Fermi, label it illustrative, then say “I would take the order-week count from the WBR denominator on day one.”

You have 10 minutes left – convince me you are L7 not L6

L7 is owning the **policy instrument** (k , PY, Andon, peak regime) and the **org interface** (DS owns the model, I own whether it touches Maya; Product owns the SSS toggle, I own the insurance contract). L6 is running the WBR well. I have run the WBR (Flexport). I have also flipped a socialized roadmap with a two-column table (\$5M, 66%), published tolerances that stopped go-lives (0.5%), and throttled my own launch. That is the step. Gap: I have not held a large direct-report org — I will not pretend that is the differentiator for this pod.

Section 14 – Case interviews (8-12 min scripts)

Format they expect: **clarify (60–90s)** → **framework (60s)** → **math checkpoint** → **recommendation with risks**.
Speak the clarifying questions out loud; do not dive.

Case 1 – Design the dashboard (blank page)

Situation. “You’re the new Senior Manager. There is no trusted Tuesday view. Design it.” **Clarify.** Who sits Tuesday? What is already in the lake (promise_log at display)? What floors are policy vs aspiration? Week definition Mon–Sun? **Framework.** Audience / cadence / grain / input vs output / RAG / PY / action log / refuse-list (Part 8.5). **Math.** Wk 33 PY = $0.451 \times 0.954 = \mathbf{0.430}$. Floor O1 95%. WoW RAG $\pm 5\%$ relative.

Recommendation. Ship page 1 this week even if ugly; freeze seven definitions (Flexport move); twins and BWI in days 31–60. Risk: two dashboards — I kill the side deck.

Case 2 – SSS adoption stall

Situation. I1 stuck ~40–45% of SFS GMV; I10 activations look fine; 30-day retention is the unknown. **Clarify.** Funnel drop-off: cost vs control vs scorecard distrust? What % of enrolled GMV is full-bundle vs toggle-only? Wave criteria exist? **Framework.** Insurance contract (Part 3) + Flexport wave gates (A1) + I10 paired with retention. **Math (ILLUSTRATIVE).** If unenrolled GMV is 55% and those sellers run 2× the late-ship rate of enrolled, the portfolio O4 is dominated by the unenrolled tail — enrollments that do not retain do not move O1. **Recommendation.** Rebuild SSS as AHT-equivalent + transit authority + protected label; Seller Center line “SSS would raise your GMV by \$X”; wave entry = shadow-run their orders; hold waves that miss retention. Do not buy I1 with I10 vanity. Risk: label-purchase antagonizes negotiated-rate sellers — tier protection with bundle depth (proposal 4 failure mode).

Case 3 – Peak regime

Situation. It is October. Last year O1 broke in week 2 of peak. **Clarify.** Was k frozen from summer? Did σ inflate? Any freeze window? Same-day included? **Framework.** Proposal 10 + Q4 tab: frozen summer buffers → model accuracy **91.9%** (illustrative) when $\sigma \times 1.45$. **Math.** Re-solve k at peak σ ; show speed cost vs 95% floor. Policy B vs A still holds *inside* the regime. **Recommendation.** Publish peak promise policy in October: named lane classes, k, expected O3 hit, Andon rules, change-freeze (Sephora UDC analog). Success = O1 $\geq 95\%$ with zero emergency national pads. Risk: over-padding kills the quarter’s penetration — hence solved k, not guessed.

Case 4 – One-seller fire (8% GMV, worst handler)

Situation. Seller Ops pages you. Fast-badge complaints on this seller’s SKUs. **Clarify.** Measurement: first scan vs phantom label? One node or all? SSS enrolled? Scorecard OTD vs our O1? **Framework.** Andon → surgical badge pull → evidence conversation → SSS off-ramp → WFS if structural. **Math (ILLUSTRATIVE).** 8% GMV $\times 20\%$ late-ship vs 5% floor: they can be most of a portfolio O4 miss even if everyone else is clean. Buffering the *lane* by +0.5d to hide them spends O3 on 100% of lane volume. **Recommendation.** Pull fast badge on *their* cells in 24h; do not national-pad. 30-day plan with return bar. Risk: commercial blowback — pre-wire the Marketplace GM with PY math, not adjectives.

Case 5 – Same-day metro launch

Situation. “Three metros, seller-fulfilled same-day, this quarter.” **Clarify.** Store-as-node coverage? Observed handling in *hours*? Courier windows including Saturday? Cutoff vs intraday order curve? Shadow possible? **Framework.** Gates before geography (C3 analog: 98% bar). Trial: 100 packages at full standards, badge withheld (Amazon SFP analog). Shadow 4 weeks; promote if $O1 \geq 95\%$ and $O6$ does not rise. **Math.** At same-day, buffer ≈ 0 ; eligibility *is* the promise. A 0.22σ California intra lane in the model can clear; a 1.60σ mountain lane cannot — do not pick metros by GMV alone. **Recommendation.** Pick metros where store density + proven sellers + cutoff recover same-day *on the calendar*. Fallback = next-day, never a broken same-day. Risk: Product wants the three largest GMV metros; I will show σ .

Case 6 – Accuracy vs speed executive debate

Situation. VP-A: “Accuracy is trust, raise the floor to 97%.” VP-B: “Speed is GMS, cut buffers.” You have 10 minutes. **Clarify.** Priced breach cost? Measured elasticity ($O7$)? Current k ? Mix-adjusted $O1/O3$? **Framework.** Promise Yield + Solver economics. **Math.** Illustrative: A 7.6% fast / 95.8% acc; B 39.6% / 95.0%; D 22.9% / 97.8% profit-opt at \$12 breach / \$6.50 margin. Floor 95% is slack if \$12 is right; if breach is actually \$6, D moves down and a 97% floor spends conversion. **Recommendation.** Do not pick a VP. Price the breach with Finance this quarter (contact + refund \$ + LTV discount from broken-promise A/B). Set floor at or slightly below economic optimum. Meantime: harvest BWI / Policy B — speed *without* cutting the floor. Risk: a meeting that “compromises” at 96% and +3 pts $O3$ with no mechanism — I will refuse a number unattached to k .

Section 15 – Fermi / estimation

Always: **state assumptions, compute out loud, label ILLUSTRATIVE, say how you'd replace with a WBR pull on day one.** They are testing structure, not your knowledge of Walmart GMV.

How many seller-fulfilled orders per week?

Assume (illustrative): US e-comm Walmart share, Marketplace mix, SFS vs WFS split — **do not invent a company number.** Structure: (weekly Marketplace orders) $\times$ (% SFS). Checkpoint: the WBR O1 denominator is delivered SFS orders for that week — I will use that, not a Fermi, in the job. In the 12-lane model, sessions $\sim 18.5\text{M}/\text{week} \times \text{CVR} \sim 7\% \approx$ **order of 1M orders/week** — that is the **model's** book, not Walmart's. Say that sentence.

Cost of a 4-point O1 miss at peak for one week

ILLUSTRATIVE: 1,000,000 SFS orders (model scale) $\times$ 4% extra breaches = 40,000 breaches $\times$ \$12 = **\$480K** direct in the Solver pricing, plus unpriced LTV. At real Walmart scale the dollars scale with the denominator; the *method* does not. I would rather contain 40k cell-level Andons than buy a national pad that also costs O3 $\times$ elasticity.

How many PDP views sit behind a 45% order-weighted fast share?

If converters prefer fast promises, page-view fast share $<$ order-weighted. ILLUSTRATIVE: order-weighted O3 = 45%; if fast pages convert 1.2 $\times$ slow pages, view-weighted $\approx 45\% / (0.45 \times 1.2 + 0.55 \times 1) \approx$ **41%**. Amazon cares about the view. I would report both.

How much k must rise to hold 95% if σ inflates 1.45 $\times$?

Under normals, k is in units of σ , so **k stays 1.645** if we re-measure σ . The *days* of buffer rise 1.45 $\times$. The failure is freezing *days* (Policy A days) while σ moves — Q4 tab. Speak “re-estimate σ , keep k, or re-solve k if the floor vs profit trade changed.”

Size of prize: +5 pts O3 on already-eligible volume

ILLUSTRATIVE: 1M orders $\times$ 5% = 50k newly-fast orders $\times$ \$6.50 margin $\times$ (elasticity via O7 $\sim 2.4\%$ lift on the fast cohort, not on all orders — don't multiply 2.4% $\times$ all GMV). Honest version: use O7 as CVR delta on exposed items, times sessions on those items. I will not quote a GMS number I cannot derive from O7's experiment grain.

How many sellers to shadow-onboard per week for I10 = 305 retained?

If 30-day retention is 80%, activations needed $\approx 305 / 0.80 \approx$ **381 starts/week** (illustrative). If retention is 50%, you need 610 starts to print 305 retained — that is why I10 without retention is a costume.

Section 16 – Questions they ask because they are ex-Amazon

Short answers. These words are the interview.

Who is the single-threaded owner?

Me, for Promise Yield and O1, and for the action log. Input owners are single-threaded per line (Adoption PM on I1/I10, Promise DS on I5/I6, Seller Ops on I3, Carrier PM on I4). A metric with two owners has zero.

Work backwards – what's the press release?

See §13 PR/FAQ for k. For SSS: "Sellers who give Walmart handling + transit + label get carrier-delay insurance and a GMV line in Seller Center. Everyone else keeps control and the risk."

Mechanisms vs effort?

Effort is "we coached 200 sellers." Mechanism is Andon in 24h, k in the WBR header, wave gates that can hold a cohort, mock tolerances. Flexport 7→68 was mechanisms. I do not want credit for effort.

Name three input metrics you can move *this week*

(1) I6/k on a 10-lane shadow (no customer display). (2) I3 on one seller cell — badge pull + handling reset. (3) I9 cache-hit if p99 is the display problem. I cannot honestly move I1 5 points this week; I can move the funnel artifact and the shadow.

So what / size of prize?

Always: volume × delta × \$ per unit, minus breach cost, grain named. Solver D vs A: **+\$255K/week** profit on the illustrative book (~\$13M/yr) plus fast-share movement. I will not say "huge" without that shape.

Two-way door or one-way?

Two-way: shadow k, throttle robots, pull one seller's badge, cutoff pilot on SSS nodes. One-way: JDA cutover, site-wide displayed buffer policy, legal uniform pad, peak k announced to customers. One-way gets mocks (7%→0.3%).

What are your tenets? (write 5)

1. Promise is underwriting; buffer is priced risk.
2. One owner of k; never uniform days.
3. Manage inputs, report outputs; green output + red input is not a win.
4. Scan-level grain; national averages are a last resort.
5. Gates can stop the line (98%, 0.5%, wave ETA 80%).

COE – when do you write one?

Accuracy incident ≥ published bps (I'd set with the GM), or any silent mechanism failure (G1 UAT, G2 RF devices, G3 security review). Blameless; ends in a mechanism (heartbeat, entrance criterion, kill switch), not a "try harder."

Andon – who can pull the cord?

Any input owner can pull fast badge or widen buffer on a **cell** that trips CUSUM ($n \geq 30$, $2 \times$ baseline). I cannot require a Tuesday meeting to stop a lying promise. Restore when the chart recovers. False-trip budget $< 5\%$ (proposal 5).

Disagree and commit after the room

B2 pattern: math in private, commit in public. SSS disagreement: I commit to your enrollment target; I still print retention and PY. Next Monday is the appeal, not Slack.

Dive deep until the grain is an order / lane / seller

If I am still talking “US Marketplace accuracy” after two follow-ups, I am failing the LP. Stop at the cell that explains the WoW delta. I4 stays lane-grain on purpose — I will say so rather than pretend it is P(package).

Input metrics you own vs DS owns

DS owns model MAE, distributions, recommended lag. I own whether a new model may set a *displayed* EDD (shadow bar, 4-week champion-challenger). I own k. Split is the Flexport contract: they own the model, I own the outcome, we share the WBR instrument.

What would you put in the annex, not page 1?

MAE histograms, p99 graphs, seller rank lists, weblab readouts, take-rate, NPS, GMV. Page 1 is PY, O1–O7, I1–I10, RAG, five callouts, action log. Annex is how you earn trust with Dive Deep without blinding the VP.

How do you run tenets vs a dashboard?

Tenets decide what is *allowed* (no uniform pad, no ungated I8). Dashboard decides what is *true* this week. When they conflict, tenets win and the dashboard records the PY we spent (Legal pad example).

Bar-raiser closer – “anything else I should know?”

I will not invent a marketplace or a 12-person org. I will invent nothing on the resume. What I will do in ninety days is move one output with a mechanism your WBR can audit — k, PY, or an Andon that stops a lying cell in 24 hours. If that's the job, I'm here for it.

Drill card – 30-minute HM hit list (ex-Amazon)

If you only re-read one page the morning of Sep 9: Part 8.5 wireframe + these fifteen, in likely order:

1. Tell me about yourself (§1)
2. Why this role / why Marketplace (§1)
3. Design the WBR from a blank page (§12.1)
4. Improve accuracy without slowing the promise (§2)
5. Disagree with manager / Have Backbone (§11 B2)
6. Dive deep on a metric that looked fine (§11 F2)

7. How do you move SSS (§2 / *case 2*)
8. Output green, input red (§12.3)
9. No marketplace on the resume (§13)
10. Two years supervisory (§5 / §13)
11. Set buffers / show me Solver (§12.11)
12. North star vs guardrails (§12.5)
13. Peak, accuracy –4 pts (§13)
14. Why you vs an SFP transfer (§13)
15. Your questions for me (§8 Q1–Q2)

Appendix – Solver variables and calculations

This appendix is the formula sheet for the two Excel Solvers in `Walmart_WBR_R2596147_Solver_Model.xlsx`, verified by `model_reference.py`. It does not replace Part 5. It is what you open when they say “*show me Solver*” and then “*walk the math.*”

Every number below is **canonical and illustrative** — the 12-lane book and the Solver-2 coefficients from `model_reference.py` / `build_wbr_model.py`. They are not Walmart production, not resume metrics, and not the FY27 Q2 QBR figures. Do not quote the dollar profit as a Walmart actual. Quote the mechanism.

There are **two Solvers** plus a **peak re-solve** that reuses Solver 1:

Engine	Sheet	Decision variables	Objective	Binding constraints
Solver 1 — Buffers	Solver 1 – Buffers	12 lane buffers b_i	Maximize weekly profit N25	Order-weighted accuracy $\geq 95\%$; $0.05 \leq b_i \leq 4$
Solver 2 — Allocation	Solver 2 – Allocation	7 initiative weeks w_j	Maximize Promise-Yield gain H12	$\sum w_j \leq 60$; $\sum \text{spend}_j \leq \$900,000$; $0 \leq w_j \leq w_j^{\text{max}}$
Q4 Peak re-solve	Scenario – Q4 Peak	same b_i , new σ and sessions	Same profit objective, new regime	Same floor; $\sigma_{\text{peak}} = 1.45 \sigma$; sessions $\times 2.2$

Excel dialog (type once; `openpyxl` cannot persist Solver parameters): Solver 1 uses **GRG Nonlinear + Multistart** (the 2-day badge term creates local optima). Solver 2 is concave — GRG is global.

Solver 1 – Buffer optimization

What it is. Price the delivery-promise as insurance. The decision is how many extra days b_i to add on each of 12 illustrative lane cohorts. The engine is Problem 2 in `model_reference.py` (maximize profit subject to an accuracy floor). Problem 1 (minimize average promise days subject to the same floor) is the closed-form sibling — policy C — not what Excel GRG runs.

Spoken one-liner. “Buffer is priced risk. I do not pad. I solve b_i so the last day of protection on every lane is worth what it costs.”

S1.1 The physical identity (one order)

The displayed promise on Maya’s blender page is

$$EDD = t_{\text{order}} + h + \hat{t} + b$$

The promise is **kept** if and only if the combined execution error is no larger than the buffer:

$$\varepsilon = (h_{\text{actual}} - h) + (t_{\text{actual}} - \hat{t}) \leq b$$

Model $\varepsilon \sim N(0, \sigma^2)$ per lane cohort, so

$$P(\text{kept}) = \Phi(b / \sigma)$$

where Φ is the standard-normal CDF (`NORM.DIST(b, 0, σ, TRUE)` in Excel; `NormalDist().cdf` in Python).

****Worked identity — California Intra / Same-Day, policy D.**** $h = 0.30$ d, $\hat{t} = 0.55$ d, $b^* = 0.54$ d. Promise days $d = 0.30 + 0.55 + 0.54 = \mathbf{**1.39**}$ d (fast). $z = 0.54 / 0.22 = \mathbf{**2.45**}$. $P(\text{kept}) = \Phi(2.45) = \mathbf{**99.3%**}$. Same lane under naive $b = 1.10$: $d = 1.95$ d, $z = 5.00$, $P(\text{kept}) \approx 100\%$ — you bought four extra sigmas of protection that conversion already paid for.

S1.2 Every variable (Solver 1)

Assumptions (yellow cells B3:B9 – editable; Solver re-prices off them)

Symbol	Name (sheet)	What it is	Units	Canonical value	How it is calculated / where it comes from	How it is used
m	margin_per_order	Contribution margin per marketplace order	\$ / order	\$6.50	Assumption. Not a Walmart P&L line.	Profit per order = $m - (1 - \Phi(z))c_b$

Symbol	Name (sheet)	What it is	Units	Canonical value	How it is calculated / where it comes from	How it is used
c_b	breach_cost	Cost of a broken promise: WISMO contact + refund/return risk + trust decay	\$ / breach	\$12.00	Assumption to be priced with Finance. Bundle, not a ticket average.	The premium side of the insurance. Raises optimal accuracy.
β	beta (speed elasticity)	How fast conversion decays as the promise gets slower	1 / day	0.16	In $CVR(d) = CVR_0 \cdot \exp(-\beta \cdot (d-1)) \cdot (1 + \gamma \cdot badge(d))$.	One extra promise-day multiplies CVR by $\exp(-0.16) \approx 0.852$ (before the badge term).
CVR_0	CVR0 (at 1-day promise)	Conversion at a 1-day EDD, before badge	fraction	0.072 (7.2%)	Baseline of the CVR curve.	Anchors orders $O_i = S_i \cdot CVR(d_i)$.
γ	gamma (2-day badge bonus)	Extra conversion from the soft ≤ 2 -day badge	dimensionless	0.11	Multiplies CVR by $(1 + \gamma \cdot badge(d))$.	Makes the 2-day cliff economically real. Creates local optima — hence Multistart.
τ	tau (badge softness)	Logistic width around the 2-day cliff	days	0.15	$badge(d) = 1 / (1 + \exp((d-2)/\tau))$	Softens a hard 2-day step so GRG can climb. Smaller $\tau \rightarrow$ sharper cliff.
A^*	accuracy_floor	Portfolio promise-accuracy constraint	fraction	0.95	Policy floor. Same number as O1's 95% line.	Constraint: order-weighted $\Phi \geq A^*$. At policy D it is slack*.

Lane inputs (columns B-E, rows 13-24 – not decision cells)

Symbol	Name	What it is	Units	How it is calculated	How it is used
S_i	Weekly PDP sessions	Product-page views on that cohort in a week	sessions / week	Canonical 12-lane table (below). Exogenous — Solver does not choose demand.	$O_i = S_i \cdot CVR(d_i)$
h_i	Handling h	Configured handling time (Dallas taping the box)	days	Lane table	Enters $d_i = h_i + \hat{t}_i + b_i$
$\hat{t}_i$	Pred. transit $\hat{t}$	Model's predicted transit	days	Lane table	Enters d_i . Error vs actual sits inside ϵ .
σ_i	σ error	Std. dev. of combined handling + transit error	days	Lane table; $\sigma \in [0.22, 1.60]$	Unit of risk. $z_i = b_i / \sigma_i$. Garbage σ makes k confidently wrong.

Canonical 12-lane book (LANES in `model_reference.py`):

Cohort i	S _i (sessions/wk)	h _i (d)	f _i (d)	σ _i (d)	h+f (d)
California Intra / Same-Day	1,100,000	0.30	0.55	0.22	0.85
Northeast Metro / 2-Day	2,100,000	0.45	0.85	0.28	1.30
West Metro / Ground	2,550,000	0.50	0.95	0.30	1.45
Southeast Metro / Ground	2,300,000	0.55	1.05	0.42	1.60
Midwest Metro / Ground	1,900,000	0.60	1.10	0.45	1.70
Florida Metro / Ground	1,620,000	0.65	1.25	0.55	1.90
Great Lakes / Ground	1,420,000	0.70	1.45	0.62	2.15
Pacific NW / Ground	1,100,000	0.75	1.70	0.70	2.45
Southeast Regional / Ground	1,380,000	0.90	2.10	0.95	3.00
Texas Rural / Ground	1,060,000	1.00	2.60	1.20	3.60
Northeast Rural / Ground	960,000	1.10	2.90	1.35	4.00
Mountain Rural / Ground	1,020,000	1.20	3.40	1.60	4.60

Decision variables (column F – yellow, seeded at 1.10)

Symbol	Name	What it is	Units	Bounds	How Solver sets it	How it is used
b _i	Buffer b (DECISION)	Extra days Walmart adds because h and f̂ will miss	days	0.05 ≤ b _i ≤ 4.0	GRG changes F13:F24 to max N25	d _i , z _i , Φ, profit

Column G is **not** a decision cell. It is the pre-computed policy-D vector from `model_reference.py`, used only to verify Solver landed:

$$b^D = (0.54, 0.58, 0.61, 0.90, 0.96, 1.14, 1.25, 1.37, 1.72, 2.03, 2.20, 2.46)$$

Policy C (analytic min-days) for the same book:

$$b^C = (0.49, 0.59, 0.62, 0.80, 0.84, 0.96, 1.04, 1.12, 1.33, 1.46, 1.51, 1.53)$$

Derived per-lane quantities (columns H-Q – formulas, not typed)

Sequence is left-to-right. Excel refs are the live sheet (\$B\$3 = margin, ...).

Symbol	Sheet column	Formula	Units	What it means
d _i	H Promise days	d _i = h _i + f̂ _i + b _i (=C+D+F)	days	The printed EDD length. Fast iff d _i ≤ 2.
badge(d)	I Badge factor	1 / (1 + exp((d-2)/τ)) (=1/(1+EXP((H-2)/\$B\$8)))	0-1	Soft 2-day badge. ≈1 when d < 2; ≈0 when d > 2.

Symbol	Sheet column	Formula	Units	What it means
CVR(d)	J CVR	$\text{CVR}_0 \cdot \exp(-\beta \cdot (d-1)) \cdot (1+\gamma \cdot \text{badge}(d))$	fraction	Sessions that become orders.
O _i	K Orders / wk	$O_i = S_i \cdot \text{CVR}(d_i) \text{ (} =B*J \text{)}$	orders / week	Endogenous. Faster d → more orders.
z _i	Q z=b/σ	$z_i = b_i / \sigma_i$	sigmas	Protection in risk units, not calendar days.
Φ(z _i)	L P(kept)	$\Phi(z_i) = \Phi(b_i / \sigma_i)$ (=NORM.DIST(F,0,E,TRUE))	fraction	Lane accuracy.
φ(z _i)	(inside P)	$\phi(z) = \exp(-z^2/2) / \sqrt{2\pi}$ (NORM.DIST(...,FALSE))	1 / sigma	Normal PDF. Marginal accuracy per sigma.
φ(z _i)/σ _i	P φ(z)/σ	$\phi(b_i / \sigma_i) / \sigma_i$	1 / day	KKT statistic. Constant across lanes at policy C only (0.1549).
π _i	M Profit / order	$\pi_i = m - (1-\Phi(z_i)) c_b \text{ (} =\$B\$3 - (1-L)*\$B\$4 \text{)}$	\$ / order	Expected profit of one order on this lane.
Π _i	N Weekly profit	$\Pi_i = O_i \cdot \pi_i \text{ (} =K*M \text{)}$	\$ / week	Lane contribution to the objective.
F _i	O Fast ≤2d	$F_i = 1 \text{ if } d_i \leq 2, \text{ else } 0$	0/1	Feeds fast-share and Promise Yield.
ε	(model, not a column)	$\varepsilon \sim N(0, \sigma_i^2)$	days	The random miss. Kept iff ε ≤ b.

Portfolio KPIs (rows 25–32 – what you speak)

Let $T = \sum_i O_i$ (cell K25).

Symbol	Sheet	Formula	Units	Canonical policy D	How used
Π	N25 / B28 objective	$\Pi = \sum_i \Pi_i$	\$ / week	\$6,074,391	Solver 1 maximizes this.
A	B29 constraint	$A = \sum_i O_i \Phi(z_i) / T$	fraction	97.78%	Must be ≥ A* = 95%. Slack at D.
$\bar{d}$	B30	$\bar{d} = \sum_i O_i d_i / T$	days	2.927	Order-weighted average promise.
f	B31 Fast share	$f = \sum_i O_i F_i / T$	fraction	22.9%	O3 analog on this book.
PY	B32 Promise Yield	$PY = A \times f$	fraction	22.4%	Un-gameable joint KPI.
T	K25	$\sum_i O_i$	orders / week	974,492	Denominator of every portfolio rate.

Dual / policy instruments (not Excel changing cells, but you must be able to name them)

Symbol	Name	What it is	Units	Canonical	Calculation	Use in the room
k	Portfolio buffer dial	Single number: $b_i = k \cdot \sigma_i$	dimensionless	$k = \Phi^{-1}(0.95) = 1.64485$	Excel <code>NORMSINV(\$B\$9)</code> ; Python <code>NormalDist().inv_cdf(0.95)</code>	Policy B. The 60-second free lunch. Re-solved per regime.
λ	Accuracy shadow price	Lagrange multiplier on $A \geq A^*$	\$ per accuracy-point (P2); days per accuracy-point (P1)	P1: $\lambda = 6.46$; P2 / Solver: $\lambda = 0$	Bisection in <code>solve_p1</code> / <code>solve_p2</code> until the floor binds or $\lambda=0$ is feasible	$\lambda=0$ at D is “the floor is slack.”
$\phi(z)/\sigma$	Equimarginal residual	Last buffer-day’s accuracy per lane	1 / day	0.1549 on every lane at C; not constant at D	$\phi(b_i/\sigma_i)/\sigma_i = 1/\lambda$ at C	If this is not flat, you are not at the min-days optimum.

S1.3 Calculation sequence (what GRG actually walks)

1. **Read assumptions** $m, c_b, \beta, CVR_0, \gamma, \tau, A^*$ and the 12 ($S, h, \hat{t}, \sigma$).
2. **Guess** a buffer vector $b \in [0.05, 4]^{12}$ (seed: all 1.10).
3. **For each lane, in order:** $d \rightarrow \text{badge} \rightarrow CVR \rightarrow O \rightarrow z \rightarrow \Phi \rightarrow \pi \rightarrow \Pi \rightarrow F$.
4. **Aggregate:** $\Pi, A, \bar{d}, f, PY, T$.
5. **Feasible?** $A \geq 0.95$ and bounds hold.
6. **Step b** (GRG) to raise Π . Multistart because $\text{badge}(d)$ is S-shaped around 2 days.
7. **Stop** when Π cannot rise without breaking the floor. Verify column $F \approx$ column G .

Python’s `solve_p2` does the same job by pricing: each lane maximizes $CVR(d) \cdot \pi + \lambda \cdot CVR(d) \cdot (\Phi - A^*)$, then bisects λ until $A \geq A^*$. At the canonical economics, **unconstrained** ($\lambda = 0$) already clears 97.78%, so Solver and Python both return the unconstrained profit optimum.

S1.4 Worked numeric example – California Intra, policy D vs A

Inputs (canonical): $S = 1,100,000, h = 0.30, \hat{t} = 0.55, \sigma = 0.22, m = 6.50, c_b = 12, \beta = 0.16, CVR_0 = 0.072, \gamma = 0.11, \tau = 0.15$.

Policy D ($b = 0.54$):

1. $d = 0.30 + 0.55 + 0.54 = 1.39$
2. $\text{badge} = 1 / (1 + \exp((1.39-2)/0.15)) = 1 / (1 + \exp(-4.0667)) = \mathbf{0.983}$
3. $CVR = 0.072 \cdot \exp(-0.16 \times (1.39-1)) \cdot (1 + 0.11 \cdot 0.983) = 0.072 \cdot \exp(-0.0624) \cdot 1.1081 = \mathbf{0.07496}$
4. $O = 1,100,000 \times 0.07496 = \mathbf{82,456}$ orders/wk

$$5. z = 0.54 / 0.22 = 2.4545; \Phi(2.4545) = \mathbf{99.29\%}$$

$$6. \pi = 6.50 - (1 - 0.9929) \cdot 12 = \mathbf{\$6.415}$$

$$7. \Pi = 82,456 \times 6.415 = \mathbf{\$528,985 / wk}$$

$$8. \text{Fast? } 1.39 \leq 2 \rightarrow \mathbf{yes}$$

Policy A (same lane, $b = 1.10$):

$$1. d = 1.95; \text{badge} = 0.583; \text{CVR} = 0.06581; O = \mathbf{72,392}$$

$$2. z = 5.00; \Phi \approx 100\%; \pi = \$6.50; \Pi = \mathbf{\$470,545 / wk}$$

3. Fast? $1.95 \leq 2 \rightarrow \mathbf{yes}$, but you burned conversion ($82,456 - 72,392 = \mathbf{+10,064}$ orders) to buy protection you already had at $z=2.45$.

Decision use. This is the free lunch on a tight lane: shrinking b from 1.10 $\rightarrow$ 0.54 keeps the promise honest and **raises** weekly profit on this cohort by $\$528,985 - \$470,545 = \mathbf{\$58,441}$. Uniform padding is a conversion tax.

S1.5 Worked numeric example – Mountain Rural, policy D (the opposite lane)

$$S = 1,020,000, h = 1.20, \hat{t} = 3.40, \sigma = 1.60, b^D = 2.46.$$

$$1. d = 1.20 + 3.40 + 2.46 = \mathbf{7.06} \text{ d — never in the fast band}$$

$$2. \text{badge}(7.06) \approx 0; \text{CVR} = 0.072 \cdot \exp(-0.16 \times 6.06) = \mathbf{0.02731}$$

$$3. O = \mathbf{27,851 / wk}$$

$$4. z = 2.46 / 1.60 = 1.5375; \Phi = \mathbf{93.8\%} \text{ — below the 95\% floor on this lane alone}$$

$$5. \pi = 6.50 - (1 - 0.938) \cdot 12 = \mathbf{\$5.755}; \Pi = \mathbf{\$160,282 / wk}$$

Decision use. High- σ lanes get **fewer** sigmas ($z = 1.54$ vs California's 2.45), not more. Protecting Mountain into 95% on its own would require a thicker b , a slower d , and a conversion hole. The portfolio still clears 97.8% because California / Northeast Metro over-perform. That is the underwriting: do not spend the book's risk capital making a chaotic lane look like a metro lane. Pull it from the fast band or route it to WFS (proposal 9).

S1.6 Four policies – the six numbers, with the arithmetic underneath

Computed by `model_reference.py` on this exact book (full precision; Part 5 rounds for the room):

Policy	How b_i is set	Accuracy	d	Fast share	Orders/wk	Profit/wk	PY
A Naive uniform	$b_i = 1.10$ for all i	95.78%	3.086	7.6%	948,619	\$5,685,617	7.3%
B σ -proportional	$b_i = 1.64485 \cdot \sigma_i$	95.00%	2.753	39.6%	1,006,947	\$5,940,986	37.7%
C Min-days (P1)	KKT: $\phi(z_i)/\sigma_i = 0.1549$; $\lambda = 6.46$	95.83%	2.807	22.4%	1,000,211	\$6,001,032	21.5%
D Profit opt (P2 = Excel Solver)	$\lambda = 0$; $b = b^D$ above	97.78%	2.927	22.9%	974,492	\$6,074,391	22.4%

Deltas you may say (do not mix them):

- **B vs A (the free lunch):** fast share +32.0 pts at the **same** 95% floor; profit +\$255,369/wk. This is the +\$255K line in Part 5.
- **D vs A (what Solver actually finds):** accuracy 95.78% → 97.78%; fast +15.3 pts; orders +25,873/wk; profit +\$388,773/wk ≈ **\$20.2M/yr** on this million-order book. This is the “Solver-1 ~\$20M/year” line in the Q-bank.

****HM lens.**** If they ask “what did Solver do?”: it moved F13:F24 from a flat 1.10 to b^D , raised N25 from \$5.69M to \$6.07M, and ****left the 95% floor unused****. Either Finance’s $c_b = \$12$ is high, or the floor is slack insurance. Forcing that question is the Senior Manager job.

S1.7 Problem 1 (analytic) – every extra symbol

Excel does not run Problem 1. You still need it: it is why $\phi(z)/\sigma$ exists.

$$\min_b \sum_i v_i b_i \text{ s.t. } \sum_i v_i \Phi(b_i/\sigma_i) \geq A^* \sum_i v_i$$

KKT: $\phi(z_i) / \sigma_i = 1/\lambda$ (constant). Solve $\phi(z_i) = \sigma_i / \lambda$ by bisection on z_i , then $b_i = z_i \sigma_i$. Bisect λ until the floor binds. Verified: $\phi(z_i)/\sigma_i = 0.1549$ on every lane, spread $< 10^{-15}$.

Corollary you speak: higher $\sigma \rightarrow$ fewer sigmas of protection (z falls). Past a σ threshold the interior solution vanishes — corner $b \rightarrow 0$, pull the lane.

Solver 2 – Investment allocation

What it is. Allocate a scarce engineering budget across seven initiatives to maximize **Promise-Yield gain**, with diminishing returns. This is not a buffer model. It is the “where does the next week go?” model.

Spoken one-liner. “First dollars go where adoption compounds. Equal-split is a committee tax.”

Objective (cell H12):

$$\begin{aligned} \max_w \sum_j a_j \cdot (1 - \exp(-c_j \cdot w_j)) \text{ s.t.} \\ \sum_j w_j \leq 60, \\ \sum_j \text{cost}_j w_j \leq 900,000, \\ 0 \leq w_j \leq w_j^{\max} \end{aligned}$$

Excel: Set Objective H12 → Max; Changing F5:F11 ; F12 <= B14 ; G12 <= B15 ; F5:F11 >= 0 ; F5:F11 <= B5:B11 . GRG Nonlinear.

S2.1 Every variable (Solver 2)

Symbol	Name (sheet)	What it is	Units	Canonical	Formula / sequence	How used
w_j	Weeks allocated (DECISION)	Engineering-weeks given to initiative j	weeks	Seed $\approx 60/7$; opt in column I	GRG changes F5:F11	The only decision vector
$w_j^{\max}$	Max weeks	Capacity ceiling for that workstream	weeks	See table	Input	Constraint $w_j \leq w_j^{\max}$
cost_j	Cost / week	Fully-loaded cost of one week on j	\$ / week	See table	Input	$\text{Spend}_j = w_j \cdot \text{cost}_j$
a_j	a (pts at ∞)	Asymptotic Promise-Yield gain if you staff j forever	PY points	See table	Input (illustrative ranking, not a measured Walmart A/B)	Saturation level of the gain curve
c_j	c (curvature)	How fast initiative j saturates	1 / week	See table	Input	Higher c → fill it early, then stop
Spend_j	Spend	Dollar cost of the allocation	\$	=F*C	$\text{Spend}_j = w_j \cdot \text{cost}_j$	Sum constrained to $\leq \$900k$
g_j	Yield gain (pts)	Promise-Yield points this initiative adds	PY points	=D*(1-EXP(-E*F))	$g_j = a_j (1 - \exp(-c_j \cdot w_j))$	Sum is the objective
$g_j'(w)$	Marginal gain	Extra PY pts from one more week	pts / week	$a_j c_j \exp(-c_j \cdot w_j)$	Derivative of g_j	GRG equalizes this (where not at a bound)

Symbol	Name (sheet)	What it is	Units	Canonical	Formula / sequence	How used
W	Week budget B14	Total weeks you may spend	weeks	60	Editable	$\sum w_j \leq 60$ — binds at the optimum
B	Dollar budget B15	Total dollars you may spend	\$	\$900,000	Editable	$\sum \text{Spend}_j \leq B$ — slack at the optimum (\$102.5k left)
$\sum w$	F12	Total weeks used	weeks	60.0 at opt	=SUM(F5:F11)	Constraint check
$\sum \text{Spend}$	G12	Total spend	\$	\$797,500 at opt	=SUM(G5:G11)	Constraint check
G	H12 objective	Total PY gain	pts	7.50 at opt	=SUM(H5:H11)	What Solver 2 maximizes

Seven initiatives (INITS in build_wbr_model.py) and the pre-computed optimum S2_OPT :

j	Initiative	w_j^{max}	cost_j (\$/wk)	a_j (pts)	c_j	w_j^*	Spend_j^*	g_j^* (pts)	$g_j'(w^*)$
1	SSS default-on (new sellers)	16	12,000	2.40	0.12	12.5	\$150,000	1.8645	0.0643
2	Transit-model refresh cadence	12	18,000	1.40	0.15	8.0	\$144,000	0.9783	0.0633
3	Buffer recalibration program	8	8,000	1.80	0.18	8.0 (at max)	\$64,000	1.3735	0.0768
4	Inventory-feed accuracy onboard	14	15,000	1.60	0.10	9.0	\$135,000	0.9495	0.0651
5	Cutoff-time extension, top nodes	10	14,000	1.20	0.14	7.0	\$98,000	0.7496	0.0631
6	Promise-engine cache / latency	8	20,000	0.80	0.16	4.0	\$80,000	0.3782	0.0675
7	Speed-eligibility assortment expand	14	11,000	2.00	0.08	11.5	\$126,500	1.2030	0.0638
	Total					60.0	\$797,500	7.50	

Interior weeks sit at a common marginal gain $\approx 0.063\text{--}0.067$ pts/week. Buffer recalibration is **at its 8-week max** with a still-high marginal 0.0768 — you would buy more if the cap allowed. Cache is cut at 4 of 8 because infra saturates.

S2.2 Calculation sequence

1. Read (w^{max} , cost, a, c) for seven rows.
2. Guess w_j (seed: $\min(w_j^{\text{max}}, 60/7) \approx 8.57$, capped at 8 for the two max-8 rows $\rightarrow 58.86$ weeks).
3. $\text{Spend}_j = w_j \cdot \text{cost}_j$; $g_j = a_j (1 - \exp(-c_j \cdot w_j))$.
4. Sum weeks, spend, gain.

5. GRG steps w to raise G without breaking 60 weeks, \$900k, or per-row caps.
6. Verify $F \approx I$ within 0.5 week.

S2.3 Worked numeric example – SSS default-on at the optimum

$a = 2.40$, $c = 0.12$, $w^* = 12.5$, cost = \$12,000.

1. Exponent: $c w = 0.12 \times 12.5 = 1.50$
2. Saturation: $1 - \exp(-1.50) = 1 - 0.22313 = 0.77687$
3. Gain: $g = 2.40 \times 0.77687 = \mathbf{1.8645}$ PY points
4. Spend: $12.5 \times 12,000 = \mathbf{\$150,000}$
5. Marginal week: $g' = 2.40 \times 0.12 \times \exp(-1.50) = 0.288 \times 0.22313 = \mathbf{0.0643}$ pts/week

Same formula on cache at $w=4$: $g = 0.80 \times (1 - \exp(-0.64)) = 0.80 \times 0.4727 = \mathbf{0.378}$ pts for \$80k — and the *next* week is worth only 0.0675 pts, which SSS still beats after 12.5 weeks once you also count that SSS unlocks every other lever. That is the ranking, not the decimal.

S2.4 Equal-split vs optimum (the committee tax)

Workbook seed $w = \min(w^{\max}, 60/7)$:

$(8.57, 8.57, 8.00, 8.57, 8.57, 8.00, 8.57)$, $\Sigma w = 58.86$, $\Sigma \text{Spend} = \$824,000$, $G = \mathbf{7.26}$ pts

Optimum: $G = 7.50$ pts, spend \$797,500, weeks **60**. Equal-split leaves **0.24** PY points on the table and spends **more** dollars to get less yield — a committee tax.

Decision use (the sentence): first weeks to SSS default-on and speed-eligibility (adoption compounds); fill buffer recalibration to its 8-week cap (highest curvature); starve promise-engine cache at 4 of 8 (infra saturates). Do not split 60 seven ways.

Q4 Peak – same Solver, new regime

Not a third Solver. The same profit program with two regime multipliers, then a re-solve. Sheet **Scenario – Q4 Peak**.

S3.1 Every additional variable

Symbol	Name	What it is	Units	Canonical	Formula	How used
μ_σ	peak_sigma_mult	How much error spread inflates when carriers saturate	dimensionless	1.45	Editable B3	$\sigma_i^{\text{peak}} = \sigma_i \cdot 1.45$
μ_S	peak_vol_mult	How much PDP sessions scale	dimensionless	2.2	Editable B4	$S_i^{\text{peak}} = S_i \cdot 2.2$
σ_i^{peak}	σ_{peak}	Peak error spread	days	e.g. CA 0.22 $\times$ 1.45 = 0.319	=Solver1!E * \$B\$3	Re-enters $\Phi(b/\sigma_{\text{peak}})$
S_i^{peak}	Sessions (peak)	Peak sessions	sessions / week	e.g. CA 1.1e6 $\times$ 2.2 = 2,420,000	=Solver1!B * \$B\$4	Re-enters O = S-CVR
b_i^D	Buffer (frozen D)	Summer profit-optimum, not re-solved	days	b^D vector	Linked to Solver 1 col G	The failure case: July policy in November
$A_{\text{peak}}^{\text{frozen}}$	Peak accuracy (frozen D)	Portfolio Φ at peak σ , summer b	fraction	91.91%	Same A formula on peak O, Φ	Hard red. Floor is 95%.
$\bar{d}_{\text{peak}}$	Peak avg promise days	Order-weighted d (buffers unchanged $\rightarrow$ same as summer $\bar{d}$)	days	2.93	Same $\bar{d}$ formula	Speed did not save you; the tail moved.
f_{peak}	Peak fast share	Fast share at frozen b	fraction	23.0%	Same f formula	Still looks “fine” while A is red.
$A_{\text{peak}}^{\text{re-solved}}$	Re-solved peak accuracy	Profit opt re-run at $\sigma \times 1.45$	fraction	~96.8%	Pre-computed note on the sheet	Floor recovered.
$\bar{d}_{\text{peak}}^{\text{re-solved}}$	Re-solved avg days	After buying tail protection	days	~3.24	Same	You spent speed to buy the tail.

Symbol	Name	What it is	Units	Canonical	Formula	How used
$f_{\text{peak}}^{\text{re-solved}}$	Re-solved fast share	After the re-solve	fraction	~8.7%	Same	Fast share collapses. Announce in October.

S3.2 Worked numeric example – California Intra at peak, frozen D

Summer: $b = 0.54$, $\sigma = 0.22$, $z = 2.45$, $P(\text{kept}) = 99.3\%$.

Peak, **same buffer**:

1. $\sigma_{\text{peak}} = 0.22 \times 1.45 = \mathbf{0.319\ d}$
2. $S_{\text{peak}} = 1,100,000 \times 2.2 = \mathbf{2,420,000}$
3. $z_{\text{peak}} = 0.54 / 0.319 = \mathbf{1.693}$
4. $P(\text{kept}) = \Phi(1.693) = \mathbf{95.48\%}$ on this *easy* lane — already down ~4 pts
5. Across the book, order-weighted accuracy lands at **91.91%** (verified), **3.1 pts under the floor**

Mountain at peak with frozen $b=2.46$: $z = 2.46 / (1.60 \times 1.45) = 2.46 / 2.32 = 1.06$, $\Phi(1.06) \approx 85.5\%$ — the tail is where the floor dies.

Decision use. A July-calibrated b^D is a different policy in November. Re-solving holds ~96.8% accuracy, slows $\bar{d}$ to ~3.24 days, and drops fast share to ~8.7%. That trade is a memo in October (proposal 10). It is a firefight on Black Friday. The k-dial is seasonal; the floor is not optional.

What you do with this on Tuesday

1. **Open Solver 1.** Point at the four-policy block. Speak B-vs-A (+32 pts fast, +\$255k/wk, same 95%). Then D-vs-A (97.8% accuracy, floor slack, +\$389k/wk $\approx$ \$20.2M/yr on the illustrative book).
2. **If they say “walk a lane.”** California Intra, §S1.4 — eight lines, no slide.
3. **If they say “what about peak?”** Flip to Q4. Frozen D $\rightarrow$ 91.9%. Re-solve $\rightarrow$ 96.8% at 3.24 days. Publish k in October.
4. **If they say “where do you put sixty weeks?”** Solver 2 ranking: SSS and eligibility first, fill buffer-recal to 8, starve cache at 4. Equal-split costs 0.24 PY pts.
5. **Never quote \$6.07M as a Walmart actual.** Quote: *“on a million-order illustrative book, at \$12 breach and \$6.50 margin, the economics already want 97.8%.”*

Source of truth for every digit in this appendix: `model_reference.py` (run it) and the yellow/verify cells on Solver 1 – Buffers , Solver 2 – Allocation , and Scenario – Q4 Peak .